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The Horizon Europe Programme

A strategic assessment of selected items

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The Horizon Europe Programme: A strategic assessment of selected items

Horizon Europe is the EU's key funding programme for research and innovation, containing ambitious commitments to scientific progress, climate neutrality, and improving the EU's competitiveness and growth. This study evaluates the following selected items in Horizon Europe: the evolution of calls and funding, adoption of the Common Model Grant Agreement (CMGA), implementation of the strategic plan, discontinuation of the Future and Emerging Technologies (FET) Flagship, the participation barriers faced by SMEs, and stakeholders' views on the evaluation system. Drawing on surveys and interviews, the study aims to identify both the key strengths and shortcomings of Horizon Europe in terms of these selected items. Lastly, the study suggests ways to mitigate the shortcomings highlighted by stakeholders.

AUTHORS

This study has been written by Haya Al-Ajlani, Vladimir Cvijanović, Marta Mirambell Huguet and Helena Vanzeller Novo (all affiliated with Ecorys) at the request of the Panel for the Future of Science and Technology (STOA) and managed by the Scientific Foresight Unit, within the Directorate-General for Parliamentary Research Services (EPRS) of the Secretariat of the European Parliament.

ADMINISTRATOR RESPONSIBLE

António Vale and Aleksander Väljamäe, Scientific Foresight Unit (STOA)

To contact the publisher, please e-mail stoa@ep.europa.eu

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Executive summary

Rather than adopting a holistic analysis of Horizon Europe, this study delves into six distinct aspects of the Programme, primarily through the lens of stakeholders' opinions and perspectives. The study commences with an analysis of the **evolution of calls and funds from FP7 up to Horizon Europe**. It then undertakes an in-depth examination of selected novelties of Horizon Europe, namely the adoption of the **Common Model Grant Agreement**, implementation of the **strategic plan**, and the **discontinuation of the Future and Emerging Technologies (FET) Flagship**. On a more general note, the study conducts an analysis of the **barriers encountered by SMEs** when applying for funding. **Stakeholders' perceptions of the existing evaluation system** are also examined, shedding light on its effectiveness and areas for improvement.

While these six facets may appear disparate, they are united by a common theme: a focus on areas witnessing transitions from Horizon 2020 to Horizon Europe, with an endeavour to assess the implications of these transformations. By examining these selected dimensions, the study aims to provide insights into the evolving landscape of Horizon Europe and its impact on various stakeholders. Through this targeted analysis, the study seeks to offer a nuanced understanding of the Programme's trajectory and shed light on areas for enhancement and refinement.

The analysis mainly relies on input gathered via an online survey engaging with various stakeholders involved in Horizon Europe and the FET Flagship. Stakeholders of interest included beneficiaries, National Contact Points (NCPs), proposal evaluators, and SMEs. An appropriate number of stakeholders from each category were contacted, aiming to build a representative sample. Stakeholders were asked to fill an online survey, with tailored questions based on respondents' roles and experiences. However, challenges arose in certain stakeholder categories, where poor response rates impeded the statistical significance of the results. To address this limitation and enrich our understanding, targeted interviews were conducted with key stakeholders, including SMEs, FET Flagship beneficiaries, private consultancies offering support in funding applications, and NCPs.

In terms of how **calls have evolved since FP7**, our findings suggest that, despite increased funding, concerns persist about Horizon Europe's capacity to meet the expanding scope of European research goals and address disparities in Member States' R&I performance. While average application success rates have improved compared to Horizon 2020, challenges remain when it comes to equitable distribution of funding, leading to high-quality proposals facing rejection. H2020 and Horizon Europe have grown in terms of participating organisations, although the proportions per type of organisation have not experienced significant shifts. Notably, there is a dedication to social sciences and humanities (SSH), integrated across all pillars and Clusters, to address societal challenges comprehensively. However, Horizon Europe's emphasis on higher technology readiness levels (TRLs) may sideline early-stage research crucial for breakthrough innovations. Nevertheless, the lack of publicly available data on the distribution of funded projects' TRLs presents a significant challenge, making it difficult to draw definitive conclusions regarding the innovation levels achieved through Horizon Europe projects.

To establish a uniform framework for all projects receiving funding under Horizon Europe, a single, universally applicable **Common Model Grant Agreement (CMGA)** was introduced. The primary objective of this CMGA was to simplify the complexity of grant agreements so that the number of articles were reduced and simpler language was used. By standardising grant agreements, the expectation is that projects will be able to collaborate more easily, irrespective of their specific funding programme. Our results show that, according to most of the stakeholders, the CMGA does make it easier to understand the terms and conditions but its ability to simplify administrative burdens and enhance transparency remains debatable. Additional positive changes in the CMGA mentioned by survey respondents include the obligatory communication and dissemination plans, and the data sheet.

Horizon Europe calls are now guided by a **strategic plan** that sets out priorities for the European R&I landscape. Although it marks a significant effort to enhance adaptability, promote policy learning and engage stakeholders effectively, about half of the surveyed stakeholders were unaware of the strategic plan. Furthermore, many respondents expressed uncertainty about its impact on launched calls, with some believing that relevant topics may have been excluded from funding. Nonetheless, those who participated in co-design activities reported positive experiences, emphasising the effectiveness of incorporating valuable input from citizens. These findings highlight the importance of increasing awareness of future co-design activities to ensure comprehensive participation from stakeholders across various disciplines.

Another change witnessed under Horizon Europe is the **discontinuation of the FET Flagship** and the integration of the FET thematic focuses into Horizon Europe's Missions and the European Innovation Council (EIC). The termination of the FET Flagship prompted concerns among stakeholders interviewed and surveyed, who feared disruptions in collaboration, hindrances in developing applications, and potential loss of investment. Although the FET Flagship strived to translate scientific breakthroughs into market innovations, its success in this endeavour remains uncertain. While some projects managed to transform their research into tangible products, establish companies, generate products and file patent applications, others encountered obstacles in achieving market readiness. Interviewed and surveyed FET beneficiaries argue that the current Horizon Europe landscape appears fragmented, making it challenging to replicate the same level of community and strategic alignment without the FET Flagship. Regarding recommendations to address the absence of the FET Flagship, stakeholders emphasised the importance of creating initiatives resembling its long-term research communities. This entails prioritising the establishment of networks and maintaining dedication to specific technological domains to facilitate knowledge exchange. Furthermore, stakeholders propose the implementation of new initiatives within Horizon Europe that encourage interdisciplinary and international collaboration, align research strategies, and effectively tackle scientific and technological challenges.

Efforts to support SMEs' R&I endeavours have been made through initiatives like the EIC Accelerator programme, which provides funding and business acceleration services. However, SME participation in such programmes is hindered by general lack of awareness of funding opportunities and by administrative burdens. Our findings show that SMEs often struggle with a lack of awareness about available funding opportunities or encounter difficulties in crafting competitive proposals. Thus, addressing the fragmented nature of European funding information is imperative. Introducing navigational tools or platforms resembling a compass, as recommended by some stakeholders, could streamline the process for SMEs to identify suitable funding opportunities aligned with their expertise. Furthermore, the crucial role of NCPs in bridging the gap between SMEs and European funding is frequently emphasised by stakeholders. Collaborative efforts between national authorities and NCPs are perceived as essential to improve the dissemination of information about European funding, thereby enhancing accessibility for SMEs across various sectors and regions.

In terms of **perceptions of the evaluation system**, respondents widely agree on the increased emphasis on results within Horizon Europe, signalling a positive shift toward more impactful research. Nevertheless, challenges persist in evaluating projects, particularly in selecting non-mainstream and multi-disciplinary endeavours, and addressing biases favouring certain stakeholders and disciplines. Discrepancies in perceptions of the evaluation process based on country of origin suggest potential inequities in funding distribution. These concerns echo those expressed by interviewees, who highlight the existing bias towards well-known organisations and researchers, and a general lack of transparency in the evaluation and communication of results. Key recommendations include improving the quality of feedback given to non-selected projects in order to increase transparency, especially in programmes such as the EIC Accelerator and Pathfinder; better matching of experts to projects; and streamlining the application process to simplify it and shorten the gap between submission of proposals and communication of results.

The consensus among stakeholders interviewed highlights a prevalent **lack of awareness regarding available funding opportunities**. While beneficiaries of Horizon Europe feel confident applying for funding due to their prior engagement in Horizon 2020 projects or collaboration with experienced consortia, they frequently lack knowledge of funding opportunities beyond the R&I framework programmes. This lack of awareness may also account for the concerns expressed by FET Flagship beneficiaries regarding its discontinuation.

Similarly, approximately half of the consulted stakeholders are **unaware of the modifications introduced under the CMGA or the strategic plan**. This finding suggests that the envisaged streamlining benefits of the CMGA are not being fully realised, particularly in terms of fostering synergies across diverse projects. Furthermore, the lack of awareness regarding the strategic plan and its collaborative co-design initiatives excludes potential stakeholders from actively shaping the orientation of Horizon Europe.

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Introduction

Horizon Europe stands as the European Union's primary funding initiative for research and innovation, boasting a budget of €93.4 billion[1] for its seven-year span (2021-2027). The programme has the ambition to extend beyond mere scientific exploration, aiming to combat climate change, align with the UN's Sustainable Development Goals (SDGs), and fortify the EU's competitiveness and growth trajectory.

In comparison to its predecessors, Horizon Europe introduced several new elements. For example, in an effort to streamline and harmonise the grant agreement process across the various funding programmes of the European Union, Horizon Europe adopted a Common Model Grant Agreement, standardised across different EU action grants (2021-2027). The primary objective of this CMGA is to simplify the complexity of grant agreements such that the number of articles was reduced, and the language used was simplified. By standardising the grant agreement, projects are anticipated to experience improved ease in collaboration, irrespective of their specific funding programme.

Horizon Europe calls are now guided by a strategic plan, which set the priorities for the European research and innovation landscape. The programme also introduced dedicated calls under five missions, each targeting a distinct challenge, ranging from fighting cancer, adapting to climate change, protecting oceans and waters, ensuring soil health and food, to managing climate-neutral and smart cities. The missions are supposed to encourage innovative solutions, research, and technologies to address these pressing issues and drive sustainable development. The Future and Emerging Technologies (FET) Flagship was discontinued under Horizon Europe, with certain of its thematic focuses now included in the missions and the European Innovation Council (EIC).

The EIC supports innovative projects with scale-up potential, particularly those considered too risky for private investment. A substantial portion of its budget, 70%, is allocated specifically for small and medium-sized enterprises (SMEs).

In this report, we examine the impact of these newly introduced items as well as barriers to participation in Horizon Europe, and the perception of the existing evaluation system. This analysis aims to shed light on the strengths as well as the shortcomings of Horizon Europe with the hope of identifying areas for improvement.

This study has been multi-faceted, encompassing the following steps:

- Examination of the evolution of calls and funding under various Framework Programmes
- Assessment of the perceived impact of selected changes introduced under Horizon Europe, including:
- Adoption of the Common Model Grant Agreement
- Implementation of the strategic plan
- Discontinuation of the FET Flagship
- Analysis of the barriers SMEs face when applying to funding
- Delving into stakeholders' perspectives on the existing evaluation system.

[1] As of 1 February 2024, Horizon Europe will have a budget cut of €2.1 billion https://sciencebusiness.net/news/horizon-europe/horizon-europe-budget-be-cut-eu21b-defence-research-gets-eu15b-boost#:~:text=Horizon%20Europe%20will%20have%20its,a%20meeting%20in%20Brussels%20today.

The analysis relies heavily on the input gathered from an online survey, which was specifically designed to engage key stakeholders including beneficiaries, National Contact Points, and evaluators. The survey is further complemented by interviews with key stakeholders to gain further insights into the perceived challenges of Horizon Europe and to explore potential solutions.

The report also presents the methodology employed to identify stakeholders and to administer the online survey and interviews. Subsequently, it provides an overview of the results obtained, categorised by the scope of focus.

1. Methodology and resources used

Given the study's focus, the stakeholders of interest encompassed beneficiaries of Horizon Europe (including SMEs), beneficiaries of the FET Flagship, National Contact Points (NCP) for Horizon Europe, and Horizon Europe proposal evaluators. Approximately 100 individuals from each stakeholder category were contacted for the survey. However, recognising the importance of SME engagement, we extended our outreach efforts to approximately 300 enterprises in order to enhance the response rate of this stakeholder category. The sample selection process of stakeholders was guided by randomisation principles while maintaining proportional representation across countries. Using randomisation in the sample selection process ensures that each stakeholder has an equal chance of being selected, which helps to minimise bias and increase the reliability and validity of the study's findings. Additionally, maintaining proportional representation across countries ensures that the sample is representative of the diverse perspectives and experiences within the target population. This approach enhances the generalisability of the study's results and allows for more robust conclusions to be drawn. Annex 1 further details the selection criteria process.

Contact information for Horizon Europe and FET Flagship beneficiaries was obtained from the publicly available data on The European Research Results CORDIS website[2]. For SMEs, we established contact with the beneficiary via a contact form on Cordis.

An invitation to an online EU survey was sent to these stakeholders. In order to encourage transparent and honest responses, we avoided asking for personal data in the survey, such as requesting personal names, allowing respondents to contribute anonymously. Instead, respondents were asked to provide their country of residence, occupation, and their role in relation to Horizon Europe. This last category played a crucial role as it determined the survey routing logic, thereby specifying the set of questions each respondent would receive. This approach ensures that only relevant questions are presented to each stakeholder, optimising the survey experience. Invitations included a support letter from the STOA Panel of the European Parliament, as well as a privacy notice. Whenever possible, personalised invitations were sent.

Following the closure of the online survey, to account for the low response rates from certain target groups, particularly SMEs and FET Flagship beneficiaries, focused interviews were conducted. These interviews also included private consultancies that aid stakeholders with funding applications and NCPs. Private consultancies provided insights into challenges faced by applicants, while NCPs offered perspectives on the impact of discontinuing the FET Flagship and the current Horizon Europe evaluation process.

Conducted in a semi-structured format, these interviews provided flexibility in questioning. While maintaining a general framework of topics, interviewers could adapt and delve deeper into specific areas based on the interviewee's responses. This approach facilitated a dynamic and responsive exchange of information, enabling a thorough exploration of participants' opinions and experiences.

An overview of the stakeholders interviewed by category and thematic focus is presented in Table 1. For reference, the questions posed in the survey and the interviews are provided in Annexes 2 and 3, respectively.

[2] European Commission, 'Cordis', website, (n.d.)

Table 1 – Overview of interviews conducted

Stakeholder type	Selection criteria	Number of interviews conducted	Thematic focus
FET Flagship beneficiary	FET Flagship projects that are considered success stories in terms of either spin-off projects facilitated through the funding or through groundbreaking innovation. 20 FET Flagship projects were identified, and invitations were sent to the project coordinators.	4	Beneficiaries of three flagship initiatives: Battery 2030+, Quantum technologies Flagship, and The Human Brain Project. Notably, two of these projects have led to the creation of spin-offs.
National Contact Points (NCPs)	National Contact Points responsible for the FET Flagships or the EIE/EIC in the top 10 FET Flagship recipient countries in terms of funded projects: Germany, France, Italy, Spain, Netherlands, Sweden, Austria, Belgium, and Greece. A total of 18 NCPs were contacted.	4	National Contact Points with experience on the European Innovation Council (EIC) and European Innovation ecosystems (EIE).
Private consultancies	Reputable consulting companies with sound experience in providing proposal writing support services to applicants, headquartered in the EU-27. A total of 24 firms were contacted.	4	Private consultancies with experience in writing proposals and supporting in application to Horizon Europe.
SMEs	SMEs from top and mid-tier beneficiaries of Horizon Europe were contacted. Associations representing SMEs were also contacted. A total of 15 invitations were sent out.	5	SMEs with experience in more than one programme as well as experience in Horizon Europe only.

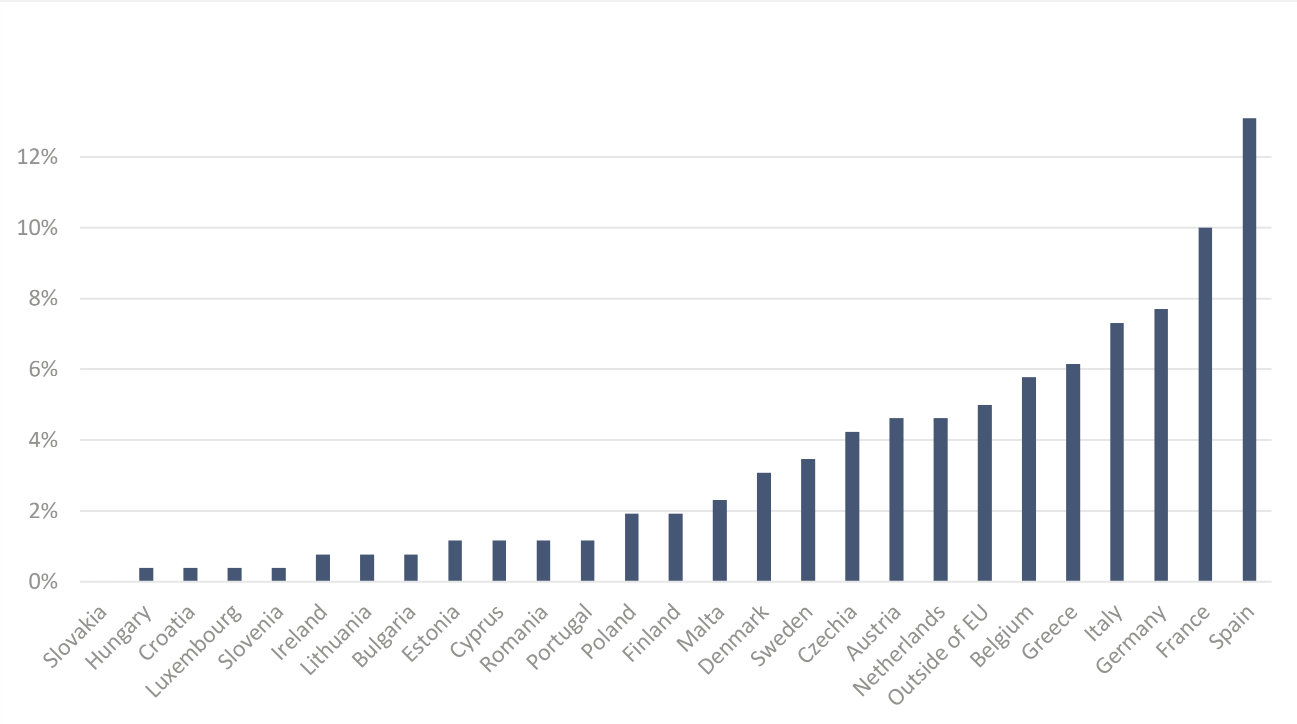
2. Synthesis of the research results and findings

In this section, we outline the outcomes of our study, beginning with an overview of the types of stakeholders who participated in the online survey. Each subsection focuses on a particular theme and contains background research to offer contextual understanding. Following this, results from both the survey and interviews are presented, where applicable. Each sub-section ends with a synthesis of the findings obtained through different means.

2.1 Stakeholder profile

The total number of survey responses received was 260 responses, representing a 31% response rate out of the total 841 stakeholders contacted. The results exhibit a dispersed pattern of responses across Member States, as depicted in Figure 2 . Spain, France, Germany, Italy, Greece, and Belgium contribute to over 5% of the responses. These countries are also the top recipients of Horizon Europe funding in terms of number of projects funded, as shown in (N=72994).

Figure 1 – Distribution of survey responses by country (N=260)



In terms of stakeholder types represented in the survey, beneficiaries of both Horizon 2020 and Horizon Europe constitute over 30% of respondents, with an additional 10% identified as beneficiaries of either programme individually. Approximately 15% of respondents[3] serve as proposal evaluators while also benefiting from Horizon 2020 and/or Horizon Europe. Evaluators solely involved in assessing proposals within these programmes represent 7% of responses. Similarly, National Contact Points (NCPs) account for 6% of survey participants. Figure 3 provides a detailed breakdown of respondent categories, where the "other" category includes stakeholders who either did not specify their affiliation with Horizon Europe or expressed interest in the programme without being direct beneficiaries.

[3] Horizon Europe evaluators can also apply for funding as long as there is no conflict of interest.

Figure 2 – Distribution of countries accounting for at least 1% of projects funded by Horizon Europe (N=72994)

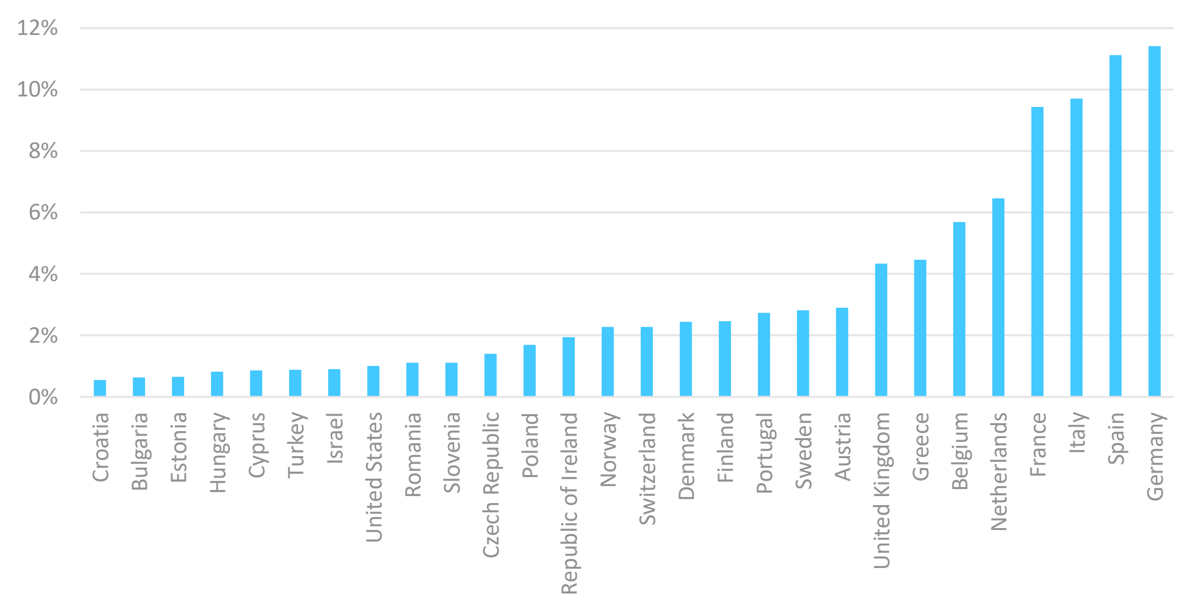
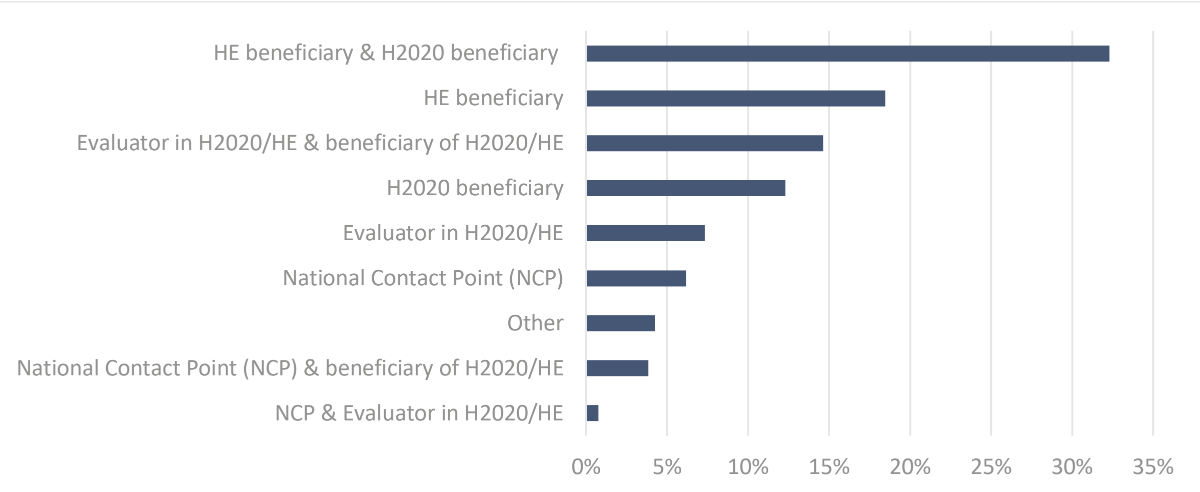


Figure 3 – Distribution of the survey respondents by type (N=260)



Regarding the interviews, a total of 17 sessions were conducted. The table below provides additional details on the stakeholders interviewed, including their country of origin, roles, and thematic relevance.

Table 2 – Overview of interviews carried out

Stakeholder type	Name (project/entity)	Country	Theme
FET Flagship Beneficiary	Bat4EVER	Belgium	Impact of discontinuing the FET Flagship
	SENSIBAT	Spain	
	French Atomic and Alternative Energies Commission (CEA)	France	
	Qurv (spin-off)	Spain	
NCP	/	Austria	Impact of discontinuing the FET Flagship & perception of HE evaluation system
		Netherlands	
		Italy	
		Spain	
Private companies assisting with proposal and application process	Euroconsultants	Greece	Challenges and participation barriers applicants/stakeholders face when applying to EU funding
	PWC	Malta	
	GrandXpert	Cyprus	
	Euronovia	France	
SMEs and associations	Greece Technology Center	Greece	SMEs Participation barriers
	European Business and Innovation Centre Network (EBN)	Belgium	
	European Builders Confederation (EBC)	Belgium	
	Multimedengineers	Italy	
	Termo line	Romania	

2.2 Evolution of calls and funding patterns across Framework Programmes: A comparative analysis from FP7 to Horizon Europe

In this section, we examine the evolution of calls and funding patterns throughout various framework programmes, with a focus on the seventh Framework Programme (FP7), Horizon 2020, and available data from Horizon Europe. We investigate the progression of funding, total application numbers and success rates, along with the composition of participating organisations, potential Technology Readiness Levels (TRLs) ascribed to the prospective funded projects, as well as an assessment of Social Sciences and Humanities (SSH) aspects. This section relies on desk research using secondary sources, including programme-specific regulatory documents and information extracted from existing databases.

2.2.1 Overview of EU Framework Programmes: From FP7 to Horizon Europe

The European Union has aimed at facilitating advancements in research, innovation, and scientific progress through its successive framework programmes. The FP7, operative from 2007 to 2013, fostered collaboration and research-driven solutions. It proved to be effective in strengthening scientific excellence, enhancing industry competitiveness, and driving economic growth and job creation. It also addressed societal challenges and promoted interdisciplinary research, facilitating Europe-wide collaboration and networking. However, shortcomings emerged regarding administrative complexity, different parts of the programme operating in isolated silos, with the various components being excessively rigid, and lacking synergies with related European funding programmes[4].

Horizon 2020 (H2020), implemented from 2014 to 2020, paid close attention to climate action, which aimed to advance scientific understanding and practical solutions, with 32% of funding allocated to this critical area. Together with its predecessor, FP7, Horizon 2020 ranked as the world's second-largest provider of climate science[5]. However, it fell short in allocating an adequate budget to fund all high-quality proposals submitted, required further simplification and a reduction of administrative burdens, as well as faced shortcomings in addressing gender balance measures and closing the EU's R&I gap between top and bottom performers[6].

Building on the successes and lessons learned from H2020, Horizon Europe (HE) emerges as the latest and most ambitious research and innovation framework programme. Launched in 2021 and spanning until 2027, HE places a strong emphasis on achieving a green, digital, and resilient future for Europe.

Based on data from the Horizon Dashboard[7], Table 3 and Table 4 provide an overview of the scale, scope, and financial aspects of each Framework Programme, showcasing the evolution and increasing investment in European research and innovation initiatives over time, providing insights into their progress at the same stage of execution.

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4 European Commission, '7th Framework Programme for Research', website 2016

5 Council of the European Union, 'Ex post evaluation of Horizon 2020, the EU framework programme for research and innovation'. Publications Office, 2024.

6 European Commission. Ex-post evaluation of Horizon 2020, the EU Framework Programme for Research and Innovation. Publications Office of the European Union, 2024.

7 European Commission, 'Horizon Dashboard', website, (n.d.). Retrieved January, 2024.

Table 3 – Overview of the three framework programmes

Framework Programmes	Signed Grants	Participation[8]	Unique Participants[9]	EU Contribution [10] (billion euro)	Total Cost[11] (billion euro)
FP7 (2007-2013)	25,809	139,241	29,649	46.09	65.91
H2020 (2014-2020)	35,426	179,111	41,827	68.33	86.27
Horizon Europe (2021-2023)	10,357	70,643	21,443	30.07	37.49

Source: Own elaboration using Horizon Dashboard data[12]

Table 4 – Overview of the three framework programmes in their three-year initial stage of implementation

Framework Programmes	Signed Grants	Participation	Unique Participants	EU Contribution (billion euro)	Total Cost (billion euro)
FP7 (2007-2009)	6,671	41,544	12,620	12.82	18.14
H2020 (2014-2016)	11,298	59,625	20,260	22.03	28.46
Horizon Europe (2021-2023)	10,357	70,643	21,443	30.07	37.49

Source: Own elaboration using Horizon Dashboard data[13]

Between each of the different framework programmes, there has been an increase in both EU contribution and total costs. The significant surge in the EU contribution from FP7 to H2020 underscores a financial commitment to advancing scientific and technological endeavours, a commitment that has continued over the first three years of Horizon Europe, as seen from Table 4.

Focusing on projects at the same stage of execution, i.e., during the first three years of their respective implementation, a similar level of funding is observed between HE and H2020. In this context, both programmes had used approximately one-third of the total budget. Figure 4 illustrates the evolution of funding accompanied by the indicators, with FP7 set as the baseline at 100, to facilitate evaluation.

A clear observation from the graph is that none of the indicators align with the funding evolution, except for signed grants from FP7 to H2020, meaning a higher average value per grant in Horizon Europe. Notably, signed grants and unique participants exhibit a deviation from the trend when transitioning from H2020 to Horizon Europe. The increase in unique participants from FP7 to H2020 suggests a broadening and diversification of entities engaged in European research and innovation

[8] Number of organisations involved in the selected projects. One organisation participating in N is counted N times.

[9] Number of distinct organisations involved in the selected projects. One organisation participating in N is only counted once.

[10] Total EU funding granted to the participants of the selected projects.

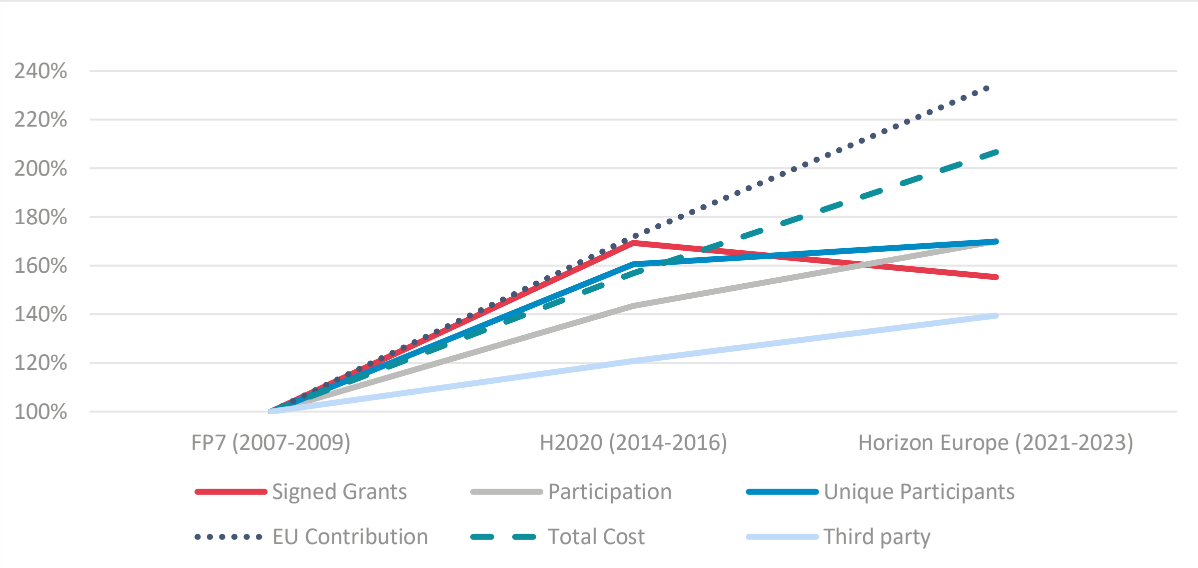
[11] Total cost of the selected projects, covered by the EU contribution and other funding sources.

[12] Data retrieved in March 2024 (data last loaded: 22/02/2024)

[13] Data retrieved in March 2024 (data last loaded: 22/02/2024)

initiatives, a trend that seems to persist in HE. However, it is crucial to account for significant external factors such as the Covid-19 pandemic, which may contribute to variations. For instance, the pandemic might have posed challenges in forming consortia with new partners, potentially limiting the rise in unique participants despite the substantial budget boost. These external factors have the potential to shape the trajectory and outcomes of HE in comparison to H2020.

Figure 4 – Overview of the three framework programmes in their initial stage of implementation (FP7=100)[14]

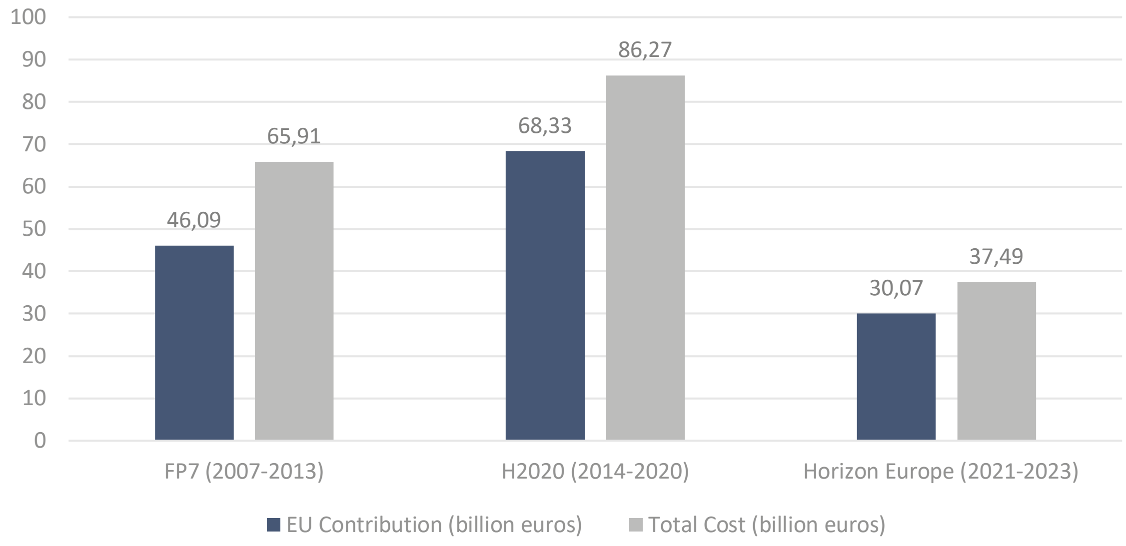


2.2.2 Evolution of the funding

Starting with FP7 and building on the strategic advancements seen in H2020, HE is considered an important instrument for transformative research and innovation across the European Union. While the programmes have demonstrated considerable effort in enhancing scientific excellence, driving technological progress, and tackling societal challenges, the budget allocation has not fully aligned with Europe's scientific capacity nor the magnitude of the challenges at hand. Figure 5 provides the evolution of EU contributions and total costs across the three framework programmes.

[14] Third-party funding is calculated as the total cost minus the EU contribution

Figure 5 – EU contribution and total cost by framework programme



Source: Own elaboration using Horizon Dashboard data[15]

FP7 represented a total budget of 66 billion euros. Among these, around 46 billion euros of EU contribution were allocated to four specific programmes: FP7-Cooperation (stimulating EU wide collaborative research), FP7-Ideas (fostering EU wide research excellence), FP7-People (building human resources, mobility and networks) and FP7-Capacities (addressing specific needs in the innovation systems). These programmes were executed through annual Work Programmes featuring competitive calls for proposals[16].

In the funding distribution, Collaborative Projects (CP) received most of the funding (53%), followed by the European Research Council (ERC) with 17%, Marie Curie Actions (MC) with 10%, and Coordination and Support Actions (CSA) and Joint Technology Initiatives (JTI) each receiving 6%. The smallest funding and grant allocations were distributed to Collaborative Projects - Coordination and Support Actions (CP-CSA) (4%), Broadening Societal Impact and Governance of Research (BSG) (3%) and Networks of Excellence (NOE) (1%).

H2020 had an overall budget of nearly 75.6 billion euro[17]. The total contribution EU funding granted to the participants of the selected projects amounted to 68.3 billion euros[18]. H2020 grants are implemented through 12 types of actions, including Research & Innovation actions, Innovation actions, Coordination & Support actions, ERC grants to support frontier research, Marie Skłodowska-Curie actions (MSCA), COFUND actions supporting joint programmes, Procurement actions (PCP and PPI), SME instrument (phase 1 - feasibility study, phase 2 - innovation projects, and phase 3 - commercialisation support), and Inducement and Recognition Prizes, along with a debt and equity facility[19]. Among these, five types of actions received most of the funding and grants: Research and

15 Data retrieved in March 2024 (data last loaded: 22/02/2024)

16 European Commission, Directorate-General for Research and Innovation, 'Ex-post-evaluation of the 7th EU Framework Programme (2007-2013)', Publications Office of the European Union, 2018.

17 European Commission. 'Horizon Europe – Performance', website, (n.d.). *This amount is the final budget, including transfers and adjustments following the annual, while the EU contribution in Table 3 is the amount allocated to research and innovation projects following calls for proposals.*

18 European Commission, 'Horizon Dashboard', website, (n.d.)

19 European Commission, 'What you need to know about H2020 calls', website, (n.d.)

Innovation Actions (28.9% of the funding), ERC Actions (19.1% of funding), Innovation Actions (16.1% of the funding), JTI (9.6% of the funding) and MSCA Grants (9.2% of funding)[20].

Horizon Europe, the current framework programme, holds the highest funding to date, with an initially agreed budget of 95.5 billion euros allocated for the period from 2021-2027. The budget was initially distributed across four pillars and 15 components to establish a comprehensive programme supporting various aspects of Research and Innovation: global challenges and industrial competitiveness (56% of the funding), excellent science (26% of the funding), innovative Europe (14% of the funding), and widening participation and strengthening the European Research Area (4% of the funding)[21].

However, according to a recent agreement reached in February 2024[22], Horizon Europe will face a budget reduction of 2.1 billion euro, with 1.5 billion euro redirected to defence research Consequently, the final budget will amount to 93.4 billion euro, potentially amplifying existing concerns regarding underfunding.

Table 8, Table 9 and Table 10 in Annex 4 provide detailed overview of the types of actions categorised by EU contributions and participation across the three framework programmes.

2.2.3 Total applications and success rates

In terms of total application numbers and success rates, FP7 saw the submission of 135,805 research proposals, with 25,163 retained proposals selected for funding[23], achieving a success rate of 19%.

Regarding H2020, 283,132 research proposals were submitted, resulting in the selection of 33,818 projects that received funding, reflecting an average success rate of 12%. Although transitioning to H2020 implied an increase in the number of projects selected due to a bigger budget, the average success rate decreased due to a more competitive environment. In fact, 46% of the eligible proposals were assessed as being of high quality[24] but funding could not be granted to all, resulting in an oversubscription rate[25] of 74%[26].

A measure put in place to mitigate the loss of effort invested in unfunded proposals is the Seal of Excellence (SoE), a quality label awarded under Horizon 2020, which serves as a recognition for successful yet unfunded proposals. This initiative captures the inherent value of such proposals submitted by mono-beneficiaries with the aim to facilitate their resubmission to alternative funding programmes. Among the 97,403 high quality proposals not selected for funding, 20,890 received a SoE certificate under H2020. This recognition was applicable across various programmes, including

[20] European Commission. 'Interim Evaluation of Horizon 2020', website, 2017.

[21] European Commission, Directorate-General for Research and Innovation, 'Horizon Europe, budget – Horizon Europe - the most ambitious EU research & innovation programme ever', Publications Office of the European Union, 2021.

[22] European Council, 'MULTIANNUAL FINANCIAL FRAMEWORK 2021-2027'. 2024

[23] Proposals selected for funding as a result of the evaluation process, in other words, eligible proposals that where shortlisted for funding.

[24] Proposals are considered as being of high quality when the expert evaluators give them a score above the quality threshold.

[25] The proportion of eligible proposals evaluated as above the quality threshold and which were not retained due to budget constraints, out of all eligible proposals evaluated by experts to be above the quality threshold.

[26] European Commission. Ex-post evaluation of Horizon 2020, the EU Framework Programme for Research and Innovation. Publications Office of the European Union, 2024.

the SME Instrument (later known as the EIC Accelerator), the MSCA, Teaming actions, and the ERC Proof of Concept[27].

Horizon Europe has also included the Seal of Excellence scheme, extending its reach across several EU programmes. Currently, this distinction is awarded within various initiatives such as the Digital Europe Programme, LIFE programme, Erasmus+, and Connecting Europe Facility – Digital[28].

Moving forward to Horizon Europe, 61,713 eligible proposals have been submitted for grants until 2023. Out of these, 10,346 proposals have been chosen to receive funding, resulting in an average success rate of 17%. These figures indicate a notable increase in the success rate from H2020 to Horizon Europe. Although increased funding could be a contributing factor, notably, the primary reason for the lower success rate in H2020 appears to be the volume of proposals received, as illustrated in Figure 7. Moreover, the non-participation of the UK and Switzerland in the programme might has boosted success rates, as they were major beneficiaries of H2020.

Overall, while the rise in success rates could be interpreted as a positive sign, a higher success rate is not necessarily entirely beneficial. Instead, under the Horizon Europe programme, it suggests that a larger amount of funding has resulted in fewer projects being funded, primarily benefiting the same group of individuals. This raises concerns about the equitable distribution of resources, particularly as Horizon Europe still leaves many outstanding proposals unfunded. Key data from 2021-2022 show that seven out of 10 high quality proposals cannot be funded[29]. Figure 6 and Figure 7 summarise these findings.

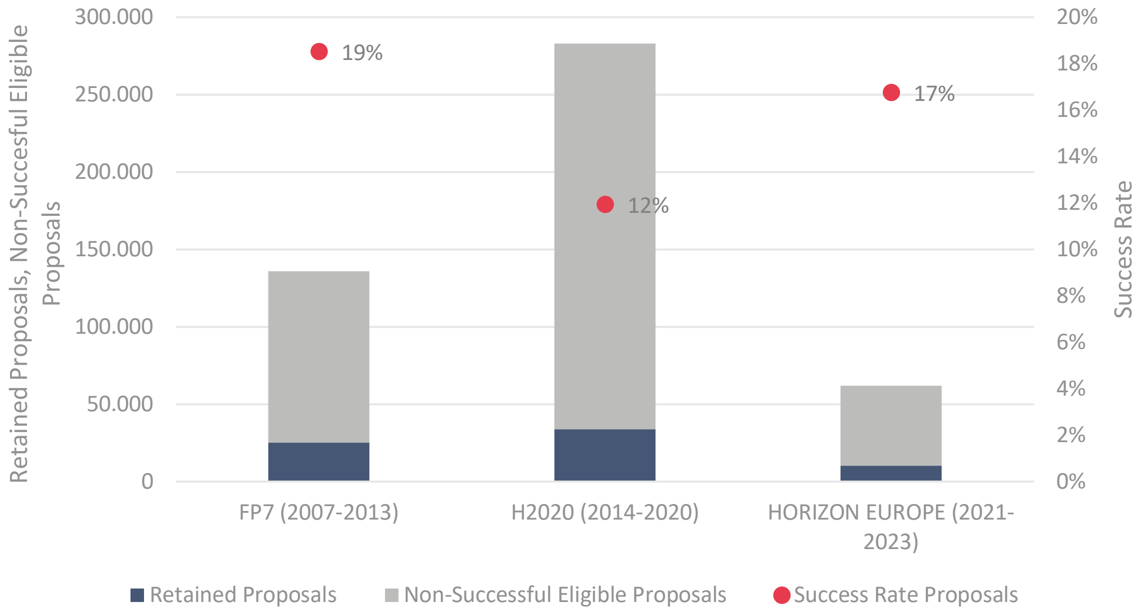
However, it is crucial to view these results as preliminary, given that Horizon Europe has only been operational for three years. Hence, comparing results can be challenging, as initial success rates typically tend to be high across most EU research programmes. As shown in Figure 7, FP7 experienced higher success rates during its initial three years. Hence, the recently announced cuts to the budget of Horizon Europe could potentially lead to a decrease in the success rate in the coming years, albeit a minor one, given the relatively small magnitude of the reductions in comparison to the overall budget.

27 European Commission. Ex-post evaluation of Horizon 2020, the EU Framework Programme for Research and Innovation. Publications Office of the European Union, 2024.

28 European Commission. 'Seal of Excellence', website, (n.d.).

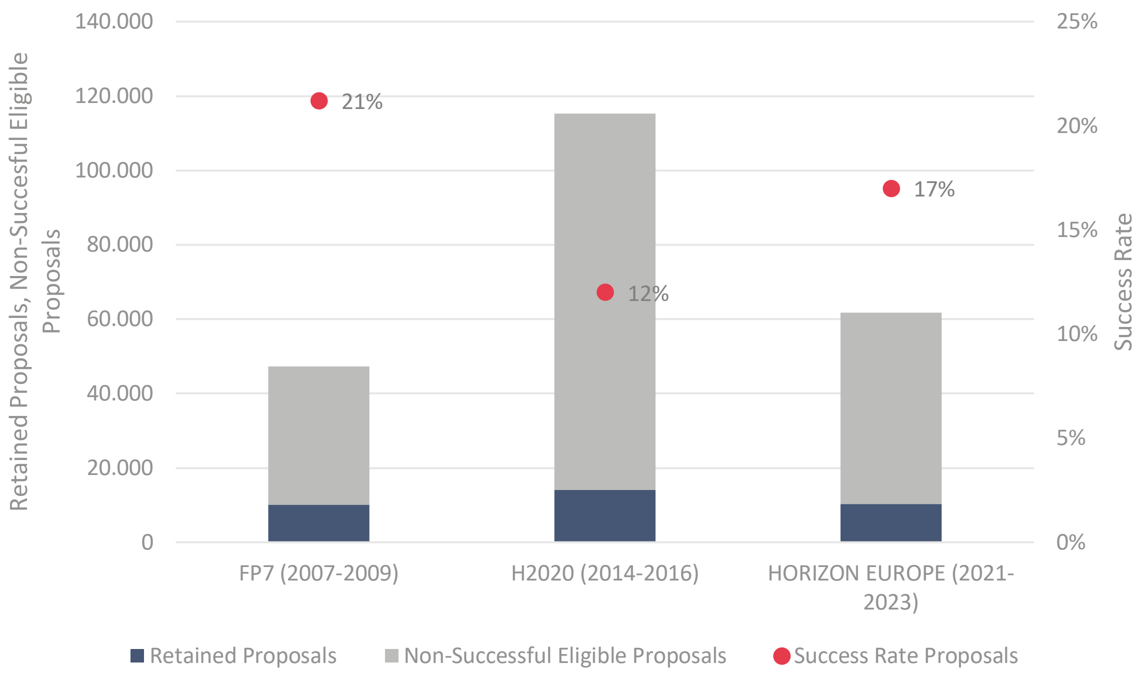
29 European Commission, Directorate-General for Research and Innovation, 'Horizon Europe implementation – Key data for 2021-2022', Publications Office of the European Union, 2023.

Figure 6 – Eligible and retained proposals by framework programme



Source: Own elaboration using Horizon Dashboard data

Figure 7 – Eligible and retained proposals by framework programme in their initial stage of implementation



Source: Own elaboration using Horizon Dashboard data

Detailed success rates by thematic priority/subprogrammes across the three framework programmes can be found in Table 11, Table 12 and Table 13 in Annex 4 – Statistics on the evolution of calls, FP7 – Horizon Europe

2.2.4 Size and composition of consortia

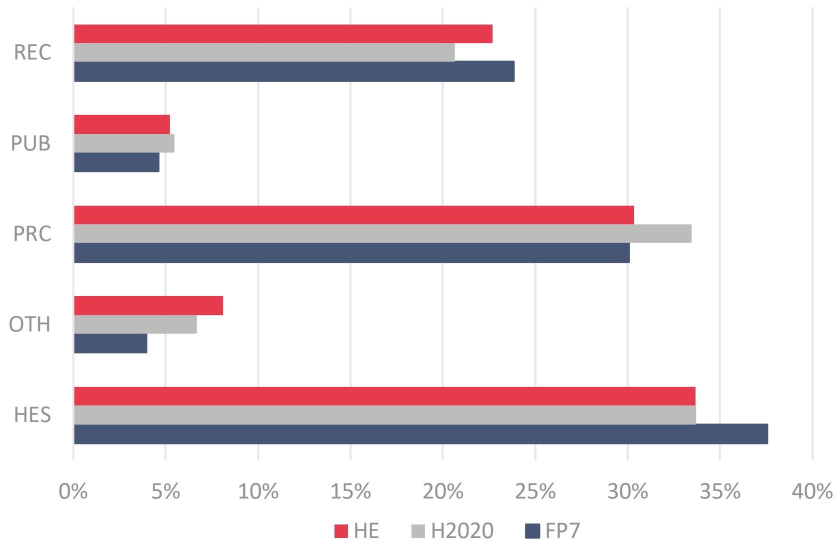
The dynamics of consortia size and composition varied across framework programmes and their respective subprogrammes and calls. FP7 involved 139,241 organisations and beneficiary entities across 25,809 projects. In FP7, diverse subprogrammes required research consortia from at least three countries to apply for funding, while others invited individual researchers to submit their proposals. FP7 funded research involved organisations categorised into four main groups. Universities (HES) were the primary focus, comprising 38% of the total number of participants. Private for-profit entities (PRC) represented 30%, Research and Technology Organisations (REC) made up 24%, and Public Authorities (PUB) accounted for 5%.

Transitioning to Horizon 2020, there was an expansion in the number of organisations engaged, encompassing 178,311 beneficiary entities across 35,131 projects. Universities (34% of the total) led in participation together with private for-profit entities (33% of the total). Research and Technology Organisations represented 21% of the total, while Public Authorities accounted for 5%.

In Horizon Europe, the consortia dynamics continued to evolve, engaging 70,643 beneficiary entities across 10,357 projects. Higher or Secondary Education Establishments (34%), and Private for-profit entities (30%) continued to be major players. Research and Technology Organisations represented 23% of beneficiaries across all calls and Public Bodies accounted for 5%.

As illustrated in Figure 8, despite the growth in the total number of beneficiaries, the proportions per type of organisation have not experienced significant shifts. For instance, universities consistently maintained their position, although with slight fluctuations in their relative percentages. However, while universities consistently maintained a significant presence as primary beneficiaries across all frameworks, there was a notable transition in H2020 where private for-profit entities emerged as leading participants alongside universities. This reflects a growing involvement of the private sector in EU-funded research initiatives, which could further be maintained under Horizon Europe.

Figure 8 – Participation by type of organisation across the three framework programmes[30]



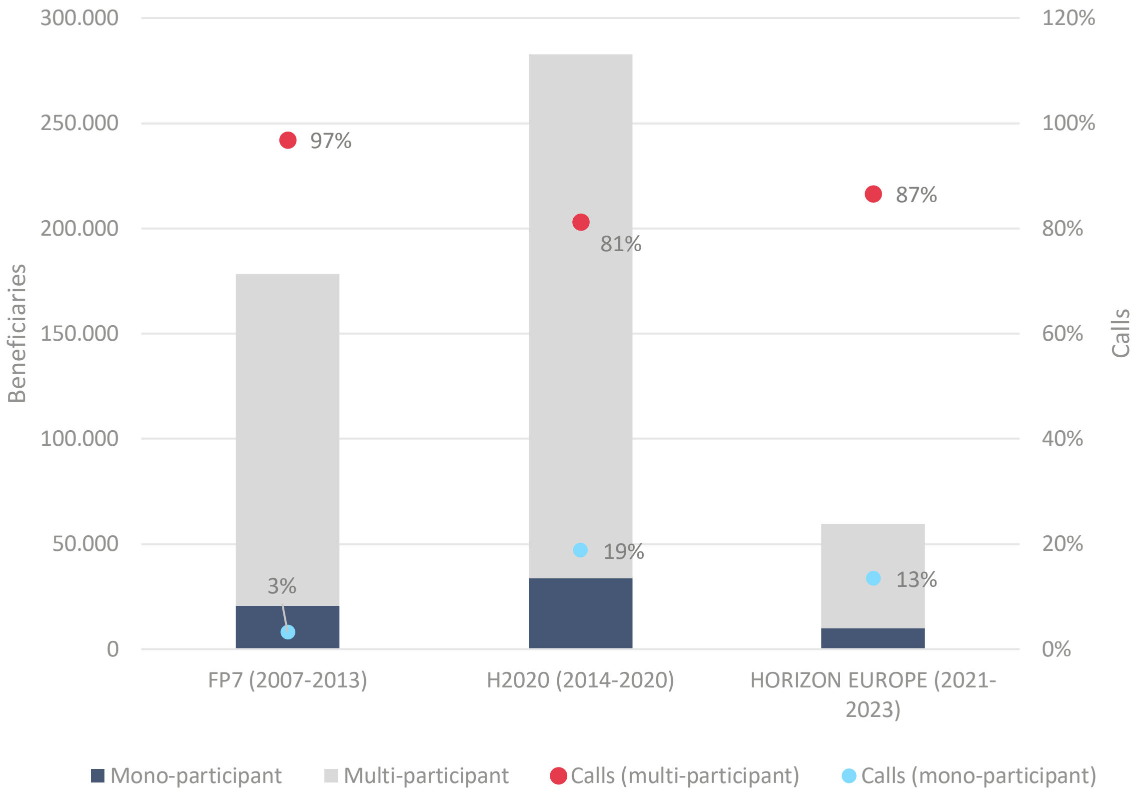
Source: Own elaboration using Horizon Dashboard data

[30] Public Bodies (PUB), Research Organisations (REC), Private for-profit entities (PRC), Higher or Secondary Education Establishments (HES), Other (OTH).

In terms of SME participation, this share is mainly classed under the research and technology organisation (REC) grouping, or private-for-profit entities (PRC), or other (OTH) referenced in Figure 8. During the initial three years of framework programme implementation, SME participation figures are as follows: 6,690 (FP7), 10,610 (H2020), and 10,965 (HE). This trend indicates a modest increase in SME involvement across successive funding programs, suggesting the potential for its continuation. More information on SMEs is provided in Section 2.5.

Figure 9 illustrates the evolution of EU framework programmes in terms of mono-participant[31] and multi-participant[32] calls. The data show that collaborative research calls have become more prominent compared to mono-participant projects across FP7, H2020, and the first three years of HE. Most calls are aimed at multi-beneficiary consortia, as the aim is to foster collaboration among organisations across different countries, strengthen the quality of applications, and ensure equitable competition and opportunities throughout the EU. However, there is an escalating trend in mono-participant types of calls from FP7 to H2020, which seems to be persisting in the first three years of HE.

Figure 9 – Number of beneficiaries and calls by framework programme



Source: Own elaboration using CORDIS database

Table 14, Table 15 and Table 16 in Annex 4 provide a comprehensive breakdown of the types of calls within FP7, H2020 and Horizon Europe, featuring detailed statistics on beneficiary entities, projects, and the number of calls.

[31] Actions that fund projects with one beneficiary only.

[32] Actions that fund projects with more than one beneficiary.

2.2.5 Type of action and associated technology readiness level

Technological Readiness Levels (TRLs)[33] are employed to categorise technological research, product development, and demonstration activities as well as to define the types of actions available in calls for proposals.

The extent of innovation within projects in FP7 varied with the nature of the specific programme. For instance, the COOPERATION specific programme predominantly concentrated on TRLs up to the demonstration phase. Notably, energy-related projects often reached TRL 7-8, showcasing a complete and qualified system[34].

Compared to FP7, H2020 had a stronger focus on higher TRLs. Outside the Excellent Science pillar under H2020, which focused on more fundamental research and the stage of an experimental proof of concept, the rest of the programmes under the Industrial Leadership pillar and Societal Challenges pillar were mainly focused on higher TRLs. However, this focus may have limited the integration of innovation from projects with lower TRLs. Stakeholders have expressed concerns, suggesting that allocating more resources to lower TRL collaborative research in the Societal Challenges and Leadership in Enabling and LEIT pillars could have been beneficial. Lower TRL research is also considered crucial for fostering breakthrough innovations aligned with societal needs[35].

Moreover, one of the primary objectives of the financial instruments in Horizon 2020 was to support research and innovation projects closer to market readiness. However, analysis of Technology Readiness Levels, which measure the maturity level of innovation, reveals that the financial instruments financed an equal share of projects at various levels of maturity: low (TRL 1-3), medium (TRL 4-6), and high (TRL 7–8)[36].

Building upon the lessons from H2020, Horizon Europe aims to continue the TRLs approach established in its predecessor. While HE grants cover TRL 1-8, TRL 5/6-9 projects can access repayable advances, equity, and/or loan guarantees[37]. Despite Horizon Europe experimenting with an increase in TRLs, there is a noticeable imbalance between applied and basic research financing across maturity levels. It tends to prioritise applied, market-orientated research. Hence, there is advocacy for a better balance within the framework programme's funding. Particularly relevant to Pillar II is the need to achieve a true balance across all categories of research and development activities, from fundamental research to experimental development, when addressing global challenges.

Examining the different calls under Horizon Europe 2023-2024 work programme[38], each specific pillar addresses TRLs differently across the Clusters. For instance, EU Missions focus on calls up to

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33 Technology Readiness Levels are indicators of the maturity level of particular technologies. This measurement system provides a common understanding of technology status and addresses the entire innovation chain: TRL 1 – basic principles observed; TRL 2 – technology concept formulated; TRL 3 – experimental proof of concept; TRL 4 – technology validated in lab; TRL 5 – technology validated in relevant environment; TRL 6 – technology demonstrated in relevant environment; TRL 7 – system prototype demonstration in operational environment; TRL 8 – system complete and qualified; TRL 9 – actual system proven in operational environment.

34 Peter, V., 'Evaluation report of the FP7 COOPERATION Specific Programme. European Liaison office of the German Research Organisations', 2016

35 European Commission, 'Interim Evaluation of Horizon 2020', website, 2017

36 European Commission, Directorate-General for Research and Innovation, A new horizon for Europe – Impact assessment of the 9th EU framework programme for research and innovation, Publications Office, 2018, https://data.europa.eu/doi/10.2777/194210

37 European Commission, Directorate-General for Research and Innovation, 'A new horizon for Europe – Impact assessment of the 9th EU framework programme for research and innovation', Publications Office, 2018

38 European Commission, 'Horizon Europe work programmes', website, (n.d.).

TRL 8, while in Cluster 1 (Health), only one call specifies TRLs, emphasising the progression of existing technologies from TRL 3-4 to at least TRL 7. Cluster 3 (Civil Security for Society) requires proposals to outline plans for further development towards higher TRLs, up to 7. Cluster 4 (Digital Industry and Space) features multiple calls covering the entire range of TRLs, with one aiming for TRL 8-9. Clusters 5 (Climate, Energy and Mobility) and 6 (Food, Bioeconomy, Natural Resources, Agriculture, and Environment) both encompass TRLs from 2 to 8 and up to 7-8. In Cluster 2 (Culture, Creativity, and Inclusive Society), MSCA and WIDERA specific TRL references are not provided in the work programmes. Research Infrastructures does reference specific TRLs on the work programme expect for only three calls specifying at least level 6.

Projects under the European Innovation Council's (EIC) calls are expected to be similar to HE Innovation Actions (TRL 6-8) and should focus on the development and/or deployment of technologies and innovations, including breakthrough and disruptive technologies, through cooperation between research and innovation actors from the participating regions. Particularly in the EIC component, TRL is integrated into the programme architecture. The sister funding schemes Pathfinder, Transition, and Accelerator aim to advance technological concepts up the TRL ladder, facilitating their transition towards commercialisation:

- EIC Pathfinder: science-to-technology, proof-of-principle (from TRL 1 to TRL 4).
- EIC Transition: maturation, validation, business activities (from TRL 3-4 to TRL 5-6).
- EIC Accelerator: rolling out the market introduction strategy, final technical tweaks, business development work (from TRL 5-6 to TRL 8)[39].

However, it is important to note that there is no publicly available data on the TRLs of funded projects, making it challenging to determine the TRL achieved after project finalisation across the different funding programmes.

2.2.6 Social sciences and humanities (SSH)

The EU promotes a unique approach that combines economic growth with strong protection and inclusion, including democracy, human rights, gender equality, and cultural diversity. Addressing global challenges requires interdisciplinary collaboration, as complex issues often transcend single disciplines, and Social Sciences and Humanities (SSH) disciplines are crucial in addressing these intricate challenges. Therefore, EU framework programmes devote resources to integrate pertinent contributions from SSH to effectively tackle these issues[40].

In FP7, two subprogrammes, Socio-Economic Sciences and Humanities (SSH) and Science in Society (SiS), tackled issues of substantial societal and citizen importance. Nonetheless, both subprogrammes had comparatively modest budgets and funded a limited number of projects[41].

Socio-Economic Sciences and Humanities (SSH) was part of FP7-Cooperation and represented the smallest budget share and funded the fewest projects across all themes within FP7-Cooperation. It accounted for 20 calls that 253 funded projects, with a budget comprising only 1.3% of the total FP7 budget. Moreover, SSH saw the highest proportion of high-quality proposals that could not secure funding compared to other FP7-Cooperation themes. Out of the proposals above threshold criteria, only 18% received funding, compared to an average of 39% across all FP7-Cooperation themes.

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[39] University of Ghent, 'Horizon Insights - Technology Readiness Level: An important measure of research maturity in Horizon Europe,' website, (n.d.).

[40] European Commission, 'Social science and humanities', website, (n.d.).

[41] European Commission, Directorate-General for Research and Innovation, 'Ex-post-evaluation of the 7th EU Framework Programme (2007-2013)', Publications Office of the European Union, 2018.

Most organisations conducting SSH research were universities, accounting for 63% of the participating organisations. Economics and other social sciences were the most frequently reported main or associated disciplines, while humanities disciplines represented a smaller share[42].

Science in Society (SiS) was a subprogramme of FP7-Capacities accounting for 0.65% of the FP7 budget and funding 183 projects. Although the budget allocated was small, it seemed reasonable as 52% of proposals scoring above threshold received funding. In terms of organisations, universities received the highest share of SiS funding (48%), followed by Research and Technology Organisations (20%) and Civil Society Organisations (18%).

While SiS projects were comparably strong in terms of influencing EU policies, media visibility, and generating success stories, they were weaker in scientific breakthroughs and knowledge production. Nonetheless, SiS demonstrated a strong European added value through integrating fragmented approaches and ensuring technical interoperability on pertinent issues such as science education, gender studies, open access, public engagement and research ethics[43].

In H2020, SSH was addressed both as a specific part of the programme (SC6 Europe in a Changing World: Inclusive, Innovative and Reflective Societies), and as a cross-cutting issue. H2020 was thus the first framework programme into which SSH was integrated as part of the programme objectives[44]. Approximately 20% of the total call budget was allocated to topics specifically flagged for SSH. This allocation indicates an enhanced role for SSH disciplines in EU-funded R&I activities under H2020. However, despite this progress, limitations arose in measuring SSH activities due to the complexity of existing instruments and methodologies. Challenges included input-focused indicators, the use of proxies, and somewhat arbitrary thresholds for assessing SSH integration quality[45].

SC6 funded projects across various themes, showing strong performance and delivering high-quality results, particularly in producing immediately usable outputs. Meanwhile, SC7's civil security research had significant societal impacts by influencing policy uptake and facilitating the development of end-products or services, notably in areas like crime prevention, travel facilitation, and disaster resilience, as recognised in EU security policy documents[46].

Within Horizon Europe, the integration of SSH across all Clusters, Missions, and partnerships is identified as a crucial cross-cutting priority, as explicitly outlined in the Horizon Europe strategic plan 2021-2024[47]. SSH plays a pivotal role in research and innovation, particularly in relation to the twin green and digital transitions. The overarching goal of Horizon Europe is to fully integrate SSH into each of its pillars, notably under the pillar 'Global Challenges and European Industrial Competitiveness', where SSH should be fully integrated across all Clusters.

Although SSH research aspects are present on all parts of Horizon Europe, most of the calls are prominently featured in Cluster 2: Culture, Creativity, and Inclusive Society. This Cluster aims to bolster European democratic values, encompassing the rule of law and fundamental rights, while

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42 European Commission, Directorate-General for Research and Innovation, 'Ex-post-evaluation of the 7th EU Framework Programme (2007-2013)', Publications Office of the European Union, 2018.

43 Stroyan, J., Franke, J., Wilkinson, C., 'Ex-post evaluation of science in society in FP7 – Final report', Publications Office, 2016.

44 European Commission, 'Integration of Social Sciences and Humanities in Horizon 2020: Participants, Budgets and Disciplines 2014 – 2020', Publications Office of the European Union, 2023

45 European Commission, 'Integration of Social Sciences and Humanities in Horizon 2020', website, (n.d.)

46 European Commission, 'Integration of Social Sciences and Humanities in Horizon 2020', website, (n.d.)

47 European Commission, Directorate-General for Research and Innovation, 'Horizon Europe Strategic plan (2021 – 2024)', Publications Office of the European Union, 2021.

also safeguarding our cultural heritage and promoting socio-economic transformations that foster inclusion and growth[48].

In terms of the number of calls, based on information from Funding & Tender Opportunities, under H2020, there were 978 calls categorised under the cross-cutting priority of SSH. As for Horizon Europe, up to now, there have been 597 calls closed under this priority, with 17 calls open for submission and 25 more expected in the future. Table 5 summarises these findings.

[48] European Commission, 'Cluster 2: Culture, Creativity and Inclusive Society', website, (n.d.).

Table 5 – Calls for proposals on the SSH cross-cutting priority

Framework Programmes	Calls		
	Forthcoming	Open for submission	Closed
H2020 (2014-2020)	NA	NA	978
Horizon Europe (2021-2024)[49]	25	17	597

Source: Own elaboration using the Funding & Tender Opportunities – Single Electronic Data Interchange Area (SEDIA)[50]

2.2.7 Results synthesis on the evolution of calls and funding

The desk research conducted shows an evolution from FP7 through Horizon 2020 to Horizon Europe towards addressing complex societal challenges and enhancing administrative efficiency. Nonetheless, concerns about administrative complexity and inclusivity remain.

Despite increased budgets, there are pressing concerns regarding whether the level of funding aligns adequately with the balance of resources and the goals of the Horizon Europe programme. This prompts questions about whether the funding is sufficient to accomplish the programme's ambitions.

The increasing number of applications related to funded projects highlights a competitive and potentially discouraging landscape for researchers. While Horizon Europe has made strides in enhancing success rates, it is important to note that the number of applications has decreased from H2020 to HE in the first three years, suggesting that the improved success rate may be primarily attributed to less competition. However, the persistent issues in funding distribution continue to present a significant challenge.

While the number of beneficiaries in EU framework programmes has grown, the distribution among different types of organisations remains relatively stable. Notably, universities have consistently been major beneficiaries. H2020 witnessed a notable increase in the involvement of private for-profit entities, indicating an increasing participation of the private sector in EU-funded research, a trend likely to persist in Horizon Europe.

Horizon Europe builds on H2020, adopting a holistic approach to Technological Readiness Levels (TRLs). While H2020 instruments aimed to finance projects closer to the market, TRL analysis revealed balanced financing across maturity levels. HE prioritises research and innovation spanning TRL 1-9. However, there is concern that this focus on higher TRLs may limit the integration of more foundational, early-stage research, which is essential for breakthrough innovations. The lack of publicly available data on the achieved TRLs of funded projects under the research and innovation framework programmes pose another challenge in terms of assessing the innovations Horizon Europe, or any future funding programme, is facilitating.

EU's framework programmes demonstrate an evolving commitment to Social Sciences and Humanities (SSH), acknowledging its role in tackling societal challenges. Horizon Europe integrates them across all pillars and Clusters to address societal challenges comprehensively and inclusively.

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49 Data retrieved in March 2024.

50 European Commission, 'EU Funding & Tenders Portal', website, (n.d.). Data retrieved in March 2024.

2.3 Perceived impact of selected changes under Horizon Europe

In this section, the perceived impact of selected items introduced under Horizon Europe, namely the Common Model Grant Agreement and the strategic plan, is examined. Additionally, the perceived effect of discontinuing the FET Flagship under Horizon Europe on funding opportunities and more broadly on the European R&I landscape is considered. Each sub-section provides background information based on desk research, followed by the presentation of the survey results, and when applicable, the interview results.

2.3.1 Evaluating the impact of the Common Model Grant Agreement approach on synergies and administrative efficiency in European Union programmes

To provide a standardised framework for all Horizon Europe-funded projects, Horizon Europe introduced a single, universally applicable Common Model Grant Agreement (CMGA). This CMGA is also used for other programmes directly managed by the European Commission[51].

The aim of this CMGA is to reduce the complexity of the Grant Agreements and facilitate collaboration between projects funded by different Union programmes. Consequently, in the newly introduced CMGA the number of articles was streamlined from 58 to 44. Another key change was with regard to financial provisions, where a standardised calculation formula was introduced along with greater flexibility regarding the reporting of direct and indirect costs. By standardising the Grant Agreement across different European funding programmes, projects are expected to find it easier to collaborate between one another regardless of their funding programme[52].

In terms of exploitation and open science, communication and dissemination plans are now mandatory in the project's early stages, and research data management is obligatory. Encouraging third-party exploitation of R&I results, beneficiaries must use the Horizon Results Platform if no exploitation occurs. Open access to scientific publications and specific criteria for eligible publication fees have been defined, promoting accessibility and transparency[53].

All these changes in the Horizon Europe Model Grant Agreement seek to streamline administrative processes, and promote transparency and openness, ultimately hoping to facilitate smoother participation and thus a greater impact in Horizon Europe-funded initiatives.

The objective is to gauge stakeholders' perceptions regarding the changes introduced in the CMGA. Insights are gathered through an online survey that focuses on assessing whether the new CMGA is indeed simpler, more transparent, and conducive to fostering synergies across projects funded by different Union programmes.

Results of the online survey

The online survey developed included dedicated questions on the CMGA. These questions targeted NCPs, all evaluators and all beneficiaries[54]. Given their close involvement in the Horizon Europe programme, this group of stakeholders is expected to be aware of the CMGA and therefore is able to provide insights.

To ensure the reliability of responses from participants well-versed in the new grant agreement, the survey began with an assessment of their awareness and familiarity with the CMGA changes

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[51] The CMGA is used for the EU action grants which fund a specific action intended to help achieve one of the European policy objectives. https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/common/guidance/aga_en.pdf

[52] Cruz, A., 'From H2020 to Horizon Europe – a closer look at the new Grant Agreement', 2021.

[53] European Commission, 'Common Model Grant Agreement', 2021.

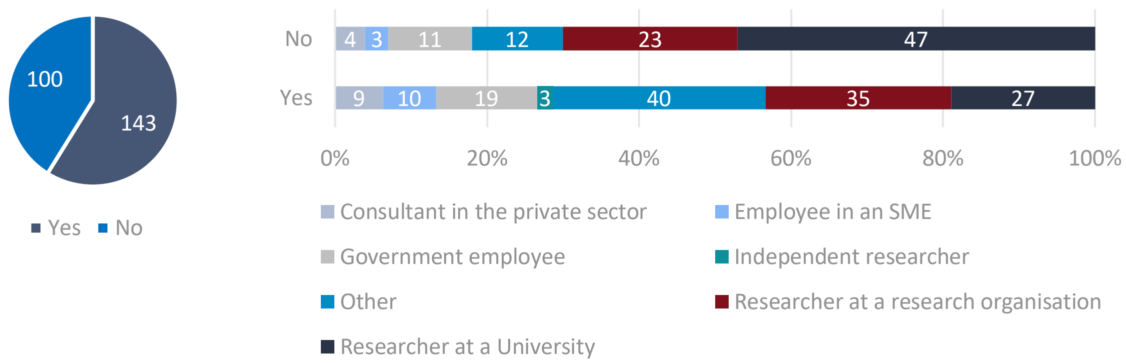
[54] All beneficiaries except FET Flagship beneficiaries.

compared to Horizon 2020. Following this, respondents who indicated a high or moderate level of familiarity were presented with a series of questions addressing impact on contract comprehension, fostering synergy, streamlining administrative processes, and promoting transparency. Furthermore, through an open-ended question, respondents had the chance to provide their opinions on the overall impact of CMGA on Horizon Europe projects, along with examples illustrating their viewpoint. Lastly, the survey seeks to gauge participants' belief in the effectiveness of mandatory communication and dissemination plans in enhancing the visibility of project results. The specific set of questions asked can be found in Annex 2.

The results indicate that a 60% of the surveyed stakeholders, totalling 143 respondents, are at least aware of the recent changes introduced under the CMGA. Of those aware of these changes, approximately 70% are independent researchers, researchers at universities, researchers at organisations, or work in other research-related roles categorised as "other" in Figure 10. This "other" category includes employees of research organisations, project managers/coordinators, and administrative support staff. The remaining portion of informed stakeholders consists of individuals employed in either the public or private sectors.

At the same time, as illustrated in Figure 10, researchers at universities/organisations account for 70% of respondents who are not aware of these changes. This result is intuitive since universities and organisations typically delegate administrative responsibilities, such as contract management, to dedicated departments. This line of reasoning can also explain why the "other" category, encompassing research organisation employees, project managers/coordinators, and administrative support personnel, constitutes the larger share of stakeholders who are aware of the changes under the CMGA.

Figure 10 – Breakdown of stakeholder type by awareness of the changes introduced under the CMGA (N=243)

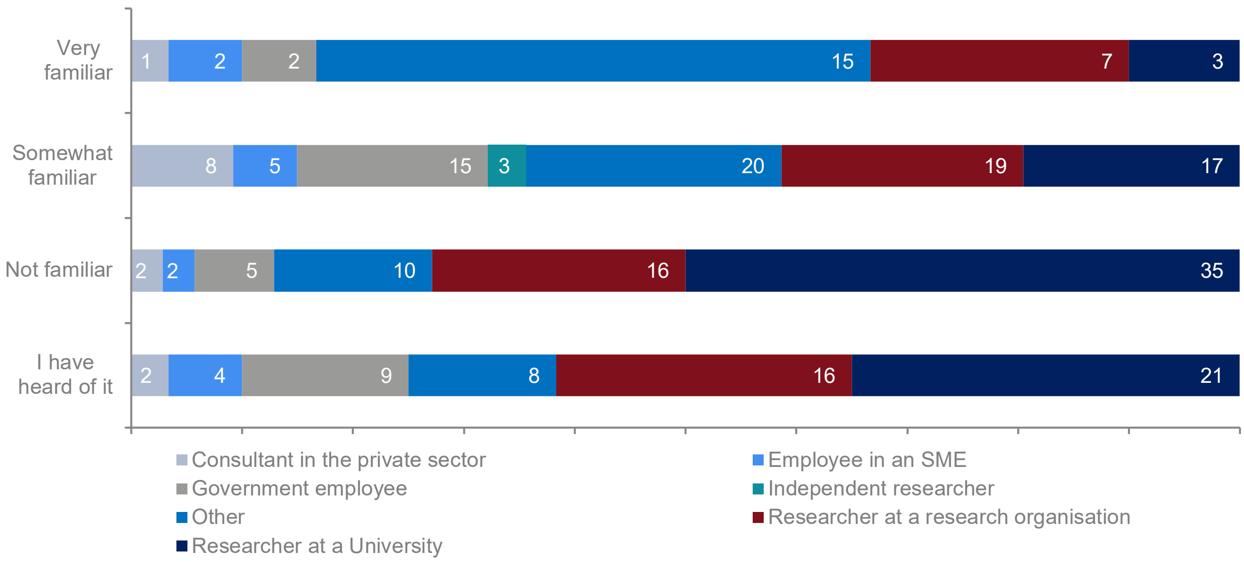


When examining familiarity with the CMGA, the survey results show varying trends among different stakeholder groups. Half of the respondents reporting to be "very familiar" with the CMGA represent the "other" category which includes employees of research organisations, project managers/coordinators, and administrative support staff (Figure 11). Approximately 60% of respondents who report to be "somewhat familiar" with the CMGA are affiliated with research activities ("other" category) and are researchers at a university/organisation.

Researchers at a university/organisation make up the larger share of respondents who are not familiar with the CMGA. This result is in line with what is illustrated in Figure 10 on awareness of changes brought under the CMGA, where researchers account for 70% of respondents who are not aware of these changes. This outcome is unsurprising, as universities and organisations commonly assign administrative tasks, including contract management, to specialised departments.

Ultimately, 48% of respondents, amounting to 117 individuals, are "somewhat familiar" or "very familiar" with the changes of the CMGA. This group of respondents were then asked a series of questions on the CMGA, as discussed in the following sub-sections.

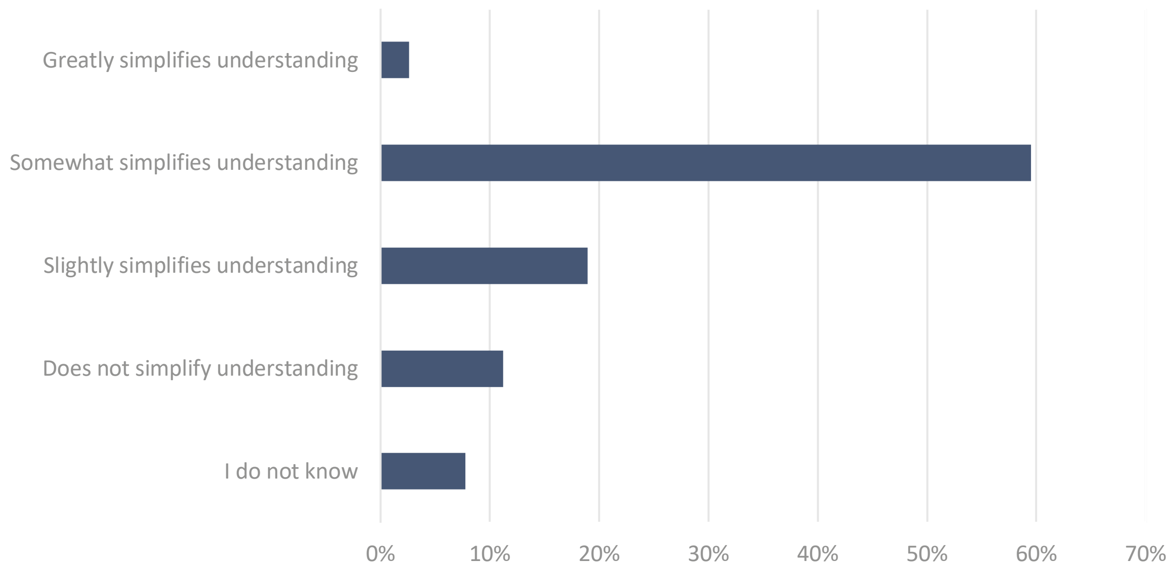
Figure 11 – Breakdown of stakeholder type by familiarity with the changes introduced under the CMGA (N=247)



The effect of CMGA on administrative process

Based on the respondents who are at least somewhat familiar with the CMGA, we find that approximately 60% of them indicate that it somewhat simplifies their understanding of the terms and conditions, as illustrated in Figure 12. This outcome is encouraging, particularly since one of the primary objectives of the CMGA was to enhance clarity and simplify comprehension.

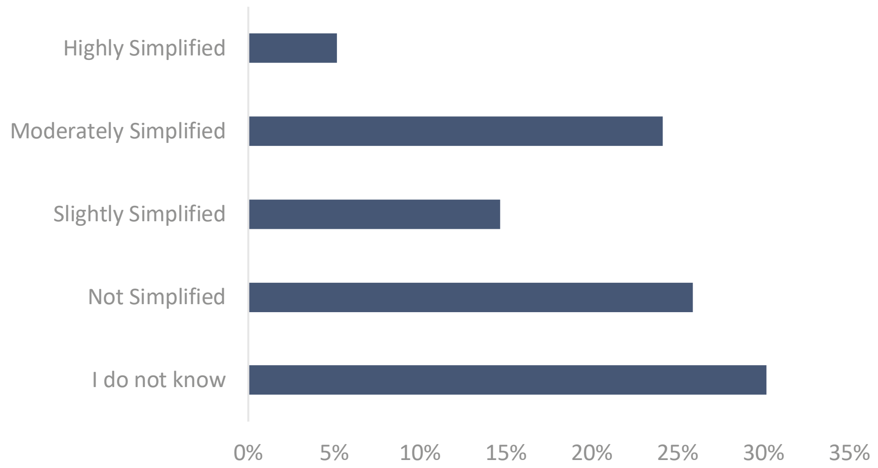
Figure 12 – Respondents' perception of the CMGA simplification of the terms and conditions compared to the H2020 grant agreement (N=116)



Nevertheless, there is considerable uncertainty among respondents regarding the level of administrative simplification provided by the CMGA compared to the previous grant agreements.

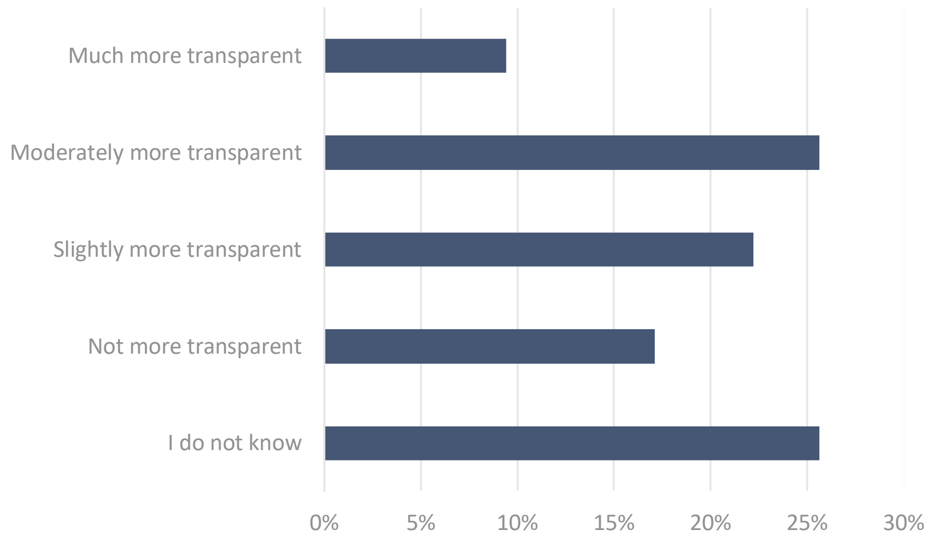
Around 30% of respondents (33 individuals) indicated that they were unsure about the simplification level, while only a small minority (5%) considered the administrative procedure under the CMGA to be highly simplified. The remaining proportion of respondents (41%) rated the CMGA as either moderately, slightly, or not simplified. This result illustrated in Figure 13 suggests a lack of consensus on its effectiveness in streamlining administrative processes, suggesting further room of improvement.

Figure 13 – Perceived simplification under CMGA in terms of administrative processes (N=116)



Similarly, in terms of improved transparency of terms and conditions under the CMGA, respondents reported mixed perceptions. Approximately 50% of respondents report that they either "do not know" if the new CMGA is more transparent or "moderately more transparent". This result could potentially be justified by the fact that the full annotation of the CMGA is not yet available. As shown in Figure 14, a combined 40% of respondents report it to be moderately more transparent or not more transparent.

Figure 14 – Perceived improved transparency under the CMGA (N=117)



The CMGA, synergies and the effect of mandatory communication and dissemination activities

As for the synergies encouraged across projects funded by different programmes by implementing a single CMGA, the survey results show mixed results. A total of 21% of respondents (24 individuals) perceive that a single CMGA has no discernible impact on the creation of such synergies. Likewise, a similar percentage of respondents are uncertain about the CMGA's influence on synergies or anticipate only a slight improvement. On the other hand, a relatively higher proportion of respondents, comprising 26%, believe that a CMGA moderately enhances the creation of synergies. Given these results, we cannot conclude that the CMGA satisfied one of its objective of encouraging synergies across different projects as illustrated in Figure 15.

Turning to the mandatory communication and dissemination (C&D) activities introduced within the CMGA under Horizon Europe, a majority of survey participants, 70%, express the belief that these obligatory activities will be of substantial help, ranging from very helpful to somewhat helpful. These results are illustrated in Figure 16.

Figure 15 – Perceived impact of CMGA on synergy creation (n=117)

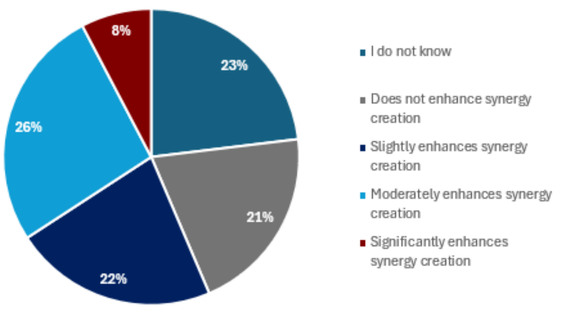
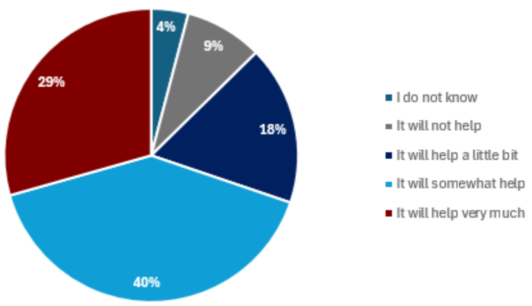


Figure 16 – Perceived impact of C&D activities on visibility of project results (n=245)



N= 245

Open feedback on improvements under the CMGA

Respondents were also given the opportunity to provide open feedback on specific improvement they noticed under the CMGA. Based on 44 written feedback responses, results suggest that many respondents noted that the introduction of a Data Sheet as part of the CMGA is a significant improvement, as it simplifies the management of projects by summarising all essential information[55]. The digitalisation of all grant phases, from proposal management to reporting, was also highlighted as a positive change, making it more efficient and streamlined. The use of lump sums for financial reporting was seen as a simplification, particularly benefiting SMEs.

Additionally, several respondents appreciated clearer explanations of personnel costs and a more straightforward approach to handling ethics issues. The ability to invoice internally sourced goods and services, as well as the inclusion of income and depreciation costs of equipment, were also mentioned as improvements. However, some respondents pointed out challenges related to the validation of organisations and the complexity of cascading grants and financial support to third parties.

[55] The Data Sheet includes keywords associated with the project, the project's unique identification number, full project title, project acronym, call identification details, topic identification details, the type of action undertaken by the project, the granting authority responsible for funding (e.g., European Commission), whether the grant is managed through the EU Funding & Tenders Portal, the project's official start and end dates, and its duration in months.

Respondents also provided open feedback on their opinion regarding the impact of the CMGA on projects of Horizon Europe in a general context. Many respondents appreciated the clearer and more transparent structure of the CMGA, as well as its datasheet, which simplifies the management of projects by summarising key information. Some respondents mentioned that the CMGA promotes harmonisation of procedures and cost calculations across different programmes, facilitating cooperation between projects. However, no concrete examples of this were referenced.

However, few challenges were noted as well. For instance, a stakeholder pointed out concerns about differences in interpretation between executive agencies, especially regarding the participation of subsidiaries of universities. Several respondents also called for a fully annotated CMGA to ensure full and comprehensive understanding of the terms and conditions.

Results synthesis on the impact of introducing the CMGA

Horizon Europe introduced a universally applicable Common Model Grant Agreement (CMGA) across various programmes managed by the European Union. The aim of the CMGA is to reduce the complexity of the grant agreement, enhance transparency and promote collaboration across projects.

Our survey result show that there is a general lack of familiarity across researchers regarding the elements of the newly introduced CMGA. While this finding is understandable given that grant agreements are typically managed by administrative staff rather than researchers themselves, it hints at the potential underutilisation of the CMGA's simplification, particularly in terms of fostering synergies between projects.

Therefore, we cannot conclude that the CMGA succeeded in satisfying all its objectives. While 60% of respondents who are familiar with the CMGA, find it to have simplified understanding of the terms and conditions compared to the previous grant agreements, its effectiveness in administrative simplification and promoting synergy remains uncertain. However, respondents believe the mandatory communication and dissemination activities incorporated in the CMGA will boost project visibility.

The introduction of the Project Data Sheet is seen as a key improvement, aiding in better project understanding and management. Similarly, the digitalisation of all grant phases, from proposal management to reporting, was also highlighted as a positive change under Horizon Europe. Several respondents also called for a fully annotated CMGA as was the case in H2020 to ensure full understanding of the terms and conditions.

2.3.2 Assessing stakeholder views on strategic planning in Horizon Europe: Inclusivity, relevance, and timing of consultations

The strategic planning process was a novelty introduced in Horizon Europe. It aims to improve the framework programme's reflexivity, i.e., to enable policy learning and allow for adaptations during its lifetime, directly introducing new instruments and revisions[56], as well as to promote the programme's impact and relevance, while deepening stakeholder engagement. The strategic plans cover periods of three to four years, for which they define the strategic orientation of research and innovation investments and set the guiding principles to better align with the political priorities of

[56] Cavicchi, B., Peiffer-Smadja, O., & Ravet, J., 'The transformative nature of the European framework programme for research and innovation: analysis of its evolution between 2002-2023', Publications Office of the European Union, 2023.

the Commission. In this sense, the strategic plan aims to promote a fruitful interface between these programme activities and the research and innovation projects funded by Horizon Europe[57].

The co-design of Horizon Europe's first strategic plan (2021-2024) involved a wide range of stakeholders who contributed their perspectives through remote and face-to-face activities, resulting in four strategic orientations supported by 15 impact areas. A public web consultation received 6,806 responses, while around 4,000 stakeholders engaged with the European Commission during the EU Research and Innovation Days. Similarly, the co-design process for the second strategic plan (2025-2027) involved stakeholders and citizens in shaping the strategic orientations, supported by an extensive online consultation and position papers to inform the analysis document of the strategic plan. The Plan was officially adopted in 2024, with the integral contribution of both Member States and the European Parliament.

Under this sub-section, we aim to gather insights from key stakeholders on the effectiveness of the co-designing process and its outcomes, particularly in what concerns its inclusivity, relevance and timing. Particularly, we tried to understand to what extent they were aware of the strategic plan of Horizon Europe, the web-based and face-to-face tools used to collect stakeholders' views for the 2019 co-design activities, if they participated in the web-based co-design activities, and how impactful they thought this new element might be on the topics of calls. The survey also addresses how the strategic plan impacted the type of calls launched under Horizon Europe.

Results of the online survey

To achieve the goal of this sub-task, a survey including dedicated questions on the strategic plan was conducted. It targeted NCPs, evaluators and beneficiaries[58] of Horizon 2020 and/or Horizon Europe as well as stakeholders who were not a beneficiaries, but interested in the framework programmes. Firstly, the survey intended to select the respondents that were aware of the strategic plan of HE, to whom a set of five to eight follow-up questions was asked. Namely, these sought to gather information on whether the respondents were aware of both co-design activities – remote and face to face – that took place in 2019, if they participated in any of them and, if so, what was their perception on the consideration of their contribution and the key takeaways of such events. Lastly, stakeholders were asked about their impression of the impact that the strategic plan has on the topics of calls launched and also had the opportunity to provide some examples on possible overlooked topics. The complete set of questions can be found in Annex 2 – Survey questions.

Awareness of the strategic plan and participation in the co-design activities

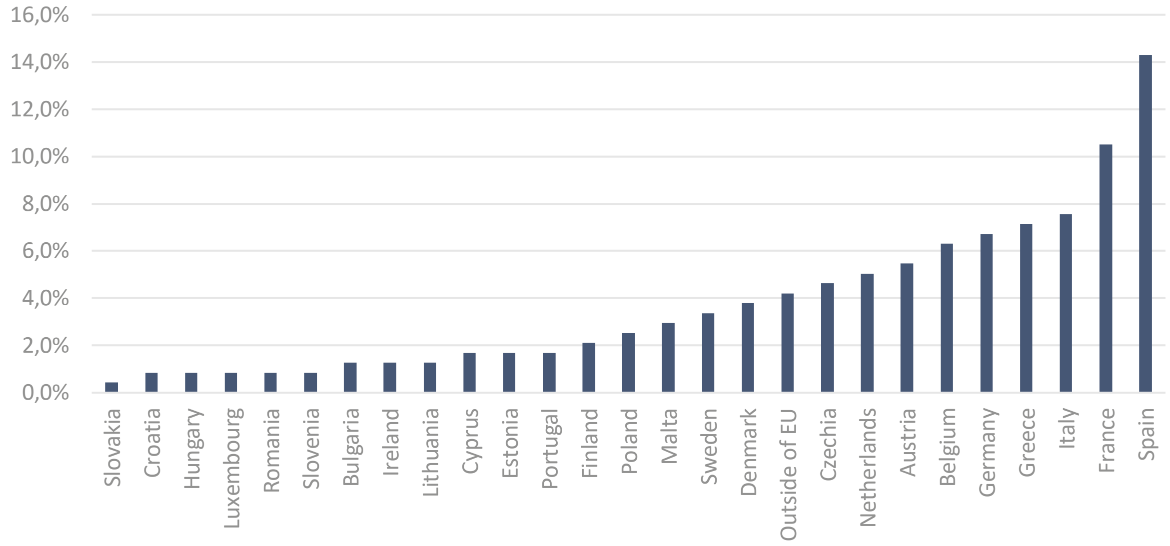
A total of 238 stakeholders answered the first question on their awareness of the strategic plan of Horizon Europe. The distribution of respondents per Member State is shown in Figure 17. Only approximately half (51%, 122 respondents) were aware of its existence. When taking a closer look at the break down of responses per occupation type, the figures indicate that mostly researchers were unaware of the existence of the strategic plan, namely 62% (32 out of 52) of researchers at a research organisation and 58% (42 out of 72) of researchers at a university. Together, these constitute 64% of the total of respondents that are not aware of the strategic plan. When considering responses by type of experience, it is mostly beneficiaries that do not know of the existence of this new element – 70% (19 out of 27) of enquired beneficiaries in Horizon 2020 and 56% (69 out of 123) of enquired beneficiaries in Horizon Europe. In fact, 58 out of these 88 beneficiaries across both framework programmes are researchers. Surprisingly, even if not the majority, 25% (10 out of 40) of evaluators in HE are also unaware of the existence of the strategic plan.

[57] European Commission, Directorate-General for Research and Innovation, 'Horizon Europe Strategic plan (2021 – 2024)', Publications Office of the European Union, 2021.

[58] All beneficiaries except those of FET flagships.

The 122 respondents aware of the strategic plan were asked if they were also aware of the remote co-design activities that took place in 2019, in order to gather stakeholders' views on Horizon Europe's contribution to EU policy. A relatively larger share of respondents (69 out of 122, 57) were not aware of these activities.

Figure 17 – Distribution of responses per Member State (N=238)

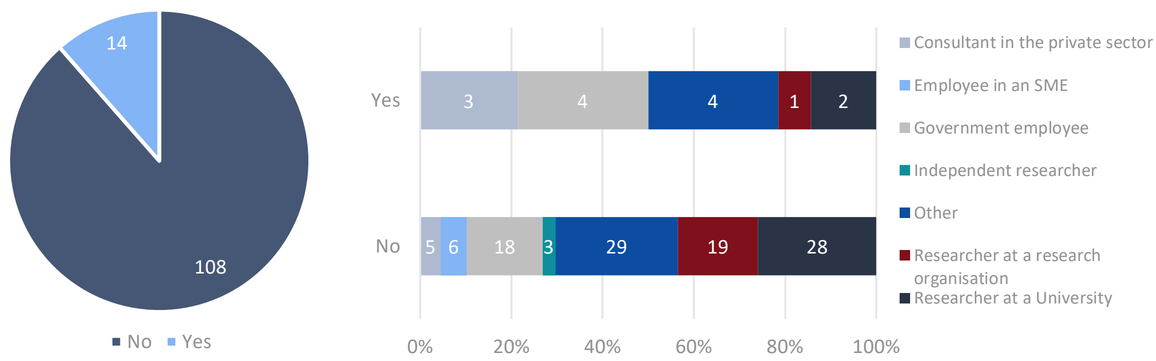


Just 14 out of the 122 (11%) respondents aware of the co-design activities also took part in them. When breaking down the answers by type of occupation, the results show that the highest share belongs to government employees and the "other" category (which includes industry workers, employees in the public sector, employees in an NGO, participants in a research council and staff of research organisations), both with 4 out of the 14 stakeholders (29%). However, these represent only 18% (4 out of 18) and 12% (4 out of 29) of the total government employee and "other" respondents, respectively. A more even share of participation is shown by consultants in the private sector, as approximately 40% (3 out of 5) state they have participated.

Certain types of occupation lack representation for this answer, in the scope of this survey, namely employees in an SME and independent researchers. In the case of SMEs, the share of answers was lower than expected, as only 13[59] participated, out of the total of 238 stakeholders that answered this section of the survey, which could explain this result. Likewise, only three independent researchers contributed to the total pool of respondents (Figure – 18).

[59] One additional SME participated in the overall survey. However, since no answers were registered for this particular section, they were not considered.

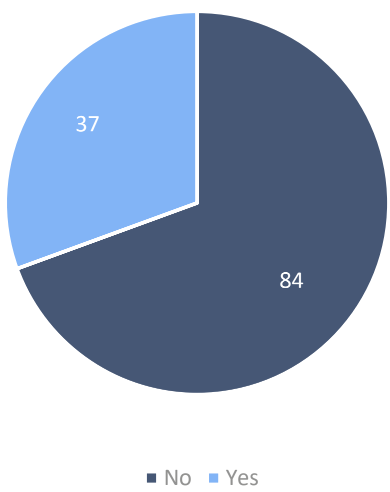
Figure 18 – Participation in the 2019 web-based co-design exercises during the strategic planning process for Horizon Europe (N=122)



Eight respondents further elaborated that they believed their contribution during the co-design activities was considered. While two respondents expressed that their contribution was not adequately considered, especially when suggesting more innovative or field-specific ideas.

When enquiring about the additional face-to-face exercise carried out to further contribute to the design of the first strategic plan of Horizon Europe, during the European Research and Innovation Days (Figure 19), the share of respondents that were aware of this activity is lower (37 out of 121 respondents, 30 %) than that of the web-based co-designing activities.

Figure 19 – Awareness of face-to-face co-design activities during the European Research and Innovation Days in 2019 (N=121)



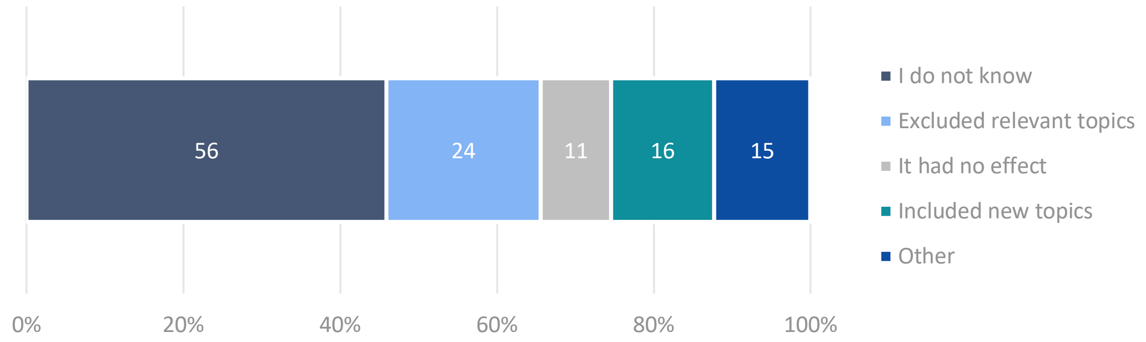
In line with this trend, the percentage of survey respondents that have participated in this exercise is also lower, totalling only 6 out of 120 (5%) – originating from Germany, Italy, Poland, Spain and outside of the EU (unspecified) – while 114 (95%) did not participate. One respondent further elaborated that the face-to-face co-design activities was a fruitful opportunity to provide input for the strategic plan. Other respondents felt the co-design activities require better organisation and are too dependent on the background of selected participants.

The findings indicate a need for enhancing awareness regarding the strategic plan co-design activities. Stakeholders who actively participated in these sessions, although comprising a small share of respondents, conveyed a positive experience, highlighting its efficacy in incorporating valuable input from citizens.

Impact of the strategic plan on the types of calls launched

A larger share of the respondents (56 out of 122 respondents, 46%) is unsure of the impact it might have had. For the other half of the stakeholders, the opinion is somewhat divided into opposite views: 24 believe it excluded relevant topics from funding, while 16 believe it included topics previously overlooked (Figure 20).

Figure 20 – Respondent's perception of the impact of the introduction of the strategic plan on the topics of calls (N=122)



When analysing the disaggregated results by stakeholder type, the share of responses per option is somewhat similar. There is only some divergence in NCPs and evaluators in HE's responses. A higher share of respondents in the NCP group (5 out of 24, 21%) believes the introduction of the strategic plan had no effect on the topics of calls, while a higher share of respondents of the evaluators in HE group (9 out of 30, 30%) believes it excluded important topics from funding.

Eight respondents further elaborated that the strategic plan provided a prioritisation of topics, which makes the allocation of resources more effective and efficient. Furthermore, some stakeholders mentioned that the roadmap provided by the strategic plan and the topics it prioritises are clearer. Respondents, however, did not specify how exactly the allocation of resources became more efficient or effective under the strategic plan.

The last question of this section of the survey sought to understand if stakeholders felt that any important topics were overlooked. Four answers were gathered, two of which supported the current selection of priority topics. The remaining two respondents mention 'human mobility and migration' and 'cardiovascular disease prevention' as key topics which are currently overlooked.

Drawing from the insights gathered, it is inconclusive to establish that the strategic plan had a detrimental impact on the nature of calls initiated within Horizon Europe. The prevailing sentiment among respondents leans towards an uncertain perception of its influence.

Results synthesis on the impact of the strategic plan

The strategic planning process in Horizon Europe is a novel approach to enhance adaptability, policy learning, stakeholder engagement, and programme impact.

The results of the online survey suggest a moderate awareness of the strategic plan. Approximately half of the surveyed stakeholders are aware of the existence of the strategic plan of Horizon Europe. From this group, approximately half were also aware of the 2019 web-based co-designing activities, and a lower share of the face-to-face exercise that took place during the European Research and Innovation Days in 2019. Consequently, there is a clear need for improved dissemination of forthcoming consultations associated with the strategic plan to ensure broader participation from key stakeholders. This proactive approach will facilitate a more inclusive and comprehensive strategic planning process within Horizon Europe.

On the impact of the strategic plan, contrasting views amongst stakeholders arose from the survey results. Over half of respondents are either unsure or believe it had no impact, whereas the other half either believes it either excluded or included relevant topics from funding. A small share of respondents believes it made the prioritisation of topics clearer and better aligned with those set by the European Commission and mirrored by the roadmap set through the strategic plan.

2.3.3 Assessing the perceived impact of discontinuing the FET Flagship on Horizon Europe: Ramifications and future prospects

Future and Emerging Technologies (FET) Flagships was a European Union initiative aimed at advancing high-risk, multidisciplinary research in cutting-edge technologies. It was part of the FET programme launched under the Excellent Science Pillar of Horizon 2020 programme. It sought to address major scientific and technological challenges through large-scale research initiatives and propel Europe to the forefront of global research and technological development by supporting visionary projects with transformative potential[60].

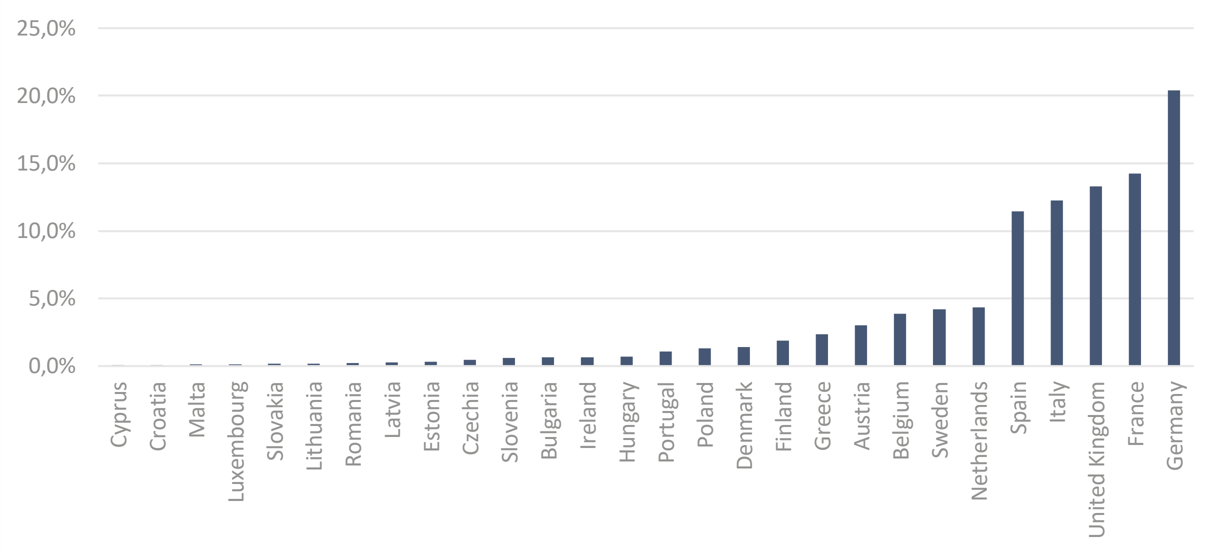
FET Flagships, with a 10-year duration and a EUR 1 billion budget each, was one of the most ambitious EU-funded research-oriented projects. The four main programmes—Graphene Flagship, Human Brain Project, Battery 2030+, and Quantum Technology Flagship— fostered cutting-edge research and innovation to address diverse challenges:

1. The Graphene Flagship aimed to unlock the full potential of graphene and related two-dimensional materials, not only advancing fundamental research but also driving practical applications across diverse sectors such as electronics, energy, healthcare, and material science.
2. The Human Brain Project sought to revolutionise our understanding of the brain, aiming to simulate its intricate functions through advanced computational models. This flagship aimed to provide insights into neurological disorders, cognitive processes, and artificial intelligence, paving the way for innovative brain-inspired technologies.
3. Battery 2030+ focused on advancing battery technology to meet the growing demand for energy storage solutions. With an emphasis on sustainability, safety, and performance, it helped to develop next-generation batteries capable of powering renewable energy systems, electric vehicles, and portable electronics.
4. The Quantum Technology Flagship aimed to harness the principles of quantum mechanics to develop transformative technologies. It explored new possibilities in computing, communication, sensing and high-precision measurement.

In terms of number of funded projects, Germany, France, Italy, Spain, and the Netherlands have been the major beneficiaries of the programme. The distribution of the FET Flagship beneficiaries per Member State can be found in Figure 21 illustrates how FET Flagship project funding is distributed among countries in relation to the total number of beneficiaries of the programme.

[60] European Commission. 'Shaping Europe's digital future', website, (n.d.).

Figure 21 – Distribution of FET Flagship-funded projects per country relative to the total FET Flagship beneficiaries



The objective of this sub-section is to analyse the impact of discontinuing the FET Flagship programme. It aims to assess the consequences on various dimensions, including technological advancement, the structure of the scientific community, research output, and the programme's ability to facilitate the transition of scientific knowledge to the market. This sub-task relies on desk research using secondary sources, such as research reports, official documents, and academic and policy papers. In addition, insights are gathered through an online survey focusing on obtaining perspectives from FET Flagship beneficiaries. To complement these findings, targeted interviews were conducted to FET Flagship beneficiaries and National Contact Points from the EIC and EIE[61]. These interviews aimed to grasp insights into how the structural absence of major initiatives affects the effectiveness of the programme.

Literature background

The FET Flagship has supported research-based innovations. The instrument's scale and size facilitated collaboration among large consortia from academia and industry, creating a network capable of extending beyond the 10-year duration of the project. In addition, it enabled the structuring of research communities around multidisciplinary themes of pan-European strategic significance. The Graphene Flagship is a notable example of the initiative's impact on innovation and economic development in Europe, as evidenced by the generation of a high number of top-quality scientific publications and numerous filed patent applications[62].

The programme has effectively fostered collaboration by bringing together numerous research organisations across Europe, creating enduring links between academia and industry. This collaboration extends across 100+ partnering organisations, contributing to an unprecedented level of collaboration and community building in Europe. Furthermore, the extended duration of Flagship programmes, which could span up to 10 years, has allowed participating research groups to build expertise and establish lasting connections[63].

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[61] National Contact Points were chosen either due to their expertise or through referrals from other NCPs.

[62] European Union, 'FET FLAGSHIPS Interim Evaluation', Publications Office of the European Union, 2017.

[63] European Union, 'FET FLAGSHIPS Interim Evaluation', Publications Office of the European Union, 2017.

Additionally, FET Flagships aimed to create and promote collaboration and international engagement across diverse research domains, shaping Europe's role in global science and technology. Their intent was to address specific challenges with notable societal and economic impacts, garnering significant public attention. Hence, they were intended as tools to enhance Europe's competitiveness against global counterparts, leveraging strategic investments in key technologies[64].

A drawback identified in their implementation was the limited effort directed towards involving SMEs and larger companies, arguing that they could have created a more entrepreneurial environment by actively engaging with companies willing to participate[65]. Nonetheless, certain Flagships demonstrated more favourable involvement. In particular, the Graphene Flagship significantly enhanced its performance in this regard, resulting in the establishment of new ventures or collaborations.

Regarding the capability to bring scientific innovations to practical applications, the FET Flagship programme envisioned the translation of scientific advancements into the market as the ultimate objective. While not all projects successfully achieved the goal due to various factors such as the complexity of the research, the novelty of the technologies involved or market readiness level, it saw varying degrees of success across its projects.

A standout example of this success is the Graphene Flagship. Over a decade, the Flagship achieved 80% of its scientific and technological targets and objectives[66], making substantial contributions to the field and catalysing commercialisation and innovation. This included the launch of 20 spin-off companies, the generation of 108 products, and the filing of 387 patent applications (84 of which have already been granted)[67].

Similarly, the Human Brain Project (HBP) showcased remarkable achievements. It produced over 160 digital tools and important advances in neuroscience, medicine, and technology, including 3D brain atlases and developments in neuromorphic computing[68].

Regarding the discontinuation of the FET Flagship programme, some initiatives under Horizon Europe emerged as potential replacements due to their alignment with similar specific challenges. For instance, EU Missions have surfaced as potential instruments, given their goals and specific objectives. These targeted initiatives address societal challenges such as cancer, climate adaptation, healthy oceans, climate-neutral cities, and healthy soils. Hence, breakthroughs from FET Flagships could contribute to Mission challenges, utilising foundational research as a basis for practical solutions[69].

FET Flagships, despite their aim of addressing societal challenges, have fallen short in defining clear goals and milestones in terms of societal relevance and giving sufficient attention to put the same emphasis on societal goals. In this regard, Missions prioritise societal relevance as a crucial

[64] Science Europe, 'Europe Policy Brief on FET Flagships', website, 2016.

[65] European Liaison office of the German Research Organisations, 'FET Flagships Lessons learned from the first 30 months of their operation', 2016.

[66] The Graphene Flagship set out 77 scientific and technical objectives and 29 system-level targets across various domains, from fundamental research to applications like energy storage and biomedical. Eleven specific roadmaps guided the initiative. After ten years, the Flagship achieved 80% of its promises, reaching 67 scientific and technological targets and 17 system-level targets.

[67] Graphene Flagship, 'Ten years of research, innovation and collaboration: the Graphene Flagship and the 2DM community'. (n.d).

[68] Human Brain Project. The Human Brain Project ends: What has been achieved. Human Brain Project, website, 2023.

[69] European Commission, 'EU Missions in Horizon Europe', website, (n.d.).

component and as one of their primary objectives[70]. In addition, while Horizon Europe Missions establish specific, time-bound goals for the 2021-2027 period, FET Flagships focused on open-ended, foundational research, aiming for transformative advances with a longer time horizon.

The European Innovation Council (EIC) also establishes some potential complementarities with the FET Flagships. It operates as a comprehensive framework within the European Commission, focused on fostering innovation and entrepreneurship. The EIC aims to propel the development of breakthrough technologies and innovations across the entire innovation spectrum and to facilitate SME development and scale-up[71].

Despite their distinct focuses, both the EIC and FET Flagship contribute uniquely to the progress of cutting-edge technologies and scientific discovery. While the EIC primarily addresses innovation broadly, supporting startups and businesses, some components of the FET Flagship could potentially be integrated into these programmes. Although there are clear differences between the two initiatives in terms of objectives, understanding the sequence of the programmes is crucial. For instance, while the EIC is designed to bolster innovations with applications closer to the market, emphasising entrepreneurship and economic impact, the FET Flagship places a stronger emphasis on fundamental research, encouraging risk-taking and exploration in scientific domains that might not yield immediate commercial outcomes but hold transformative potential.

As a result, with the flagships having fulfilled their roles, it is anticipated that the EIC could assume responsibility, benefiting from a more mature field where a less specific focus is needed. Nonetheless, this shift does not encompass other domains beyond the flagships' scope. Hence, while the EIC might adequately substitute the old flagships, it may not entirely fulfil the role typically filled by a new flagship.

Results of the online survey

The online survey developed included dedicated questions on the FET Flagship programme and its discontinuation, targeting FET Flagship beneficiaries. These survey questions assess insights from a beneficiary perspective, considering the scientific and technological advancements achieved, the ability to translate scientific knowledge to the market, and the potential for collaboration with other industry partners. Additionally, the survey aims to obtain participants' perceptions of the FET Flagship programme's discontinuation concerning the possibility of applying for calls under Horizon Europe and the alignment of other funding instruments under Horizon Europe with the intended research. Lastly, an open-ended question was also included regarding participants' perceptions of the potential impact of discontinuing the FET Flagships programme on R&I activities in Europe. The specific set of questions can be found in Annex 3.

The fact that the survey targeted beneficiaries of the programme may have introduced a potential bias, given that respondents might inherently hold a positive view towards it. However, individuals not involved in the flagship programme may lack sufficient knowledge to provide meaningful feedback. Consequently, selecting non-beneficiaries to contact would be challenging, as they may bring their own biases into the evaluation. Nevertheless, the survey was designed to extract critical feedback, aiming for a balanced perspective. It is important to note that the survey attained a response rate of only 9.5% of the total pool of beneficiaries contacted. To mitigate this and enrich the findings, a supplementary series of interviews involving both beneficiaries and National Contact Points was conducted.

[70] European Commission, Directorate-General for Research and Innovation, Mazzucato, M., Mission-oriented research & innovation in the European Union – A problem-solving approach to fuel innovation-led growth, Publications Office, 2018.

[71] European Innovation Council (EIC), 'Work Programme', 2024.

Results regarding the beneficiaries of the project can be found in Figure 22: The Graphene Flagship received the highest number of feedback (9), the Quantum Technologies Flagship and The Human Brain Project (HBP) each received 5 responses, and Battery 2030+ obtained no responses. Distribution of responses per Member State can be found in Figure 22 and Figure 23.

Figure 22 – Representation of the four Flagships (N=19)

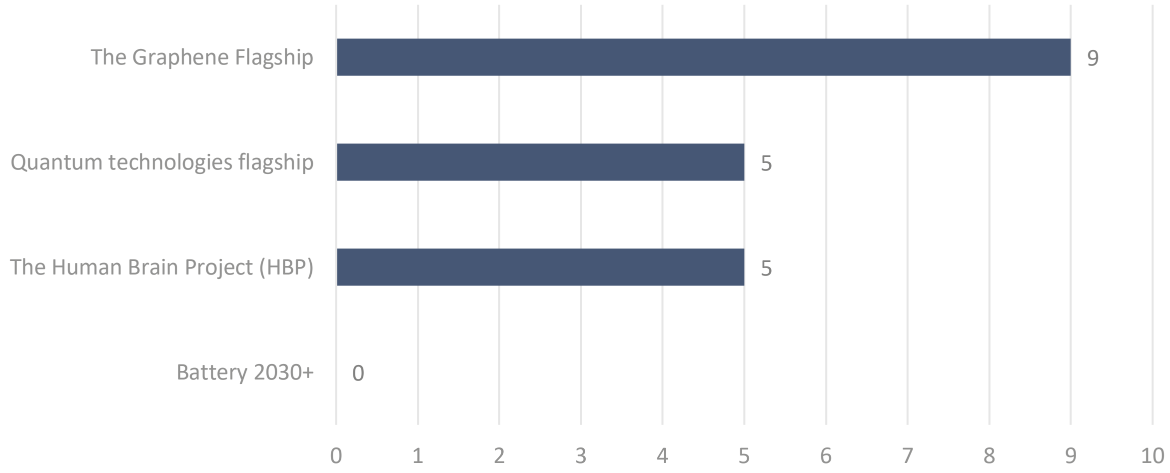
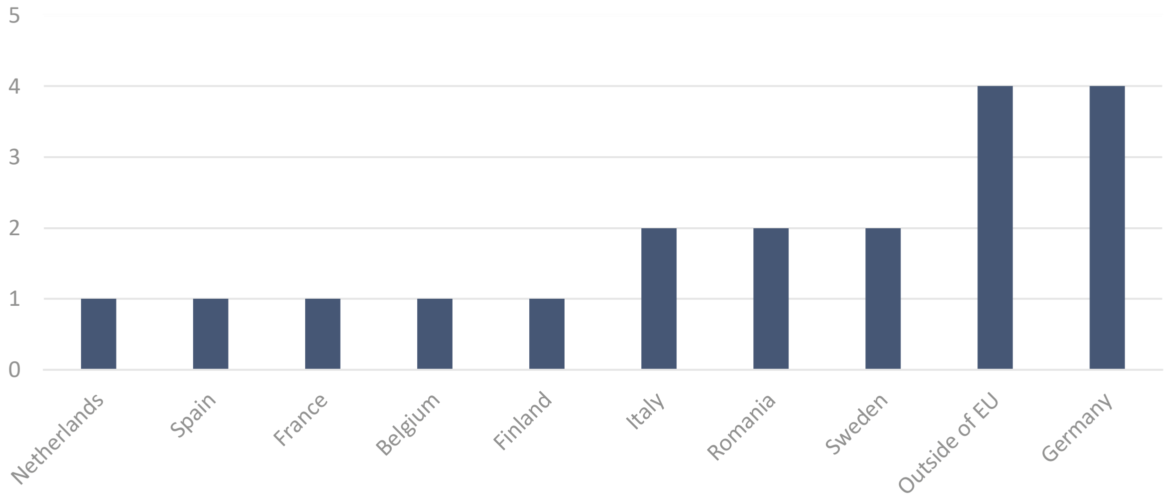


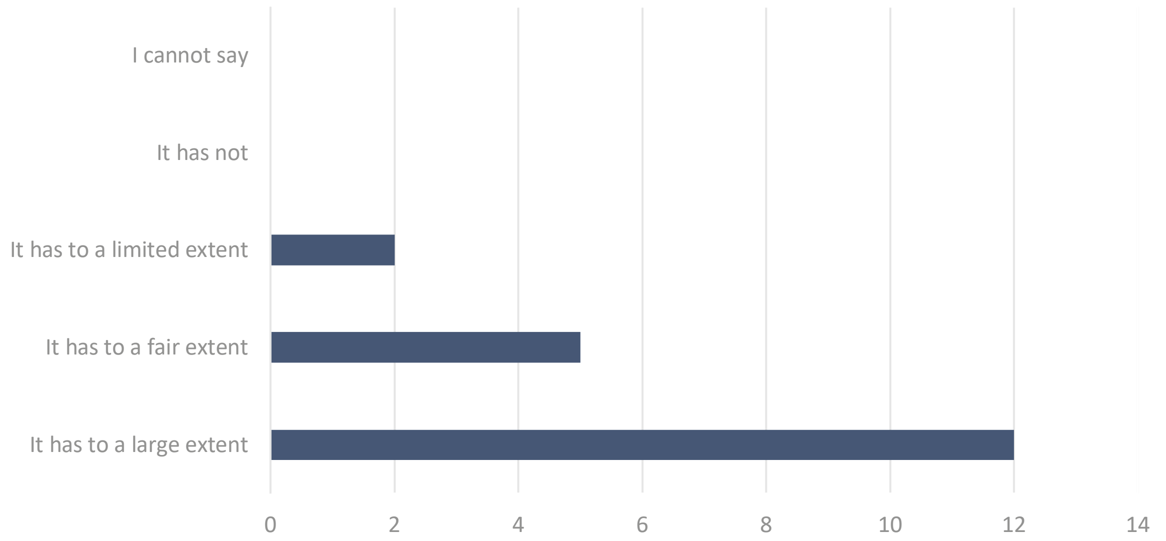
Figure 23 – Distribution of responses per Member State (N=19)



Impact on Technological and Scientific Research Community

Out of the 19 responses gathered from the online survey conducted in this study, 63% (12 respondents) stated that the funding received from the Flagship programme enabled them, to a large extent, to make significant advancements in their scientific knowledge or technological innovation. In addition, 26% (5 respondents) reported that it has enabled them to make advancements to a fair extent, while only 11% (2 respondents) reported it to a limited extent, as illustrated in Figure 24. Notably, no negative responses were received for this question.

Figure 24 – Respondent's perception of the impact of flagship funding on significant advancements in scientific knowledge and technological innovation (N=19)



Those who reported a positive impact were asked to provide open answers illustrating examples of such impact. One respondent from the Human Brain Project (HBP) highlighted the transformative impact of funding, enabling extensive, cross-disciplinary research projects. This collaboration brought together researchers from diverse fields, leading to unprecedented results and the creation of research infrastructures for neuroscience. Meanwhile, contributors to The Graphene Flagship underscored the pivotal role of long-term funding in translating early research into tangible societal impacts. They emphasised that the multidisciplinary context not only stimulated innovation in engineering approaches to device development but also created a network that facilitated the application of competencies across various fields. This, in turn, spurred the development of technological applications for innovative materials, extending beyond graphene.

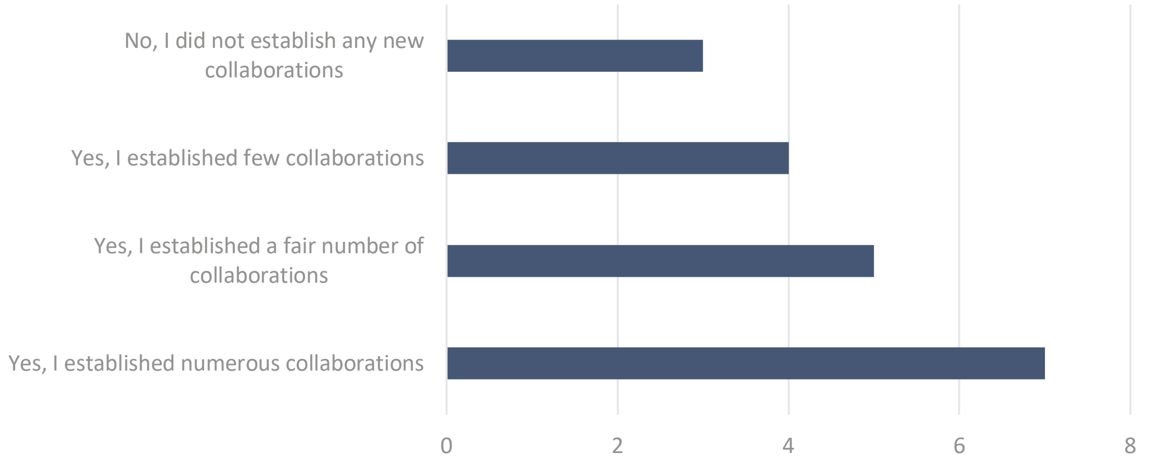
On the other hand, a beneficiary of the Human Brain Project (HBP) indicated limited impact, as the project encompassed basic research and tool creation to support that research. While successful in advancing science, its applied impact in medical and technological fields remained modest, with few early-stage clinical applications and minimal progress in IT. In addition, he stated that the potential establishment of a Research Infrastructure to share HBP tools might have limited impact due to tools being highly tailored to HBP researchers' methods and dominant partners being narrowly focused on their own research.

Collaboration and stakeholder engagement

The results of the online survey show that a majority of the respondents had established collaboration with industrial companies at least to a fair extent. Indeed, 37% (seven respondents) indicated that they had established numerous collaborations, and another 26% (five respondents) reported having established a fair number of collaborations.

Nonetheless, 21% (four respondents) stated that they had established few collaborations and 16% (three respondents) responded that they did not establish any new collaborations. This could be attributed to a potential lack of transparent and flexible governance structure. For instance, a more decentralised approach, incorporating various federated projects or regional hubs, could have ensured the inclusion of partners during their implementation, stated in the early years of the FET Flagship's implementation.[72]. Figure 25 illustrates the results.

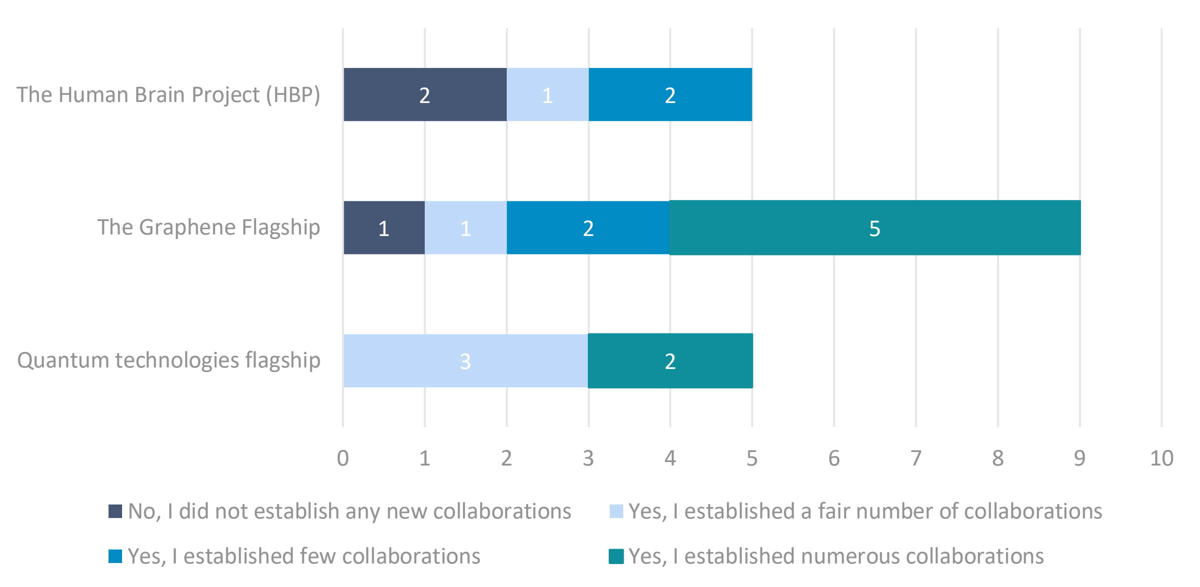
Figure 25 – Respondent's perception of creating collaboration and engagement with other industry partners (N=19)



Breaking the results down by each specific FET Flagship programme, we can see that the most cited programme was the Quantum Technologies Flagship, with two out of five respondents stating having established numerous collaborations and three respondents stating having established a fair number of collaborations. In the case of the Graphene Flagship, all options were selected, with the majority of stakeholders (five out of nine) reporting having established numerous collaborations. However, none of the five respondents from the Human Brain Project reported having established numerous collaborations. Only one respondent stated they had established a fair number, while two respondents indicated not having established any collaborations, and two respondents reported having established only a few. Figure 26 illustrates the results.

72 Science Europe, 'Europe Policy Brief on FET Flagships', website, 2016.

Figure 26 – Respondent's perception of creating collaboration and engagement with other industry partners by type of project (N=19)



Transition to market

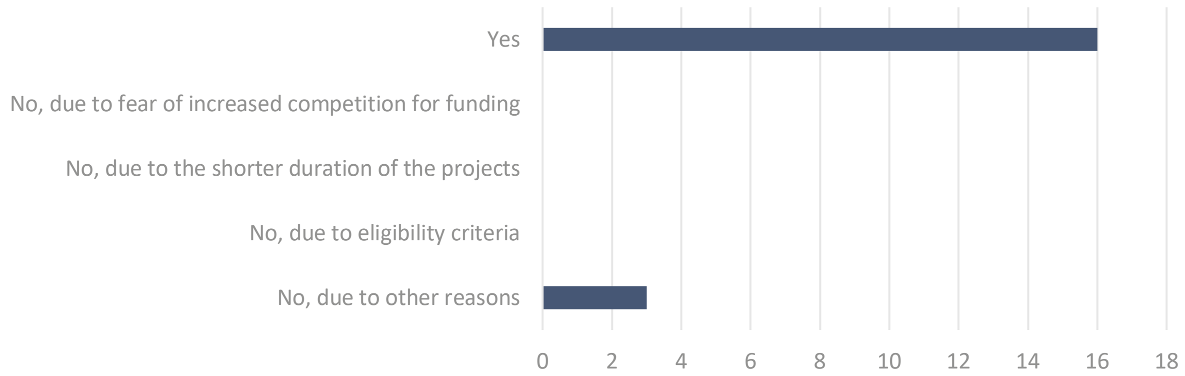
The online survey shows mixed results when it comes to the translation of scientific advancement into tangible products. Results showed that 52% of the respondents (10 respondents) reported having translated scientific knowledge into tangible, market-ready technology, while 47% (nine respondents) replied negatively.

Respondents were also given the opportunity to provide open feedback on specific examples of any scientific knowledge translated into tangible, market-ready technology. Examples include new products brought to the market, such as graphene and 2D material powders, dispersions, polymer compounds, and composite materials. Also, the development of new applications was highlighted; for instance, the use of aero graphene enabled the testing of airplanes. Moreover, four of the respondents provide specific examples of spin-offs created out of the project.

Shift from discontinued funding to new instruments

In the online survey, when asked about the discontinuation and their considerations for applying to other calls, 84% (16 respondents) expressed interest in applying for calls under HE, which implies the need for funding to sustain their projects beyond the period covered by the grant. Only 16% (three respondents) mentioned not considering other calls. Notably, no one cited eligibility criteria or project duration as reasons for not applying. This suggests that new instruments under the HE call could be seen or expected to be potential replacements for the respondents, given their apparent lack of concern over common application issues and the specific characteristics of the FET Flagship, such as the long-term duration of the projects, which are defining features of the Flagship. However, it remains unclear whether the flagships were not perceived as better options or if these issues were not considered significant. This suggests that individuals could still secure funding through Horizon Europe, implying that the discontinuation of the flagships may not be a definitive obstacle but could still pose challenges.

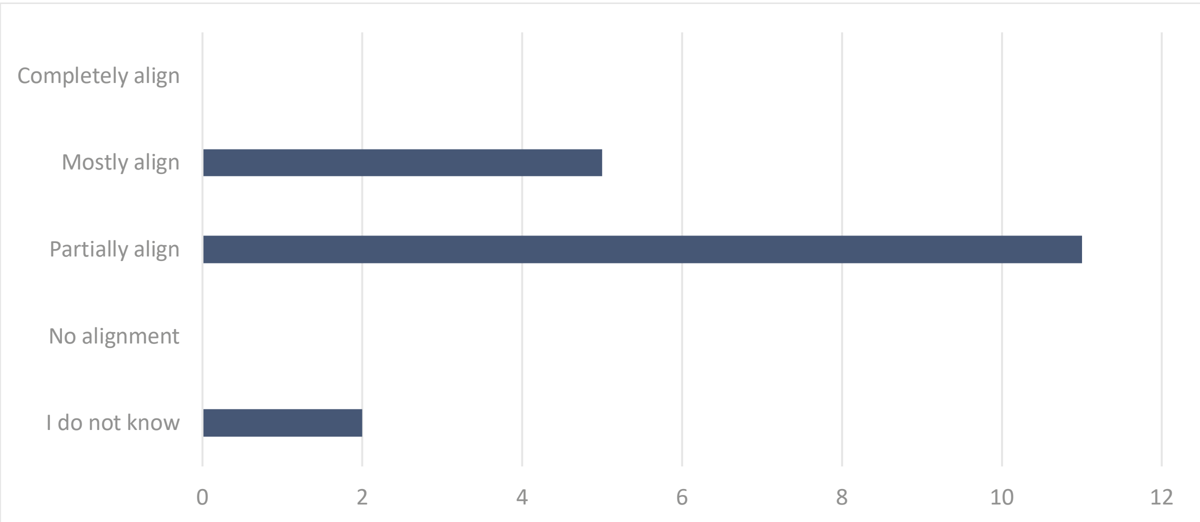
Figure 27 – Respondent's consideration of applying for calls under Horizon Europe (N=19)



However, when explicitly asked about the alignment of other funding instruments under Horizon Europe with the type of research respondents aimed to conduct, particularly in supporting research areas aligned with FET Flagship goals, the results indicated a lack of clear alignment. Out of 18 responses, only 28% (five respondents) reported mostly aligning, 61% (11 respondents) reported partial alignment, and 11% (2 respondents) reported not knowing. None of the respondents claimed complete alignment or no alignment at all, as illustrated in Figure 28.

The discrepancy in alignment could stem from the challenge of fitting the FET Flagship research area into a funding framework designed to accommodate a diverse range of research domains. FET Flagship projects, known for their high complexity, long-term objectives, and potential for transformative impacts on science and technology, may require a more tailored funding approach. Moreover, it is perhaps reasonable to consider that, having been tackled under the flagship, these topics are not as much in need of a flagship type of instrument anymore. Instead, they could benefit from more adaptable funding methods that suit their specific characteristics.

Figure 28 – Respondent's perception of the alignment of other funding instruments under Horizon Europe (N=18)



Finally, respondents were also given the opportunity to provide open feedback on their perception of the potential impact of discontinuing the FET Flagships programme on R&I activities. Overall, from the 11 respondents who provided written feedback, the responses suggest that discontinuing FET Flagship funding could have a detrimental impact.

Out of the 11 respondents who answered the open-question, six respondents underscored advantages offered by FET Flagships in comparison to other research instruments, including increased visibility, coordination, and positive impacts across technical research areas. A shared concern emerged regarding potential disruptions to collaborations, hindering application development and resulting in the loss of years of investment. Worries extended to the potential fragmentation of scientific results, emphasising the crucial multiplier effect of FET Flagships for disruptive technology development. As per one respondent: *"FET Flagships provide an additional set of advantages in contrast with other research disciplines. They provided additional visibility at many levels, which catalyse a set of new related initiatives, enhancing coordination, both at EU and MS level, and both in the research community and in the private sector […] The discontinuity of the FET Flagships might be detrimental to the development of future disruptive technologies, due to the absence of the multiplier effect provided by this instrument."*

In particular, beneficiaries of the Graphene Flagship reported potential disruptions, such as the risk of losing collaborations and partnerships in the graphene sector without core funding. In addition, Europe risks losing its technological edge as other regions, particularly the US and the Middle East, intensify their investments in graphene/2D materials. Hence, many developed technologies in Europe may be at risk of deferring to non-EU countries.

Additionally, a beneficiary of the Human Brain Project reported concerns regarding the impact on brain research, citing the possibility of reduced alignment opportunities and a retreat to smaller projects: *"Fewer possibilities for alignment and synergies in the field of brain research. The field may go back to a collection of small projects that will not have significant impact."*

On the other hand, two respondents reported that the programme posed significant challenges due to excessive bureaucracy for beneficiaries and the European Commission. The constant administrative requirements, including proposal preparation, review, reporting, and the closing and opening of project phases, were reported to affect the time researchers had available for actual research.

Results of the semi-structured interviews

The objective of these interviews is to complement the survey results by gathering additional insights on the impact of discontinuing the programme and the resulting ramifications. Specifically, they aim to examine the structural impact of the programme on the technological, scientific, and research communities, assess how the ability of Horizon Europe to bridge science and market has been affected, and evaluate the overall impact of discontinuing large-scale initiatives on the European research and innovation landscape. The concluding analysis will also discuss the extent to which new approaches and instruments available under Horizon Europe can replace some of this capability, and whether such alternative approaches enable broader participation by involving new actors who were not previously engaged in flagship programmes.

The consultation process comprised eight interviews: four with beneficiaries of FET Flagship initiatives and four with National Contact Points experienced in the European Innovation Council (EIC) and European Innovation Ecosystems (EIE). Two distinct questionnaires were developed for each stakeholder group, which are available in Annex 3.

FET Flagship beneficiaries

In terms of translating scientific and technological knowledge to market-ready technology the mixed results show that not all stakeholders succeeded in achieving this goal. Two stakeholders mentioned their inability to produce market-ready technology due to their involvement with low TRLs, thereby not expecting commercialisation. However, the funding they received played a crucial role in laying the foundation for future technological advancements, generating significant interest among various stakeholders and sparking ongoing discussions about product valorisation. In contrast, the funding received by the other two beneficiaries directly facilitated the development of tangible, market-ready technologies, as evidenced by the establishment of a spin-off company.

The FET Flagship significantly contributed to creating solid and sustainable research communities capable of aligning stakeholders and harmonising policies across different countries. There is recognition of initiatives allowing for the formation of future communities, albeit with more constrained budgets and fewer participants. Although Horizon Europe does provide avenues for research and innovation, the absence of a unifying, large-scale initiative such as the FET Flagship is felt among the research community. As noted by one respondent:

"The current landscape is fragmented, making it difficult to replicate the same level of community and strategic alignment without the flagship programme. Its absence has led to a shift towards partnerships, albeit on a smaller scale, to continue flagship initiatives. [...] While Horizon Europe maintains some coordination efforts, its structure is fragmented. This fragmentation presents challenges in maintaining a cohesive vision for ambitious, large-scale research endeavours."

This sense of fragmentation may stem from Horizon Europe's shift towards strategic planning. Under this approach, the strategic planning entails that European partnerships are formed only when they are necessary to achieve impacts that cannot be attained through other actions under the Framework Programme or national efforts alone. Consequently, research objectives that were part of the FET flagship initiative before are now redistributed across various funding instruments within Horizon Europe[73].

Regarding challenges and alternatives under Horizon Europe in the absence of the FET Flagship, stakeholders expressed concerns about the fragmented nature of current initiatives compared to the cohesive and strategic approach of the flagship programme. As per one respondent:

"The FET Flagship significantly contributed to creating solid and sustainable research communities capable of aligning stakeholders and harmonising policies across different countries. […] It aligned national authorities, stakeholders, and the industrial sector under the same umbrella."

While Horizon Europe offers calls that can fund individual projects, replacing the Flagship research, stakeholders noted the absence of the holistic and integrated approach that characterised the Flagship programme. This shift affects not only funding availability but also undermines the strategic alignment and collaborative synergy that were central to the Flagship's success.

Moreover, stakeholders reflected on the accomplishments enabled by the FET Flagship programme that are currently difficult to achieve under Horizon Europe. They highlighted that the Flagship facilitated ambitious, long-term research initiatives and fostered a strategic and coordinated approach among stakeholders. The comprehensive nature of the Flagship, offering a platform for large-scale, interdisciplinary collaborations, was particularly valued for its potential to drive groundbreaking advancements and create self-sustaining research communities.

Regarding the discontinuation of the FET Flagship impact on the European R&I landscape, mixed answers were given. Some perceive it as detrimental, emphasising the loss of strong project outcomes and global outreach, viewing its discontinuation not just as a funding loss but also as a setback for Europe's global competitiveness. Conversely, others believe the impact will not be significant due to alternative funding options.

Additionally, there are suggestions for improvements, particularly in enhancing technology transfer and industry collaboration. As cited by one stakeholder: *"The impact could have been greater if the*

[73] European Commission, Secretary-General, 'Commission Staff Working Document Impact Assessment', 2018.

funding was more towards technology transfer and able to support initiatives like ours (start-ups). More emphasis on technology transfer and industry collaboration could have enhanced its impact."

To maximise the impact of Horizon Europe and fill the gap left by the FET Flagship, stakeholders suggest the development of new, strategic initiatives that can emulate the Flagship's ability to foster large-scale, interdisciplinary collaboration and long-term research agendas. These initiatives should integrate various stakeholders, align national and European research strategies, and provide a cohesive framework for addressing grand challenges through innovative research and technology development. As highlighted by one of the stakeholders:

"The most vulnerable aspects of the HE projects include disseminating results effectively, engaging communities, enhancing public outreach of the project, and improving tracking within the know-how repository. […] Moving forward, it is essential to ensure continuity in research efforts post-FET Flagship. Each project should serve as a repository for accumulated knowledge to mitigate the risk of losing valuable insights."

In essence, while Horizon Europe offers opportunities, there is a clear desire among the research community for initiatives that can replicate the scale, scope, and impact of the FET Flagship programme to ensure Europe remains at the forefront of global scientific research and innovation.

National Contact Points

All National Contact Points reported the FET Flagship programme significantly impacted their country's research landscapes, as the programme was crucial for generating income and fostering collaborations between industrial and research sectors. The NCPs of Austria and Spain reported that their countries benefited from the programme's role in technological and research advancements, leading to practical solutions and a surge in university spin-off ventures. For instance, the Spanish NCP reported:

"The most important Flagship at the Spanish level has been, without any sort of doubts, the Graphene one […] One concrete success case in Spain is the case of the company Graphenea[74]*, a leading global graphene producer."*

In addition, two NCPs stated that their countries, the Netherlands and Austria, appreciated the programme's contribution to promoting knowledge sharing and enabling international collaboration, although it is challenging to attribute progress solely to the FET Flagship.

However, in terms of translating scientific discoveries into marketable innovations, there is scepticism about whether the FET Flagship brought advancements to that point as well as the current Horizon Europe framework. As stated by the Austrian NCP: *"The facilitation of translating scientific discoveries into marketable innovations in Horizon Europe is not yet there, but not only because of the FET Flagship; rather, it is a common problem."* Moreover, the Dutch NCP suggested that a more focused approach on lower TRLs and a commitment to long-term funding and technological specificity could enhance Horizon Europe's impact on scientific progress and innovation.

Stakeholders reported that beneficiaries currently face challenges in finding alternative funding sources, with particular concern about the creation of a network and the long-term commitment to specific technological areas that the FET Flagship once provided, citing:

"The Netherlands has experienced a lack of funding from a European perspective. […] The discontinuation has raised questions about the long-term commitment to certain technologies. As a result, there is uncertainty about how resources will be allocated across specific topics covered by the flagship, potentially leading to a reliance on national funding schemes."

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[74] Graphenea, website, (n.d.)

Moreover, they expressed concerns about important features missing in Horizon Europe that were present in the Flagship, specifically the interaction with the production community and its networking effort. The Austrian NCP mentioned: *"Although there were only a few projects being funded, making it difficult to find your place within them, it was significant as being part of this network made a difference in terms of stability."*

Regarding the potential of new instruments under Horizon Europe to fill the void left by the FET Flagship, there is a sense of ambiguity. Although the EIC Pathfinder is anticipated to be a natural successor, there are mixed feelings about its adequacy as the EIC Pathfinder instrument lacks the kind of research that was undertaken under the Flagship. While in the non-flagship aspect of FET itself, it has occurred with some differences; it is not seen as a replacement.

Specific instruments within Horizon Europe address some research topics conducted in the Flagship, covering particular research areas. For example, the Chips Joint Undertaking may address part of the Graphene Flagship scope, primarily related to semiconductors. However, scepticism persists regarding the continuation of the work done, as expressed by the Spanish NCP: *"I have serious doubts regarding the generation of cross-fertilisation and scale economies without the Flagship umbrella. A lack of coordination around the research topics of the former Flagships is more than evident."*

In order to maximise the impact of Horizon Europe in fostering scientific progress and innovation, public-private collaboration initiatives covering the main parts of the FET Flagships are still missing in Horizon Europe. Hence, promoting the aforementioned collaborations and providing EU Funds for them within Horizon Europe will be a key success factor to boost the Flagship research and facilitate the resulting applications market take up in the mid-term. Additionally, it is important to recognise that while research funding may be available from other sources, the support part would significantly benefit from EU-level coordination. Moreover, a greater connection with national initiatives needs to be considered. As highlighted by the Italian NCP: *"The FET flagship was great but exclusive, EU funding should be transparent and accessible."*

Results synthesis on the impact of discontinuing the FET Flagship

The FET Flagship initiative aimed at significantly advancing scientific and technological research in Europe. Based on the survey, 12 out of 19 respondents (63%) credited Flagship programme funding for significant advancements in scientific knowledge or technological innovation. Additionally, 29% reported achieving scientific advancements to a fair extent.

Although the programme was expected to transform scientific advancements into market innovations, reported outcomes in translating these advancements into tangible products varied. The survey revealed a success rate of only 52% (10 respondents), while 47% (nine respondents) reported no success in this translation endeavour. Similar mixed results emerged from the interviews, where two respondents successfully brought products to the market, while two stakeholders stated not having foreseen the development of market-ready technology due to their engagement with low TRLs.

The initiative fostered collaboration across Europe, creating solid and sustainable research communities. This collaboration was crucial for multidisciplinary innovation and extending the reach and impact of European research. It not only stimulated innovation in engineering approaches to device development but also facilitated the application of competencies across various fields, as confirmed by stakeholder feedback.

However, while the Flagships aimed to create lasting communities, their effectiveness in achieving this objective is uncertain and subject to debate. The discontinuation of the FET Flagships has sparked concerns regarding disruptions in collaboration, hindering application development, and resulting in investment loss, as evident from survey results and interviews. Stakeholders have pointed out the inadequacy of effective community building within the current structure of Horizon, which cannot be replicated. Nonetheless, the fact that some of the collaborations supported by the flagships diminished after their conclusion, prompts inquiries into whether the primary issue lies in coordination or if the specific funding mechanism played a significant role in fostering community cohesion.

Stakeholders noted that within the Horizon framework programme, there is only partial alignment with FET Flagship goals, lacking specific calls for new initiatives that mirror the Flagship's scale and effectiveness. With the Flagships having fulfilled their roles, there is anticipation that the EIC could assume responsibility, benefitting from a more mature field where a less specific focus is needed. However, there are mixed feelings about its adequacy, as it lacks the kind of research undertaken under the Flagship.

2.4 Fostering SME participation in Horizon Europe: Barriers, provisions, and sustainable growth

SMEs, or small and medium-sized enterprises, play a pivotal role in the European economy, contributing significantly to both employment and economic activity. In 2022, the European Union was home to approximately 24.3 million SMEs, representing a staggering 99.8% of all enterprises in the non-financial business sector[75]. These SMEs collectively provided livelihoods to 84.9 million people across the EU-27.

Despite their economic prominence, start-ups and small and medium-sized enterprises (SMEs) within the European Union have encountered challenges in commercialising innovations stemming from scientific research. Many innovative ideas with the potential for commercial success have struggled to move from the laboratory to the marketplace. This predicament can be attributed, at least in part, to a funding gap, as both basic and applied research typically receive support from public sources, including the EU, while the market-oriented phase often relies on private funding. This challenge has given rise to what is known as the "European paradox": Europe often leads in research, as evidenced by its extensive publications, but often lacks the entrepreneurial capacity to transform this knowledge into innovation, economic growth, and job creation[76].

The need to support SMEs in bridging the gap between research and the marketplace was acknowledged in the sixth Framework Programme, in the form of the “Research for the benefit of SMEs” programme. This provided non-financial support for SMEs through “knowledge providers” (mainly research organisations and universities). This support to SMEs was further strengthened in the subsequent European R&I framework programmes, especially in terms of funding provided for start-ups and SMEs.

They key financing tool under the current research framework programme Horizon Europe is the European Innovation Council (EIC) Accelerator programme, formerly the SME instrument under Horizon 2020. The EIC Accelerator programme assists primarily SMEs and startups in scaling up high-impact innovations that have the potential to create new markets. It offers funding ranging from EUR 0.5 million to EUR 17.5 million, combined with Business Acceleration Services. The programme focuses on innovations based on scientific discoveries or technological breakthroughs (known as 'deep tech') that require substantial long-term funding ('patient capital') and often face challenges in attracting financing due to high risks and extended timeframes. The EIC Accelerator aims to help these innovators attract the necessary investment for rapid scaling. It supports technology

75 Di Bella, L., Katsinis, A., Lagüera-González, J., Odenthal, L., Hell, M., & Lozar, B., 'Annual Report on European SMEs 2022/2023', Publications Office of the European Union, Luxembourg, 2023.

76 European Court of Auditors, 'EU Support for SME Innovation: The SME Instrument', 2019.

development in the later stages, requiring innovations to be tested and validated in the laboratory and other relevant environments, at least at Technology Readiness Level 5. The programme's support acts as a catalyst to attract other investors essential for innovation scale-up[77].

Despite efforts to provide SMEs with funding opportunities, statistics suggest that these funding opportunities are often overlooked by SMEs. When examining SMEs' participation in the previous research framework programme, Horizon 2020, it becomes evident that their involvement remains limited. The share of SMEs receiving funding amount to 20% of the total share of participants. This share is mainly classified under the research and technology organisation (REC), or private-for-profit entities (PRC), or other (OTH) referenced in section 2 of this report.

Several explanations for this limited engagement can be attributed to persistent participation barriers. Members of the Centre for European Policy Studies Task Force on Innovation and Entrepreneurship[78] have pointed out that SMEs frequently lack awareness of the complex system of existing support schemes and encounter significant challenges in crafting convincing proposals due to constraints on time and resources, along with limited access to essential skills and knowledge, including language proficiency. Additional barriers can include struggles to identify partners and establish and manage international consortia. Additionally, selection criterion may favour entities with an existing well-established infrastructure.

When examining the available data for Horizon Europe[79], SMEs make up 20% of the beneficiaries in the initial three years (2021-2023). This figure is in line with that of Horizon 2020 statistics, where SMEs accounted for 20% of beneficiaries[80], in its first three years. Although there is no significant change between SME participation so far in H2020 and Horizon Europe, these figures are slightly higher than the SME participation rate in the first three years of FP7, which amounted to 18%.

In Horizon Europe, there are around 7,400 unique SMEs that benefit from funding, where 47% of these SMEs were also beneficiaries in Horizon 2020. This means that approximately 4,000 new SMEs have entered the programme, accounting for 54% of the total. When examining these newcomers, it's notable that they hail predominantly from the top beneficiaries in the EU-27, including Germany, Spain, France, Italy, and the Netherlands. Specifically, Germany, Spain, Italy, and France stand out as the top SME beneficiaries among EU-27 countries (Figure). This result is not surprising given that the combined GDPs of Germany, France, and Italy account for more than half of the EU's entire economic output. Poland, on the other hand, contributing 4% to the EU's overall economy, akin to Belgium and Sweden[81], appears to rank lower in terms of newcomers and top beneficiaries per unique SMEs under Horizon Europe.

77 European Innovation Council (EIC),' Work Programme ', 2024.

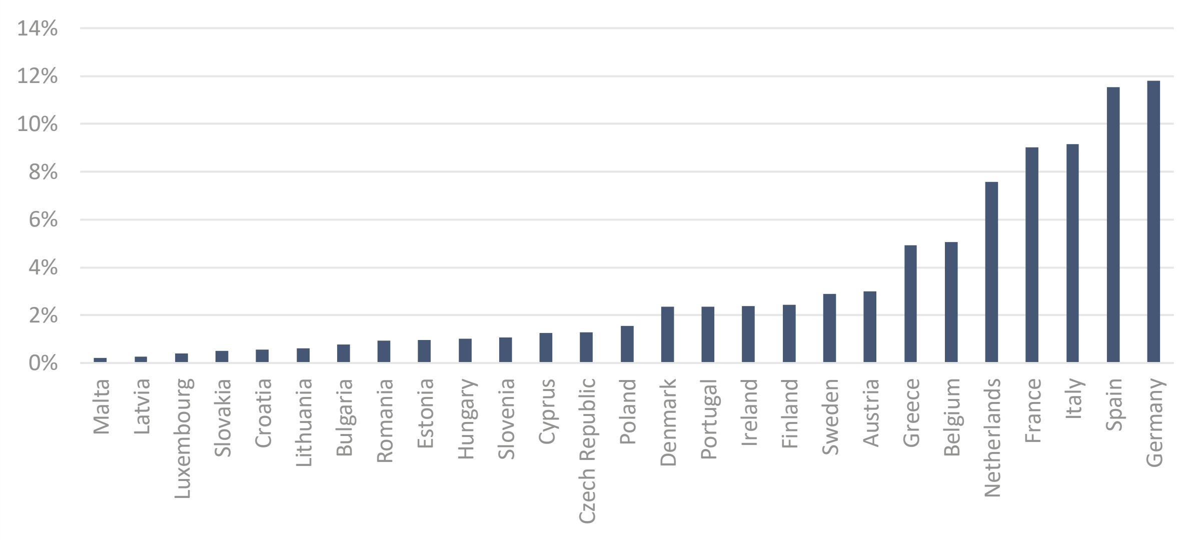
78 Simonelli, F., 'Is Horizon 2020 really more SME-friendly? A look at the figures', Centre for European Policy Studies, 2016.

79 Data accessed from EU funding &Tenders on 20 February 2024.

80 The SME share for H2020 and HE figures include duplicates because certain SMEs receive funding under different projects in which they participate.

81 World Economic Forum, 'These are the EU countries with the largest economies', website, 2023.

Figure 29 – Distribution of unique SMEs benefiting from Horizon Europe funding in the EU-27 (N=7317)



This section aims to delve into the primary obstacles encountered by SMEs when seeking European funding opportunities. To achieve this, we have incorporated a specific set of questions into the online survey, as detailed in the following section. Moreover, we have enriched the survey findings through targeted interviews with SMEs and relevant associations. The combined outcomes of these consultations are detailed below, offering comprehensive insights into the challenges faced by SMEs in accessing European funding.

2.4.1 Results of the online survey

The online survey targets employees of SMEs who received funding from Horizon 2020 and/or Horizon Europe. The set of questions begin by probing the sources of R&I funding that companies rely on for sustainability. The survey also seeks to identify barriers faced by industrial stakeholders when applying for R&I funding under Horizon Europe and gauges perceptions regarding the favourability of participation rules for large companies. Additionally, the survey examines the influence of R&I funding on SME growth and development, along with collaboration with funded partners and the resulting technological advancements. Lastly, it asks for recommendations to enhance Horizon Europe's support for SMEs in their R&I endeavours. These questions collectively aim to gain insights into SMEs' experiences and perspectives regarding R&I funding within the Horizon Europe framework. Annex 2 includes these questions.

Input from SMEs was relatively limited, with only 15 respondents, constituting a response rate of 5% in the survey. The low response rate may be attributed to the challenge of accessing direct email addresses for these SMEs. As a result, we resorted to reaching out through the contact form on CORDIS. Unfortunately, this method might not have ensured that our requests reached the intended recipients directly.

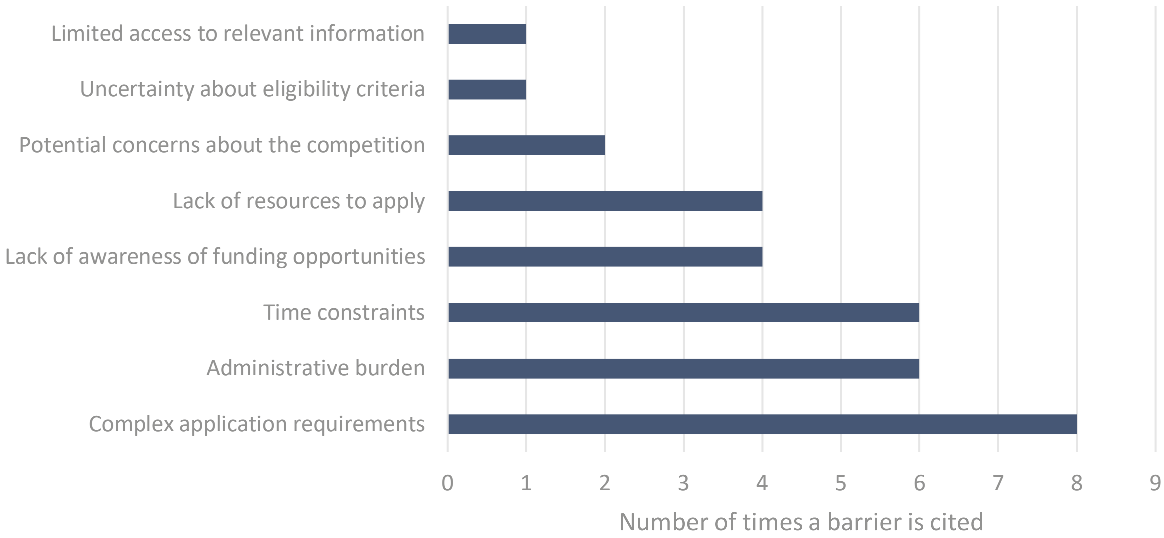
The SMEs who respondent to our survey come from Austria, Belgium, France, Germany (two responses), Greece, Italy (two responses), Outside of EU (two responses), Spain (four responses), and Sweden. Their answers are discussed in the subsections.

2.4.2 SME funding and barriers to participation

Over half of these respondents indicate that they rely on an array of funding to make their business more sustainable including internal sources; national sources; international sources, including European R&I funding. Furthermore, almost 70% of stakeholders (11 respondents) believe that additional R&I funding would help make their business more sustainable. In terms of availability of funding opportunities, around 60% of stakeholders (10 respondents) believe there are enough funding opportunities under Horizon Europe for the type of business they carry.

The majority of SME respondents, with only two exceptions, indicated that they have engaged in prior collaborations with partners who have received research and innovation (R&I) funding under Horizon 2020 or Horizon Europe programmes. When asked whether they have, to their knowledge, derived benefits from technological advancements achieved by their partner entities through European funding, a significant portion of the respondents, seven out of nine, confirmed that they have indeed experienced such benefits. When asked about the barriers SMEs face when applying for funding under Horizon Europe, the majority of respondents cite either complex application requirements, administrative burden and/or time constraints. One respondent further elaborated that drafting a Horizon Europe proposal requires a substantial level of expertise in proposal drafting, a resource often lacking within SMEs. Consequently, many SMEs find themselves compelled to engage external consulting firms to handle proposal creation, further complicating the application process. Figure provides an overview of the different barriers cited by stakeholders.

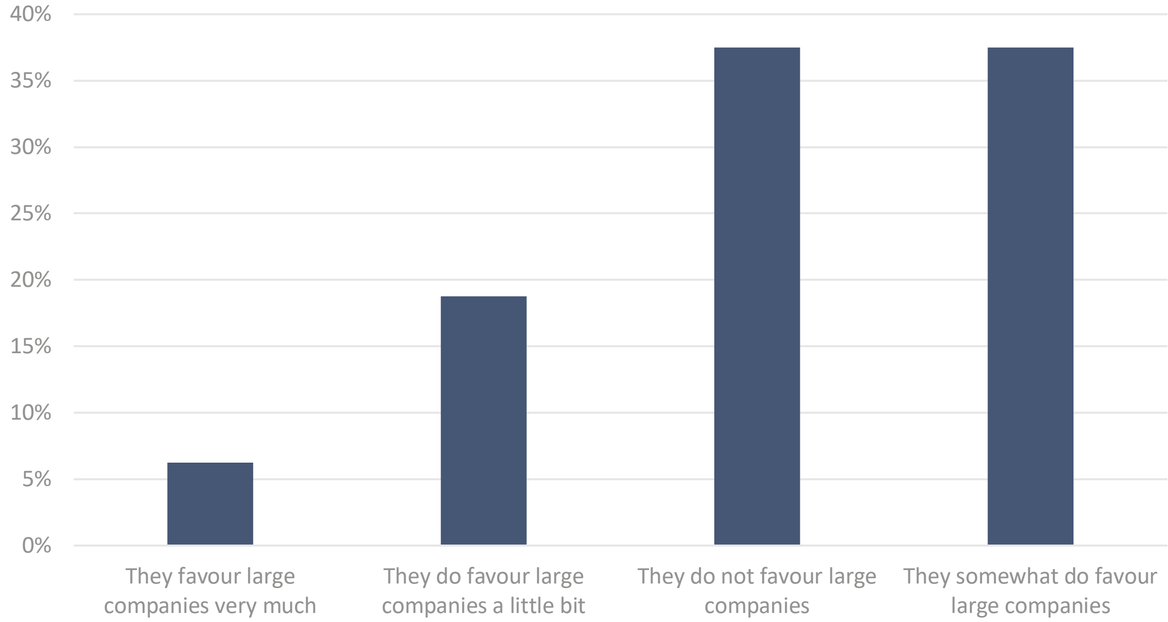
Figure 30 – Perceived barriers for industrial stakeholders applying for R&I funding under Horizon Europe (N=15)



The results indicate that opinions on whether the participation rules favour large companies are mixed. Around six respondents (38%), believe they do not favour large companies, while an equal proportion, 38%, believes that the participation rules somewhat favours them. A smaller share, 19%

(three individuals), perceives a slight favouring towards large companies, and only one person reports that the rules greatly favour large companies. These results are illustrated in Figure 31.

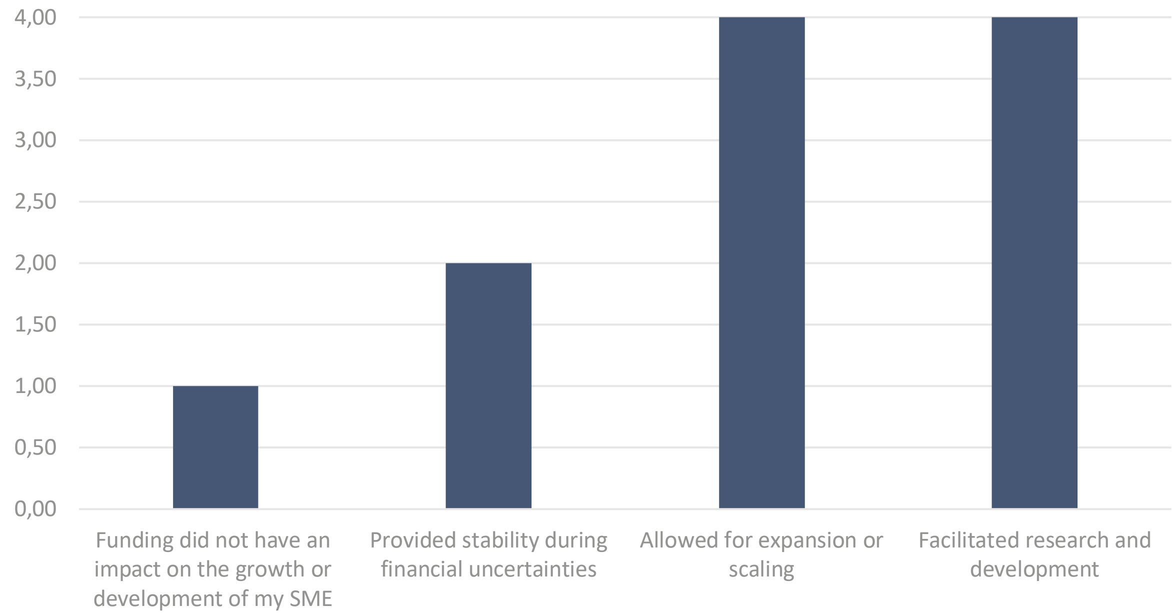
Figure 31 – Perception of rule of participation favouring larger companies (N=15)



2.4.3 European funding impact on SMEs' growth

When asked about how has the funding from Horizon Europe or any other framework programme (e.g. H2020, FP7) impacted the growth or development of SMEs, the majority of responses indicated positive effects. Examples include providing stability during financial uncertainty, allowed for expansion or scaling, and facilitated research and development as summarised in Figure 32.

Figure 32 – Perceived effect of European funding on SME growth (N=11)



Four responses from an open-ended question highlight several key concerns within regard to SMEs funding. According to one stakeholder, life science technologies requiring more capital and time tend to receive less funding compared to others, leading to potential imbalances.

Additionally, one respondent noted that new funding opportunities are relatively infrequent, resulting in extended periods between calls. Another stakeholder expressed dissatisfaction with the selection and prioritisation processes for Horizon Europe funding, citing reliance on outdated data and conventional wisdom rather than an objective evaluation of fundamentals. Consequently, this approach often results in a reduced number of R&I funding opportunities for independent fundamental research.

Eight stakeholders also provided various recommendations for enhancing Horizon Europe's general support for SMEs. These suggestions encompass a range of strategies, including implementing training and capacity-building programmes to bolster SMEs' innovation and research capabilities, ensuring the accessibility and adaptability of funding programmes for diverse SMEs, prioritising projects with clear paths to market deployment to facilitate commercialisation of SME innovations, fostering collaboration between SMEs and research institutions for knowledge transfer, conducting targeted outreach programmes to engage SMEs across regions for inclusivity, exploring interdisciplinary projects leveraging SME strengths from various domains, and partially funding preparation and post-project costs. Additionally, respondents emphasised the need for National Contact Points (NCPs) to continue their efforts in bridging the gap between industry and academia, seek more clarity on certain questions, and streamline administrative processes. Furthermore, stakeholders stress the importance of making Horizon Europe programmes world-leading, enhancing support for innovative SMEs, and aligning funding with rigorous analyses of gaps in health, science, and business fundamentals. Lastly, there are suggestions to ease the administrative burden for SMEs and improve language accessibility, along with proposals to increase SMEs' chances through additional awards, as well as enhancing funding rates in Innovation Actions.

2.4.5 Results of the semi-structured interview on the participation barriers for SMEs

To complement the findings from the survey, we conducted semi-structured interviews to gather further insights. Given the low response rate of SMEs in the online survey, these interviews became essential in obtaining additional perspectives. They were tailored to engage SMEs who have directly benefited from Horizon Europe funding, as well as representatives from SME associations and private firms providing proposal support services. This approach allowed us to delve deeper into specific experiences and perspectives, enriching our understanding of the challenges and opportunities faced by SMEs in accessing and utilising Horizon Europe funding.

The consultation process resulted in a total of nine interviews with beneficiaries of Horizon Europe (three interviews), SME associations (two interviews), private companies specialised in grant support for EU funding (four interviews). The questions posed in these semi-structured interviews are available in Annex 3.

The most significant challenge cited by HE beneficiaries that SMEs face in terms of Horizon Europe application process is lack of experience. This inexperience leads to difficulties in understanding the process, crafting effective proposals, and navigating consortium agreements. Time pressure during the application process exacerbates these challenges. To address this difficulty, all three SMEs interviewed have previously opted to enlist the assistance of external consultants to guide them through the application process. However, as they have gained experience over time, they no longer find it necessary to rely on external support.

Moreover, SMEs face stiff competition from larger research centres and corporations, which possess more resources and expertise. This puts SMEs at a disadvantage, particularly when it comes to leading consortia or competing against well-established organisations.

Furthermore, SMEs struggle to establish partnerships and gain visibility within larger consortia. Despite the support available through the proposal portal, SMEs may find it insufficient in showcasing their skills and attracting collaboration opportunities. Additionally, geographical disparities contribute to these challenges, as noted by one beneficiary:

"In Eastern Europe, SMEs rarely take the lead in consortia for Horizon Europe projects. Unless led by a university or situated in central or Western Europe, there's a reluctance to finance their proposals, often due to doubts regarding their capability to deliver."

Echoing the struggles cited by beneficiaries, private firms specialising in grant proposal writing observe a notable reluctance among SMEs to pursue European funding opportunities. Acknowledging the fierce competition inherent in securing European funding, two of the interviewed firms mentioned that they often advise SMEs to direct their efforts towards seeking national funding opportunities instead. Moreover, one stakeholder emphasised that the eligibility criteria for national funding, such as that offered in Greece, may be perceived to be clearer and more transparent compared to the criteria set by Horizon Europe.

When asked whether the rules of participation place SMEs at a disadvantage, responses from two beneficiaries and the SME associations indicate no such hindrance. However, contrasting viewpoints emerge from two other stakeholders, both SMEs currently benefiting from Horizon Europe funding, who highlight significant issues.

One concern raised is the perceived favouritism by the European Commission towards large corporations within proposals, potentially marginalising SMEs. This preference for established entities may arise from scepticism regarding SMEs' capacity to fulfil project requirements, despite their substantial contributions to the European economy, as one stakeholder argued.

Additionally, another SME expressed challenges, particularly evident during an economic downturn in their country, in securing coordinator roles within proposals. There appears to be a preference for multi-country participation, a trend that may not always be justifiable.

Nonetheless, SME associations and beneficiaries alike agree that it is rather easy to find suitable calls under Horizon Europe when it comes to the fields of technology, renewable energy, and construction. Despite this, two interviewees emphasised that the information regarding funding opportunities remains dispersed, creating uncertainty for potential applicants regarding the available funding options. One beneficiary of Horizon Europe proposed the implementation of a navigational tool akin to a compass, which would aid in pinpointing the most suitable calls aligning with one's expertise. The fragmented nature of information poses a significant burden for SMEs. As expressed by this beneficiary that while they found navigating Horizon Europe relatively straightforward due to their familiarity with Horizon 2020, understanding the available funding opportunities under the Digital Europe Programme presented challenges. This difficulty arises from their lack of experience with the programme and the fragmented nature of available information.

Regarding the present funding scheme, most stakeholders praise its structure, particularly its provision of a portion of the funding upfront, with the remainder disbursed upon project completion. This approach is commended, especially when contrasted with other national funding schemes that offer no financial support until project completion.

On the prospect of simplifying the Horizon Europe application process, a current Horizon Europe beneficiary notes that:

"The primary challenge lies in conceiving a research idea and assembling a consortium. Once achieved, all other aspects become manageable. It is worth noting that the difficulty of the application process is relative; the application process under Horizon Europe appears simpler compared to National funding application in, for example, Italy or Romania."

There was a consistent emphasis among all consulted stakeholders on the need to strengthen the role of National Contact Points (NCPs). Stakeholders articulated a need for enhanced visibility of NCPs, ensuring that SMEs are aware they can turn to them for guidance. NCPs could play a pivotal role in directing SMEs towards suitable funding opportunities and providing assistance in navigating the application process. Moreover, private firms also highlighted the potential for NCPs to promote European funding opportunities through local authorities, thereby broadening SMEs' access to vital resources and support networks. The strengthened role NCPs should play is also echoed in another study[82] on Horizon Europe support for the Green Deal. In this study, the authors highlight the fact that SMEs may not always possess proficient English language skills. NCPs can serve as vital intermediaries in overcoming this language barrier by effectively disseminating information to local SMEs.

Finally, in terms of how Horizon Europe can better support SMEs to produce market-ready innovations, interviewees put forward a few suggestions. A technology centre with strong track record in receiving European funding emphasises the importance of funding projects that are oriented towards commercialisation and market success. They suggest a focus on actual instruments and commercialisation efforts, with a call for more projects aimed at reaching higher Technology Readiness Levels (TRL) and strong dissemination efforts to facilitate cross-EU market collaboration. Another beneficiary advocates for mandatory SME involvement in consortia, given their strong ability for bringing innovations to market. Another suggestion put forward is the establishment of regional support innovation systems and networks to promote awareness of EU funding opportunities and facilitate activities like patent filing and increasing the visibility of technology transfer offices (TTOs). The Horizon Booster initiative is cited as an example of a positive effort in this direction.

2.4.6 Results synthesis on the barriers to SMEs' participation

Survey results suggest that over half of SME respondents rely on various funding sources, including internal, national, international, and European R&I funding, to enhance their business sustainability.

[82] Breuil, G., Delera, M., Diafas, I., Cvijanovic, v., 'Horizon Europe Support for the Green Deal, European Parliamentary Research Service', Panel for the Future of Science and Technology (STOA), European Parliament, 2024.

Additionally, nearly 70% of stakeholders expressed the belief that additional R&I funding would contribute to greater business sustainability.

Regarding funding opportunities, approximately 60% of stakeholders believed there were sufficient funding opportunities under Horizon Europe for their business type.

Collaboration between SMEs and partners who received Horizon 2020 or Horizon Europe R&I funding was common, with seven out of nine respondents experiencing benefits from technological advancements resulting from European funding.

Barriers faced by SMEs when applying for Horizon Europe funding included complex application requirements, administrative burdens, and time constraints. Some SMEs required external consulting for proposal drafting, further complicating the application process.

Opinions on participation rules were mixed, with some respondents believing they do not favour large companies, while an equal proportion somewhat favoured them. Finally, the impact of funding from Horizon Europe and other framework programmes on SMEs was generally positive, providing financial stability, enabling expansion, and facilitating research and development.

Despite the perception of an easier application process under Horizon Europe compared to certain national funding schemes, interviews conducted reveal that SMEs face challenges in understanding available funding options and navigating the fragmented information landscape.

Moreover, interviews highlighted concerns regarding favouritism towards large corporations in proposals, hindering SME participation. Additionally, difficulties in securing coordinator roles within proposals and the preference for multi-country participation contribute to SMEs' challenges.

Strengthening the role of National Contact Points (NCPs) was suggested to aid SMEs in accessing funding opportunities and navigating the application process. Enhanced visibility and promotion of European funding opportunities through local authorities were also recommended.

Suggestions for Horizon Europe to better support SMEs put forward by interviewees include increased funding for projects oriented towards commercialisation, and encouraging SME involvement in consortia as they are able to translate research to marketable innovations.

2.5 Stakeholders' perspectives on the Horizon Europe evaluation system

Horizon Europe (HE) introduced six Clusters, including five cross-cutting EU Missions, under the second pillar: *Global challenges and European industrial competitiveness*[83]. The clusters are focused on 'Health', 'Culture, Creativity and Inclusive Society', 'Civil Security and Society', 'Digital, Industry and Space', 'Climate, Energy and Mobility' and 'Food, Bioeconomy, Natural Resources, Agriculture, and Environment'. These Clusters were introduced envisioning better integration and synergies between the thematic areas, as well as a higher level of impact.

Along with the novelties in the structure and instruments put forward in Horizon Europe, also the proposal evaluation process underwent some changes. In the interim evaluation of H2020 published by the Commission[84], some shortcomings regarding the proposal evaluation process were identified, as the result of a stakeholder consultation. For instance, 34% of stakeholder

83 Cavicchi, B., Peiffer-Smadja, O., & Ravet, J., 'The transformative nature of the European framework programme for research and innovation: analysis of its evolution between 2002-2023', Publications Office of the European Union, 2023.

84 European Commission, Directorate-General for Research and Innovation, 'Staff Working Document on the interim evaluation of Horizon 2020', Publications Office of the European Union, 2017.

respondents regarded the feedback from evaluators as suboptimal, as it was considered not detailed enough and as too varied between evaluation panels. A trade-off between speed and quality of the evaluation reports was identified and linked to the large increase in the number of applications.

A bias in the attribution of funding linked to the size of projects was also identified, with over half of stakeholders (54%) reporting that, particularly under the 'Industrial Leadership' and 'Societal Challenges' clusters, large projects were favoured.

These constraints are in line with those mentioned in the public consultation carried out to support the co-design of the first strategic plan of Horizon Europe[85]. Additionally, other aspects were pointed out. The need for simplification of the submission and evaluation process (e.g., with a simpler and shorter proposal template) was the most voted aspect. As to the proposed changes, there was a general agreement that a 'right to react' scheme should be tested and implemented, and the simplification and clarification of the aspects considered under the three evaluation criteria were the most relevant ones. Furthermore, stakeholders also pointed out the need for a more transparent evaluation process as well as ensuring that proper experts were selected.

The impact criterion for proposal evaluation, now follows a clearer logic to link project results to expected outcomes in the medium and long term, including beyond the lifetime of the project. Clearer and easier to understand definitions, as well as a consistent use of terminology, were introduced in the application guidelines.

Regarding expert selection, besides their level of expertise, the choice of expert evaluators is made considering also diversification in terms of geography, gender and sector of activity. Furthermore, to address any real or perceived bias towards particular types of stakeholders, the Commission implemented a “blind evaluation pilot” for the Work Programme 2023 – 24, targeting the first stage of almost two-stage calls. In this scheme, evaluators are unaware of the identity of applicants[86].

Finally, the suggested 'right to react' scheme was implemented, allowing applicants to be involved in the evaluation process by responding to experts' comments. These responses were to be taken into consideration before the finalisation of the final assessment of the proposal. This element aimed at improving the quality of feedback and transparency of the process.

Following these implemented changes, the work carried out under this section ultimately aims to shed some light on their impact in the quality of Horizon Europe's evaluation system, particularly on whether it impacted its effectiveness in selecting the best and potentially most impactful research projects, regardless of their institutional or geographical origin. Furthermore, the ability to properly evaluate multi-disciplinary proposals, the existence of any perception of favouritism and the perceived impact of Clusters and Missions were investigated.

2.5.1 Results of the online survey

A survey including dedicated questions on the HE evaluation process – without focusing on a particular programme – was addressed to evaluators involved in the proposal assessment for HE, and NCPs. Specifically, the aim was to obtain insights from these key stakeholders on how and if the introduction of new elements such as Clusters, Missions, the strategic plan, and remote meetings has impacted the evaluation process, as well as if they considered that the potentially most relevant and impactful projects are selected for funding, or if they identified any kind of tendencies in project

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85 European Commission, Directorate-General for Research and Innovation, 'Report on the results of the online consultation and the European Research & Innovation days event', Publication Office of the European Union, 2019.

86 European Commission, 'Horizon Europe Proposal Evaluation, Standard Briefing', 2024.

selection. The complete set of questions can be found in Annex 2, as well as all the possible answers per question.

The survey gathered answers from 68 stakeholders[87] from these two groups. For the evaluators who participated in HE, the group can be further broken down into two subgroups: those who only participated in HE as evaluators, and those who participated both in HE and H2020 as evaluators. Table 6 shows the number of responses obtained per group and subgroup. The subgroup of evaluators involved only in proposal assessments of HE is represented by the smallest share (five respondents).

[87] The number of respondents corresponds to stakeholders that answered at least one question from this section of the survey. Since the questions did not require a mandatory answer, the number of answers varies between questions.

Table 6 – Number of respondents by stakeholder group and sub-group

Stakeholder category	Number of responses
NCP	28
Evaluator in HE	**40**
Evaluator only in HE	5
Evaluator in HE and H2020	35
Total	**68**

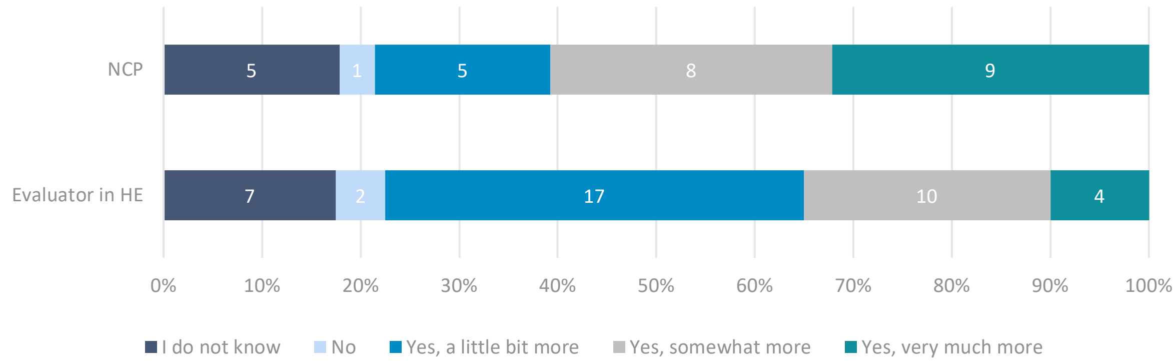
Regarding geographical representation, the survey gathered no responses from NCPs from Bulgaria, Croatia, Italy, Luxembourg and Sweden. Similarly, no answers were registered from evaluators in HE originating from Austria, Cyprus, Estonia, Hungary, Ireland and the Netherlands. There were no responses from Finland for both stakeholder groups.

2.5.2 Emphasis on impact and link to the strategic plan

Fostering a higher level of impact of projects on the thematic areas elected in the form of Clusters and missions was a priority for Horizon Europe. Having this in mind, the impact criterion for proposal evaluation assesses expected project results and outcomes, making sure they align with the priorities set in the strategic plan.

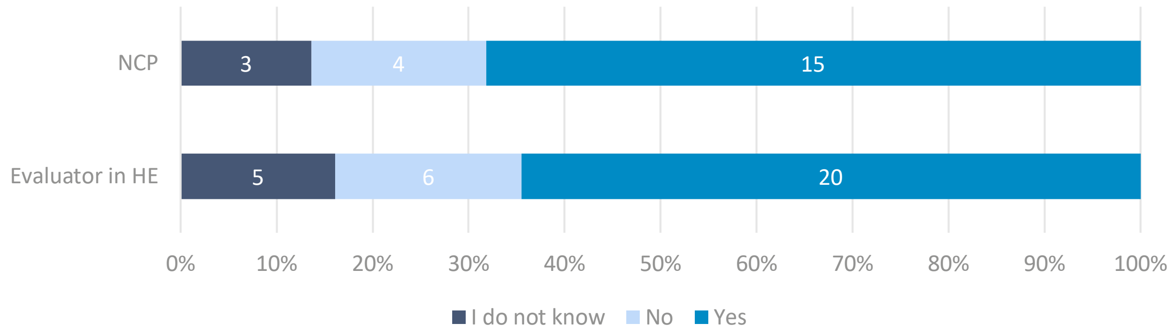
The survey results in Figure 33 show that, in general, there is consensus among the two groups of stakeholders that the evaluations carried out under HE are more results-focused. However, the perception on the extent of the focus placed on results varies between them. NCPs (nine out of 28 respondents, 32%) believe that 'they have very much become more results-focused', whereas a high share of evaluators in HE (17 out of 40 respondents, 43%) believe that 'they have become a little bit more results-focused'. When looking into the subgroups within the evaluators in HE, the results show that the four respondents that believe the evaluation process is very much more results-focused were involved in proposal assessment both in HE and H2020. Out of the five respondents that were evaluators only in HE, none selected this answer.

Figure 33 – Respondent's perception of evaluation of Horizon Europe becoming more results-focused (N=68)



For the 53 respondents that answered that the evaluations had become more results-focused to any extent (Figure 33), a follow-up question was asked (Figure 34). Namely, if they considered it to be a positive shift. Both stakeholder groups mostly considered that it was a positive shift: 15 out of 22 NCPs (68%) and 20 out of 31 evaluators in HE (65%).

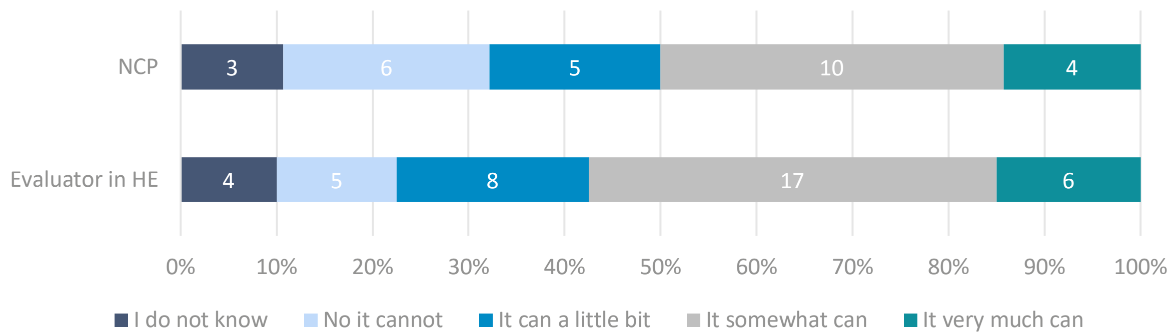
Figure 34 – Respondent's perception of positive shift in evaluation due to the strategic plan (N=53)



2.5.3 Effectiveness and transparency of the Horizon Europe evaluation process

Respondents mostly agree that the evaluation process of the current Horizon Europe is only effective to some extent in selecting non-mainstream proposals – i.e., unconventional proposals that do not align with scientific consensus to some extent – that have significant scientific merit (Figure 35). When analysing the results of the survey, it stands out that only roughly 15% of respondents (four NCPs and six evaluators in HE) view the evaluation process as very effective, while approximately an equivalent share views it as not effective at all (21%, six NCPs and 13%, five evaluators). In fact, over half of evaluators (25) and NCPs consider that the system can only select non-mainstream proposals to some or little extent, which reveals some underlying issues.

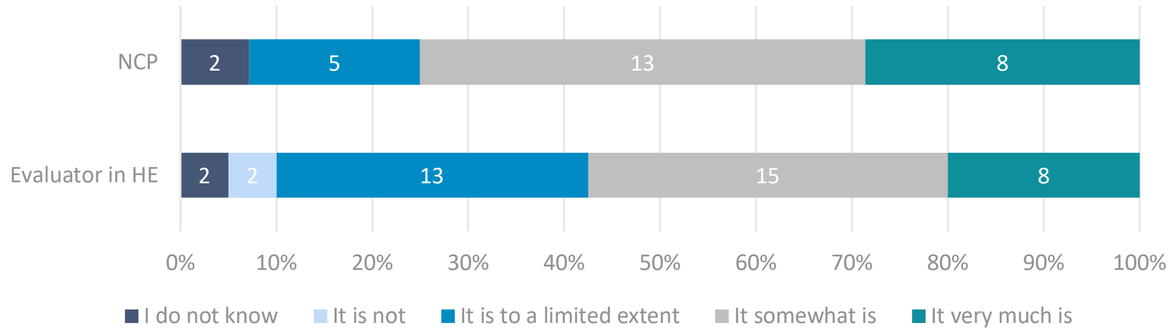
Figure 35 – Respondent's perception of the Horizon Europe evaluation process's ability to identify and fund non-mainstream proposals (N=68)



When asked about the suitability of Horizon Europe's evaluation process to assess multi-disciplinary proposals (Figure 36), in comparison with the previous question, a higher share of respondents – approximately 30% of NCPs (eight) and 20% of evaluators (eight) – considers it is very suitable to do so. Although it can be concluded from the survey that respondents believe, in general, it can assess multi-disciplinary proposals – only two out of 68 respondents in total consider it cannot –, the global feedback also indicates inherent shortcomings, as a high share of respondents reveals some reservations regarding this suitability. Around 70% of evaluators (28) and 65% of NCPs (18) think it can assess multi-disciplinary proposals only to some or a limited extent.

The 20 respondents that did not consider the evaluation process to be suitable, or only to a limited extent, were asked a follow up question. The overall feedback referred to the same key points, namely the ineffectiveness of the evaluation process for multi-disciplinary calls, as panels are too small – the required minimum of three experts[88] to constitute the evaluation panel is seen as insufficient –, the absence of a clear definition to guide the selection of what can constitute a multi-disciplinary proposal, and a tendency to favour some disciplines over others. One respondent also mentioned that there are too many non-technical constraints (gender, country, if the applicant is new to the programme) to select sufficiently multi-disciplinary proposal pools.

Figure 36 – Respondent's perception of the suitability of the Horizon Europe evaluation process for assessing multi-disciplinary proposals (N=68)

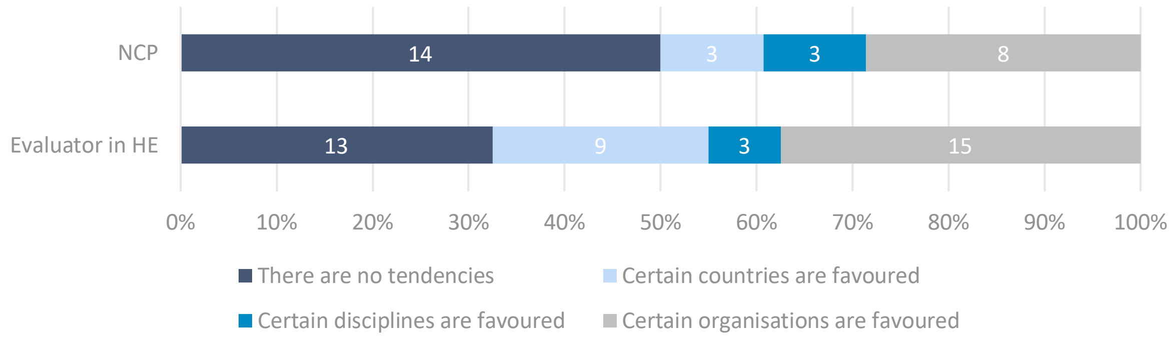


When assessing the level of bias and transparency of the process, the overall analysis shows that only half of enquired NCPs (14) and roughly 30% of evaluators (13) believe that there are no tendencies in favouring specific types of stakeholders. On the contrary, 50% of NCPs (14) and over 65% of evaluators (27) believe certain stakeholders are favoured. The most prominent figures point towards the favouring of organisations, as almost 40% of evaluators (15) and around 30% of NCPs (eight) believe this to be the case (Figure 37). Additionally, 23% (nine) of evaluators believe that certain countries are favoured. When comparing the two subgroups that constitute this group the disaggregated results show that the three evaluators who believe that certain disciplines are favoured were also evaluators in H2020.

The issue of a real or potential bias in proposal selection is a recurring one, inherited already from the previous framework programme – Horizon 2020. Despite the Commission's efforts to minimise it through the recent "blind evaluation pilot", implemented for the Work Programme 2023-24, the current survey results suggest it still is a concern. Further enquiries are needed in the future, to assess whether this initiative yielded positive results towards bias minimisation.

[88] European Commission, 'Horizon Europe Proposal Evaluation, Standard Briefing', 2024.

Figure 37 – Perceived tendencies in Horizon Europe proposal evaluations favouring specific organisations, countries, or disciplines (N=68)



When analysing the results by the country of origin of the responses, some trends arise. The respondents that believe that certain types of stakeholders are favoured originate from 21 out of the 26 represented countries in the survey. Those who believe certain organisations are favoured originate from 15 of these. Notably, 3 out of 4 respondents from Greece, 2 out of 3 respondents from Germany, 3 out of 6 respondents from Malta and 4 out of 8 respondents from Italy believe this to be the case. During the first two years of the current framework programme, Germany and Italy were the first and third biggest beneficiaries under Horizon Europe, respectively. Moreover, Germany has two institutions in the top five biggest beneficiaries, while Italy's National Research Centre is placed as the 15th biggest beneficiary[89]. The results could indicate a perceived bias from stakeholders from these countries towards these organisations.

Regarding the tendency to favour certain countries, the respondents that believe this to happen originate from eight countries (Czechia, Greece, Hungary, Lithuania, Malta, Poland, Belgium and Portugal), which all but Belgium, belong to the group of Widening countries. Notably, all respondents originating from Poland (four) – also a country with a medium to high share of responses in this survey section – believe this to be the case. Although numbers show that proposal success rates for Widening countries have improved for the first two years under Horizon Europe, in comparison with Horizon 2020, when looking at financing received, the gap is still evident between Widening and non-Widening countries. For the years 2021-2022, the total financing of all Widening countries (14.3%) falls behind that of Germany alone (15.5%)[90].

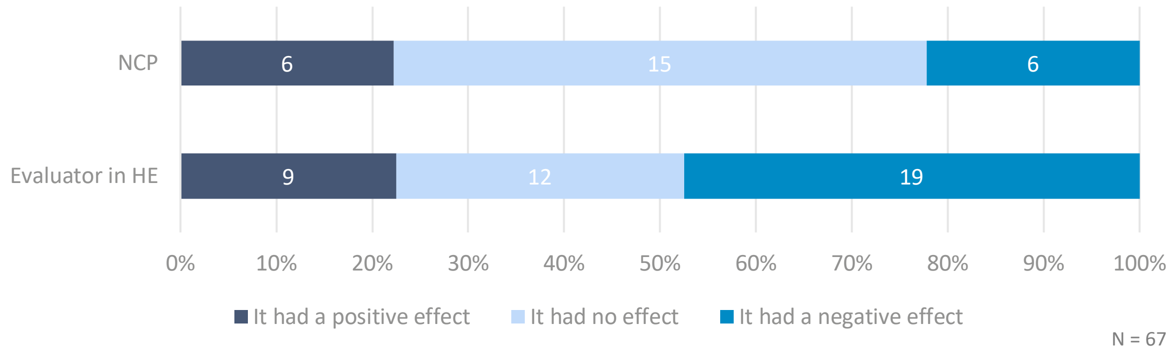
The following question of the survey regarding the evaluation process concerned the introduction of remote meetings (Figure 38). Over half of the NCPs (15 out of 27, 54 %) reported no effect on the evaluation process. Contrarily, many of the evaluators in HE (19 out of 40, 48 %) – 18 of which were also evaluators in H2020 – think that it had a negative effect on the evaluation process, while 12 (30 %) believed it had no effect.

One of the respondents further explained that, in some evaluation processes and meetings, this format does not work effectively. Examples given include panel meetings and meetings for consensus groups, such as for top-down call topics.

89 Science Business, 'Here's what the first two years of Horizon Europe look like in numbers', website, 2023.

90 Science Business, 'Here's what the first two years of Horizon Europe look like in numbers', website, 2023.

Figure 38 – Impact of the introduction of remote meetings on the evaluation process (N=67)



Based on the results of this survey, we can conclude that respondents views point towards some uncertainty as to whether the evaluation process of Horizon Europe is effective or not in selecting non-mainstream proposals with significant scientific merit. Concerns arise as well regarding its suitability for multi-disciplinary proposals due to issues such as small panel sizes, lack of clear definitions, and biases favoring certain disciplines.

A widespread and consistently raised issue concerns favoritism towards certain organisations, countries, and disciplines, particularly among evaluators in Horizon Europe. Regional trends indicate varying levels of concern, with Widening countries present in the survey expressing more doubts about favoritism. Additionally, while over half of NCPs see remote meetings as having no effect on the evaluation process, many evaluators in Horizon Europe perceive them negatively, especially in panel meetings and consensus groups.

These findings highlight both strengths and areas for improvement in the Horizon Europe evaluation process, suggesting the need for measures to enhance fairness and effectiveness.

2.5.4 Impact of Clusters and Missions in the evaluation outcome

One of the distinctive aspects of Horizon Europe is its deepened focus on integrating and creating synergies between the thematic areas, promoting a higher level of impact and enhancing the directionality of funding. This was partly attempted with the introduction of Missions and thematic Clusters.

Regarding the general impression on the impact that the introduction of these elements (Figure 39 and Figure 40) has had on the evaluation process of proposals, the hierarchy of answers per type of stakeholder does not show a strong variation.

When analysing the figures for the impact of Clusters, the results show that a fair share of evaluators –15 out of 40 evaluators in HE (38%) – believe that the introduction of Clusters made the evaluation procedure more selective. This is a somewhat positive result when considering the objective set under the current framework programme of enhancing the directionality of funding, mainly for the six thematic areas established by the Clusters.

On the contrary, a higher number of respondents from the two groups of stakeholders considers that the introduction of Missions has had no effect on the evaluation procedure: nine out of 17 NCPs (32%) and 14 out of 40 evaluators in HE (35%).

Figure 39 – Impact of Clusters in Horizon Europe on the evaluation process (N=66)

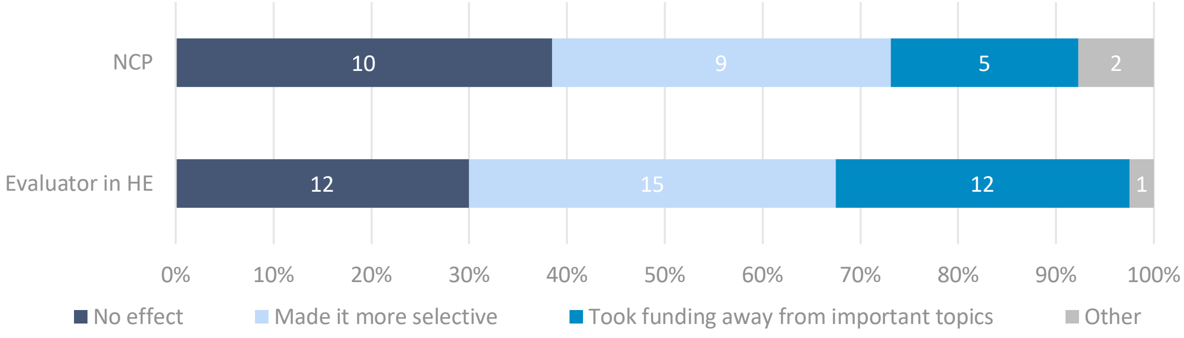
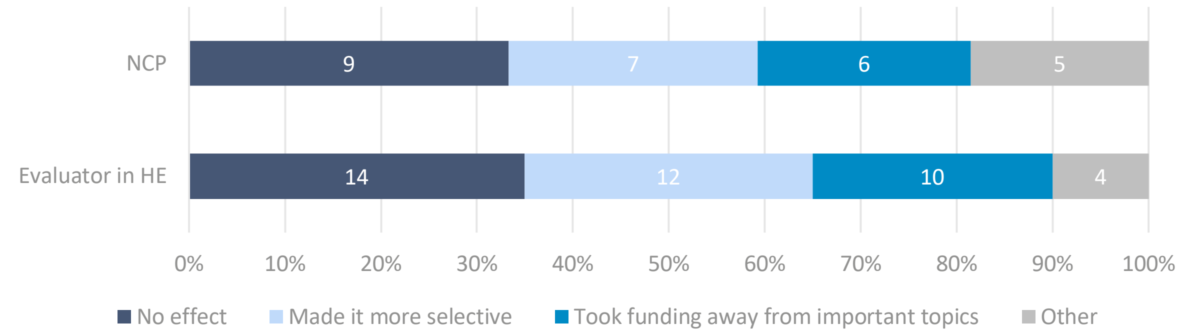


Figure 40 – Impact of Missions in Horizon Europe on the evaluation process (N=67)



Twelve evaluators in HE (30%) responded that the introduction of Clusters 'took funding away from important topics that may not fall under their thematic scope'. The same stands for the introduction of Missions, with 10 (25%) stakeholders in this group sharing the same view. These figures are higher than those shown for NCPs, for the same answer.

For the respondents that selected the answer "other" for Clusters (one evaluator involved in both framework programmes H2020 and HE, and two NCPs), further details were required in the form of open feedback. Both NCPs agree that the introduction of Clusters increased the complexity of the evaluation process – an option not available for selection in the survey –, either because the topics are considered to be 'very wide' or due to an 'increase in background knowledge' that candidates and evaluators are expected to have. Although this comes as a contradiction to the efforts put towards the simplification of the process, it represents only 7% of the total of NCP respondents[91]. The evaluator involved in both framework programmes viewed the process as 'more targeted'.

2.5.5 Results of the semi-structured interviews

In order to complement the survey results, semi-structured interviews were conducted to gather additional insights from NCPs. Specifically, these aimed to collect the views and recommendations from these key stakeholders on improvements of the evaluation and application processes, to better foster the selection of the best and potentially most impactful research projects. Furthermore, the perceived impact of the newly introduced elements – such as the Clusters, the Missions, the EIC, the

[91] The same specification was required for the question regarding Missions. However, most open answers fall out of the scope of the question, not focusing on the impact on the evaluation process specifically.

open science requirement or remote meetings in the evaluation process – both in the evaluation and application processes, was investigated.

The consultation process comprised four interviews with National Contact Points from Austria, Italy, Spain and the Netherlands. The questionnaire is available in Annex 3.

When asked about the effectiveness of the evaluation process and main points for improvement, interviewees answers align mainly on two points, notably the (1) lengthy gap taken from proposal submission to grant agreement, that can take up to eight months, which *"is still too long for many projects, especially those involving commercial partners"*; and (2) the lack of transparency in the evaluation outcomes, particularly regarding the EIC Accelerator and Pathfinder programmes. As mentioned by the Austrian NCP:

"Transparency in the evaluation process is lacking (…), the perception that people have sometimes does not align with the perception of the proposal". The Dutch NCP further recommends that *"detailed summary reports for unfunded projects, including detailed scoring criteria and content evaluation"* be provided. Top-down topics should be established to set *"clear boundaries, explaining why certain projects are deemed out of scope or do not fit the portfolio"* as the *"lack of specificity raises many questions and leaves room for improvement"*. This claim is also backed by the discontinuation of the rebuttal system for Pathfinder projects, which *"leaves certain aspects unclear"*. Moreover, in the specific case of the EIC Accelerator programme, it was mentioned that *"transparency can be challenging"* as *"after the evaluation, projects not selected based on ranking are considered based on their content"*. An additional concern from the Austrian NCP touches upon the available funding, which is claimed to be *"insufficient to fund all projects"*.

Further suggestions are put forward by the Italian and Dutch NCPs, envisioning streamlining the application process, by simplifying the evaluation criteria for all budgetary elements included in the proposals, the security scrutiny, ethical questionnaires and application appendices. Additionally, the NCP from Italy points out the need to better match experts to proposals, as *"they often lack technical knowledge"*, as well as to include interviews to projects in EIC Accelerator calls.

Regarding the impact that the introduction of new elements – such as Clusters, Missions, the EIC and the open science requirement – had in the evaluation system of Horizon Europe, generally, no changes were perceived. Similarly, the introduction of remote meetings is thought to have had no negative effect, in general. However, three NCPs agree that some elements in the evaluation process benefit from in person discussion. The NCP from the Netherlands points out that in-depth *"discussions can be challenging to conduct remotely"*. Furthermore, the Spanish NCP echoes the constraint mentioned by one of the survey respondents, agreeing that *"face-to-face consensus meetings are necessary, mainly in those cases where there are important discrepancies amongst the evaluators"*.

Lastly, the interview sought to understand if any tendencies in favouring countries, organisations and disciplines could be identified by the NCPs. Disciplinary favouritism is not evident, although certain disciplines may receive more funding in specific calls. Similarly, there appears to be no bias towards particular countries, even if under-represented nations remain a concern. On the contrary, the Austrian and Dutch NCPs agree that organisational favouring can be observed. As the Austrian NCP mentions: *"some top researchers and organisations benefit from reputation"* and when considering *"two projects with the same idea"*, if *"one comes from a renowned organisation, there might still be bias"*. The NCP from the Netherlands adds that, as such, *"SMEs or new organisations may face challenges, as experience is often valued more than the potential of the research they offer"* and, *"as a result, novel research opportunities may not always receive fair consideration"*. The view of the Italian NCP is antithetically linked to this last observation, as they believe there to be political influence on the evaluation process, particularly pushing for the *"participation of Widening countries, (…) since it* *is still limited"*, even when they are not as *"ground-breaking as needed"*, notably in the EIC Accelerator programme.

Further suggestions provided are linked to the need to promote gender and age balance, through the anonymisation of *"the identity on regard to who is writing the proposal"*, to *"ensure a gender-balanced approach"* and allow *"young researchers to conduct their work based on their merit and not experience"*. This should be accomplished through the pilot on blind evaluations of proposals, put in place since 2023.

2.5.6 Results synthesis on the Horizon Europe evaluation system

Overall, there is consensus among respondents that the evaluations under HE have become more results-focused to some extent and that this is viewed as a positive shift. It reflects the commitment made under this framework programme in selecting projects that better show how its outputs are linked to measurable societal impact.

However, there are some doubts and issues raised concerning the effectiveness and transparency of the Horizon Europe evaluation process. Respondents believe it is only effective to some extent in selecting non-mainstream proposals that have significant scientific merit.

Regarding the suitability to evaluate multi-disciplinary proposals, some concerns were raised on the insufficient number of experts in multi-disciplinary panels, a tendency to favour some disciplines over others and a lack of a clear definition on how to assess the level of multi-disciplinarity of a proposal.

Results of the online survey also suggest a strong perception of favouritism, particularly towards some organisations and countries, a concern inherited from the previous framework programme.

Stakeholders are split regarding their view on the impact that the introduction of remote meetings had on the evaluation process of Horizon Europe: most of the NCPs think it was not impactful, while most HE evaluators believe that it has negatively impacted the process.

The results of the online survey suggest that the introduction of Clusters has made the evaluation procedure more selective, and the introduction of Missions had no effect on the procedure.

Conclusions arising from the interviews conducted with NCPs align with the findings expressed in the survey, particularly regarding biases. Interviewed NCPs believe that certain organisations are favoured due to reputation, which creates a barrier for new applicants. Experience tends to be prioritised over the potential of research.

Interviewees highlight the need to reduce the time between proposal submission and communication of results, and to increase transparency, particularly for programmes such as the EIC Accelerator and Pathfinder, as the main recommendations for improvements in the evaluation process.

3. Concluding remarks

Horizon Europe represents a serious contribution towards further advancing the European research and innovation landscape, with a substantial budget of 93.4 billion euros. Despite this increased funding, concerns persist regarding its ability to meet the expanding scope of European research ambitions and to address the disparities in R&I performance among Member States.

While the programme has seen a notable increase in the average success rate of applications compared to its predecessor, Horizon 2020, challenges in equitable funding distribution persist, resulting in many high-quality proposals facing rejection. Moreover, the programme has witnessed an expansion in the size and diversity of participating organisations, particularly universities, reflecting a commitment to fostering research inclusivity. Notably, there is a discernible dedication to Social Sciences and Humanities (SSH) within Horizon Europe, with SSH integrated across all pillars and Clusters to comprehensively address societal challenges. However, according to stakeholders, Horizon Europe's emphasis on higher Technology Readiness Levels (TRLs) may unintentionally sideline foundational, early-stage research that remain crucial for driving breakthrough innovations. At the same time, the absence of publicly accessible data on the attained Technology Readiness Levels (TRLs) of funded projects within the research and innovation framework programmes presents another obstacle when evaluating the innovations facilitated by Horizon Europe, or any prospective funding programme.

The Common Model Grant Agreement (CMGA) introduced under Horizon Europe does not seem to fully satisfy all its objectives. The majority of survey respondents agree that it does simplify the understanding of terms and conditions compared to the previous grant agreements. However, there is an evident uncertainty in its ability to simplify administrative burdens, improved transparency and help in creating synergies across different projects funded by Horizon Europe. This uncertainty could also stem from the fact that grant agreements are often handled by a dedicated administrative department and not directly by the beneficiaries themselves.

The obligatory communication and dissemination plans were endorsed by respondents as they believed they improved the visibility of their project's results. Another positive change pointed out by stakeholders was the introduction of a Data Sheet summarising the key points and figures of the funded project. Similarly, the digitalisation of application process was pointed out as a positive change. Several respondents also called for a fully annotated CMGA as was the case in H2020 to ensure full understanding of the terms and conditions.

The FET Flagship, discontinued under Horizon Europe, had a key goal of significantly advancing scientific and technological research in Europe. Its emphasis on fostering collaborations, building sustainable research communities, and facilitating knowledge sharing contributed significantly to the promotion of innovation and international collaboration.

While the FET Flagship also aimed at translating scientific advancements into market innovations, whether this became a reality is debatable. Some projects successfully transitioned their research into tangible products, launching companies, generating products, and filing patent applications. However, others faced challenges in achieving market readiness due to their involvement with low TRL research.

The discontinuation of the FET Flagship raised concerns about potential disruptions to collaboration, hindering application development, and causing a loss of investment, as indicated by survey and interview respondents. Moreover, there are risks of losing established collaborations and momentum in key research areas, potentially impacting Europe's technological competitiveness.

Stakeholders noted that within the Horizon Framework programme, there was only partial alignment with FET Flagship goals, lacking specific calls for new initiatives that mirror the Flagship's scale and effectiveness. The current landscape appears fragmented, making it challenging to replicate the same level of community and strategic alignment without the FET Flagship initiative. This sense of fragmentation could be due to the fact that Horizon Europe calls are now guided by a strategic plan. In fact, the strategic planning entails that European partnerships are formed only when they are necessary to achieve impacts that cannot be attained through other actions under the Framework Programme or national efforts alone. As a result, the research objectives previously placed under the FET flagship initiative are now reallocated under other Horizon Europe funding instruments[92]. However, stakeholders' perceptions revealed a sense of ambiguity regarding the potential of these new instruments under Horizon Europe to fill the void left by the FET Flagship. While the EIC Pathfinder is anticipated to serve as a natural successor, benefiting from a more mature field, concerns persist about its adequacy. In particular, in fulfilling the role typically filled by a new flagship.

In terms of recommendations to fill in the gap of the FET Flagship, stakeholders' consultations highlighted the need to establish initiatives mirroring the FET Flagship's long-term research communities. This involves prioritising network creation and sustained commitment to specific technological areas for fostering knowledge sharing. Additionally, stakeholders advocate for new HE initiatives fostering interdisciplinary and international collaboration, aligning research strategies, and addressing grand challenges effectively.

Concerning the participation barriers encountered by SMEs in Horizon Europe, the findings suggest a significant issue: despite the programme's potential to advance innovation, SMEs frequently grapple with a lack of awareness regarding available funding opportunities or struggle with crafting competitive proposals. One could argue that the two issues are interlinked; the lack of awareness about funding opportunities may indicate a broader deficit in understanding the funding programme itself, consequently leading to a lack of insight into how to develop a competitive proposal. Therefore, a crucial initial step would be addressing the dispersed nature of funding information. Implementing navigational tools or platforms akin to a compass, as suggested by a stakeholder, could streamline the process for SMEs to identify suitable funding opportunities aligned with their expertise.

The important role NCPs could play in bridging the gap between SMEs and European funding is often highlighted. NCPs possess the expertise to efficiently guide SMEs through the application process, ensuring they are not only aware of available funding opportunities but are also equipped with the necessary resources to navigate the process effectively. Collaborative efforts between national authorities and NCPs are paramount to better disseminate information about European funding, thereby enhancing accessibility for SMEs across diverse sectors and regions.

Our semi-structured interview results also suggest that Horizon Europe should prioritise further projects that promote collaboration, facilitate technology transfer, and offer support for SMEs to engage in collaborative R&I activities. Similarly, interviewed stakeholders suggested that Horizon Europe should strive to offer a diverse range of funding opportunities, prioritise projects with clear paths to market deployment, and ensure accessibility and adaptability of funding programmes for diverse SMEs.

As for the introduction of the strategic planning process within Horizon Europe, it represents a significant effort for a step forward, aiming to enhance adaptability, foster policy learning, engage stakeholders effectively, and amplify programme impact. However, about half of the surveyed stakeholders were not aware of the strategic plan's existence – mostly researchers in occupation and with relevant experience as beneficiaries in either H2020 or HE, and a large share is unsure about the impact it had on calls launched, some believing relevant topics have been excluded from

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[92] European Commission, Secretary-General, 'Commission Staff Working Document Impact Assessment', 2018.

funding. Nevertheless, the small share that reported to have been engaged in the co-design activities expressed positive experiences, highlighting the effectiveness of incorporating valuable input from citizens. This result suggests the need to raise awareness of any future co-design activities to ensure comprehensive participation from stakeholders across different disciplines.

There is also a consensus among respondents regarding a heightened focus on results under Horizon Europe, indicating a positive shift towards more impactful research. However, challenges persist in the evaluation system's effectiveness, particularly in selecting non-mainstream, multi-disciplinary projects and addressing biases favouring certain stakeholders and disciplines. Notably, disparities in perception by country of origin highlight potential inequities in funding distribution. These concerns echo those expressed by interviewees, who highlight the existing bias towards well-known organisations and researchers, and a general lack of transparency in the evaluation and communication of results. Key recommendations include improving the quality of feedback given to non-selected projects in order to increase transparency, especially in programmes such as the EIC Accelerator and Pathfinder; better matching of experts to projects; and streamlining the application process to simplify it and shorten the gap between proposal submission and communication of results.

While the firsthand experiences of stakeholders analysed in this study are invaluable for comprehending both the merits and limitations of Horizon Europe, it's crucial to acknowledge that their encounters with obstacles may inadvertently tint their perception of the programme. Stakeholders such as evaluators who are directly involved in the evaluation process may fail to see its shortcoming. Similarly, beneficiaries of initiatives such as the FET Flagship may be predisposed to highlight only the drawbacks of discontinuing the programme, potentially skewing their perspective. Furthermore, SMEs that have faced challenges in obtaining Horizon Europe funding may harbour a more pessimistic outlook compared to those with a longer track record in EU funding.

Nonetheless, a common issue noticed among interviewed stakeholders is the lack of knowledge regarding available funding opportunities. Beneficiaries of Horizon Europe expressed comfort in applying for funding due to their involvement in Horizon 2020 projects or partnerships with experienced consortia. However, they often remain unaware of funding opportunities beyond R&I framework programmes. Similarly, the doubt FET beneficiaries cast towards Horizon Europe's ability to compromise for the discontinuation of the FET Flagship may very well stem from their lack of funding opportunities within the different programmes of Horizon Europe and even beyond.

Similarly, our results provide that approximately half of the consulted stakeholders are unaware of the modifications introduced under the CMGA or the strategic plan. This outcome implies that the anticipated streamlining benefits of the CMGA are not being fully realised, especially concerning the promotion of synergies across projects. Additionally, the lack of awareness regarding the strategic plan and its collaborative co-design initiatives excludes potential stakeholders from actively shaping the direction of Horizon Europe. Therefore, enhancing communication and dissemination efforts is key to ensure that stakeholders are well-informed to contribute meaningfully to the evolution of Horizon Europe. By bridging these awareness gaps, we can foster a more inclusive and participatory approach towards achieving the Programme's objectives and driving innovation within the European Union.

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5. Annexes

Annex 1 – Stakeholder selection criteria

This Annex provides an overview of the selection process followed to map the stakeholders. Our approach is summarised in the Table below by stakeholder type.

Table 7 – Overview of stakeholder selection process

Stakeholder Type	Number of entity/individual contacted	Contact Info Collection Method	Source of data	Selection Criteria
Mapping of SMEs[93]	200	Contact form on Cordis	Horizon Dashboard	➢ Utilised Horizon database with 14,362 entries related to Horizon 2020 and Horizon Europe projects. ➢ Conducted a Country Proportion Analysis to ensure fair and accurate representation of SMEs across Member States. ➢ Identified 2,320 unique SMEs, eliminating repetitions, and ensuring each SME in the sample is distinct. ➢ Further refined the list to include 1,982 private for-profit SMEs, aligning with the focus on SMEs in Horizon. ➢ Conducted random sampling with an emphasis on maintaining proportionality in terms of countries and Horizon Programmes.
Beneficiaries	100	Contact form on Cordis	Horizon Dashboard	➢ Utilised Horizon database to identify research organisations that participated in Horizon Europe (1,467 entities). ➢ Conducted a Country Proportion Analysis to ensure a balanced and representative sample across Member States. ➢ Randomly selected 100 beneficiaries to ensure diversity in representation.

93 While conducting the survey, we noted a lower response rate among SMEs. Consequently, we sent another round of invitations to 100 new contacts, using the same selection criteria. This result in a total of 200 invitations sent.

Stakeholder Type	Number of entity/individual contacted	Contact Info Collection Method	Source of data	Selection Criteria
National Contact Points	81 (3 per MS)	81 Email addresses collected	Horizon Europe NCP portal	➢ Selected National Contact Points (NCPs) for Horizon Europe from the Single Electronic Data Interchange Area (848 NCPs). ➢ Main selection criteria focused on identifying relevant functions such as climate, energy, environment, mobility, and industry.
Evaluators	100	Desk Research or Generic Emails	Funding & tender portal	➢ Selected experts involved in the evaluation of Horizon Europe with prior engagement in Horizon 2020 (667experts). ➢ Considered experiences from both Horizon 2020 and Horizon Europe to thoroughly assess programme evolution and effectiveness. ➢ Utilised statistical software for random sampling while ensuring a gender-balanced database during the selection process.
Coordinators	130	Desk Research or Generic Emails	Horizon Dashboard	➢ Selected entities responsible for projects under HEU, with experience as coordinators in previous framework contracts (7,094 entities). ➢ Emphasised alignment with the goals of the European Green Deal, focusing on zero emissions, environmental subjects, and sustainability. ➢ Ensured a minimum of 3 coordinators from each Member State, when possible, to maintain an accurate national distribution.
Flagship Programme Beneficiaries	130[94]	Desk Research or Funding & Tenders Portal	CORDIS database (H2020)	➢ Selected entities with experience in H2020 under the FET Flagship programme (1,616 entities). ➢ Utilised statistical software for random sampling, emphasising representation from each Member State when possible.

[94] While conducting the survey, we noted lower response rates among Flagship Programme Beneficiaries. Consequently, we sent another round of invitations to 100 new contacts, using the same selection criteria.

Annex 2 – Survey questions

Task 1 – Survey questions

BRIEF INTRODUCTION OF THE PROJECT SCOPE AND THE SURVEY'S GOAL AND CONTENT

This survey is carried out on behalf of the Panel for the Future of Science and Technology (STOA) of the European Parliament. The purpose of this survey is to collect data for an in-depth analysis of various aspects of the Horizon Europe Programme. As the European Union's key funding instrument for research and innovation, Horizon Europe plays a pivotal role in shaping the continent's scientific and technological landscape. The study to which this survey is contributing aims to offer insights into various key aspects of Horizon Europe, with a particular focus on its administrative procedures, evaluation system, strategic planning activities, barriers and challenges faced by its participants. As a stakeholder your experiences and opinions are crucial for this analysis.

The European Parliament is eager to hear your opinion, and your participation in this survey holds significance in shedding light on the strengths and areas needing enhancement. Your insights will help shape policy direction, funding approaches, and the execution of the programme.

Your profile

Country*:

[List of European countries]

Occupation*:

- ☐ Government employee
- ☐ Independent researcher
- ☐ Consultant in the private sector
- ☐ Researcher at a research organisation
- ☐ Researcher at a University
- ☐ Employee in an SME
- ☐ Other, please specify:

Please select your relevant experience (multiple responses possible):

- ☐ National Contact Point (NCP)
- ☐ Evaluator involved in Horizon Europe Proposal Assessments
- ☐ Evaluator involved in Horizon 2020 Proposal Assessments
- ☐ Beneficiary of Horizon Europe Projects
- ☐ Beneficiary of Horizon 2020
- ☐ Beneficiary of the FET Flagship Programme
- ☐ Not a beneficiary but interested in Horizon Europe funding opportunities
- ☐ Other (please specify): ___________________

[NCP, ALL evaluators, ALL beneficiaries (except FET Flagship)] Impact of the introduction of the Common Model Grant Agreement

1. Are you aware of the change in the Common Model Grant Agreement (CMGA) under Horizon Europe in Comparison with Horizon 2020?

   - ☐ Yes
   - ☐ No

2. How familiar are you with the newly introduced Common Model Grant Agreement (CMGA) in Horizon Europe?

   - ☐ Very familiar
   - ☐ Somewhat familiar
   - ☐ I have heard of it
   - ☐ Not familiar

If the answer is [somewhat familiar, very familiar], these questions will follow:

3. Does the new CMGA implemented under Horizon Europe make it easier to understand the terms and conditions of the contract? It:

   - ☐ Greatly simplifies understanding
   - ☐ Somewhat simplifies understanding
   - ☐ Slightly simplifies understanding
   - ☐ Does not simplify understanding
   - ☐ I do not know

4. Compared to the previous Grant Agreements, does the new CMGA make it easier to create synergies with other projects funded by Horizon Europe or different European programmes?

   - ☐ Significantly enhances synergy creation
   - ☐ Moderately enhances synergy creation
   - ☐ Slightly enhances synergy creation
   - ☐ Does not enhance synergy creation
   - ☐ I do not know

5. Do you find the administrative procedure under the CMGA to be simplified compared to other grant agreements? It is:

   - ☐ Highly Simplified
   - ☐ Moderately Simplified
   - ☐ Slightly Simplified
   - ☐ Not Simplified
   - ☐ I do not know

   If yes, please specify any specific improvements or simplifications you noticed (150 words)____

6. Do you believe the new CMGA is more transparent in its terms and conditions?

   - ☐ Much more transparent
   - ☐ Moderately more transparent
   - ☐ Slightly more transparent
   - ☐ Not more transparent
   - ☐ I do not know

7. In your opinion, what has been the impact of introducing a CMGA on Horizon Europe projects?

   Please elaborate with an example (max 150 words)__________

8. Do you believe that the mandatory communication and dissemination plan in any Horizon Europe funded project will help in promoting the visibility of project's results?

   - ☐ It will help very much
   - ☐ It will somewhat help
   - ☐ It will help a little bit
   - ☐ It will not help
   - ☐ I do not know

[NCP, ALL evaluators, ALL beneficiaries (Except FET flagship), not a beneficiary but interested, other]
Impact of the introduction of Strategic planning in Horizon Europe

9. Are you aware of the Strategic plan of Horizon Europe?

 - ☐ Yes
 - ☐ No

 If answered yes, the following questions will be asked]

10. Are you aware of the remote 2019 co-design activities conducted through the Horizon Europe webpage to gather stakeholders' views on the program's contribution to EU policy priorities?

 - ☐ Yes
 - ☐ No

11. Did you participate in any of the 2019 web-based co-design exercises during the strategic planning process for Horizon Europe?

 - ☐ Yes
 - ☐ No

 If you participated, did you feel that your contribution was adequately recorded and taken into account? (max 150 words)___

12. Are you aware of the face-to-face co-design activities that took place during the European Research and Innovation Days in 2019?

 - ☐ Yes
 - ☐ No

13. Did you participate in any of the 2019 face-to-face co-design activities at the European Research and Innovation Days?

 - ☐ Yes
 - ☐ No

 If you participated, what were your key takeaways or impressions from these structured and informal exchanges? (max 150 words)__________

14. What impact do you think the introduction of the Strategic plan in Horizon Europe had on the topics of calls launched?

 - ☐ It included topics previously overlooked.
 - ☐ It excluded relevant topics from funding.
 - ☐ It had no effect on the topics.
 - ☐ I do not know.
 - ☐ Other, ______

 If you believe any topics are overlooked, please provide further details (max 150 words)___________

[for ALL answering they work in an SME or a large company or a business owner] SMEs participating in Horizon Europe.

15. When it comes to R&I funding, on which sources does your company rely to make its business more sustainable (multiple answers possible)?

 - ☐ Internal sources
 - ☐ National sources
 - ☐ International sources, including European R&I funding
 - ☐ We have not invested yet in R&I
 - ☐ We have no specific R&I requirements to make the business more sustainable
 - ☐ Other, please specify (max 150 words)___________

16. Are you aware of Horizon Europe funding for SMEs?

- Yes
- No

17. Could additional R&I funding help make your business more sustainable?

- Yes, to a large extent
- Yes, to a fair extent
- Yes, to a limited extent
- No
- I do not know

18. In your opinion, what are the barriers for industrial stakeholders to apply for R&I funding under Horizon Europe (up to 3 responses)?

- Lack of awareness of funding opportunities
- Lack of resources to apply
- Uncertainty about eligibility criteria
- Administrative burden
- Time constraints
- Complex application requirements
- Limited access to relevant information
- Potential concerns about the competition.
- Other (max 150 words)_________

19. Do you believe there are enough R&I funding opportunities under Horizon Europe for the type of business you carry?

- Yes
- No
- I do not know

[If no] Please elaborate (max 150 words)__________________

20. In your experience, do the rules of participation favour large companies over SMEs?

- They do not favour large companies.
- They do favour large companies a little bit
- They somewhat do favour large companies
- They favour large companies very much
- I do not know

21. How has funding from Horizon Europe or any other Framework Programme (e.g. H2020, FP7) impacted the growth or development of your SME?

- Funding did not have an impact on the growth or development of my SME
- Allowed for expansion or scaling
- Facilitated research and development
- Provided stability during financial uncertainties
- Not applicable
- Other (please specify; max 150 words)______

22. Have you collaborated with partners who have previously received R&I funding under Horizon 2020 or Horizon Europe?

- Yes
- Not that I am aware of
- No

If yes this question will follow:

23. To your knowledge, have you benefited from a technological advancement made by this partner as a result of European funding?

24. Do you have any recommendations for Horizon Europe to better support SMEs?

    Please elaborate (max 150 words)___

[Evaluators of Horizon Europe, NCP] Horizon Europe evaluation process

25. How has the introduction of Clusters in Horizon Europe influenced the evaluation process?

    - ☐ No effect
    - ☐ Made the evaluation procedure more selective
    - ☐ Took funding away from important topics that may not fall under the Clusters
    - ☐ Other (please specify)_____

26. How has the introduction of Missions in Horizon Europe influenced the evaluation process?

    - ☐ No effect
    - ☐ Made the evaluation procedure more selective
    - ☐ Took funding away from important topics that may not fall under the missions
    - ☐ Other (please specify)

27. With the emphasis on impact and link to the Strategic plan in calls, do you think that evaluations of Horizon Europe have become more results-focused?

    - ☐ They have not become more results-focused
    - ☐ They have become a little bit more results-focused
    - ☐ They have somewhat become more results-focused
    - ☐ They have very much become more results-focused
    - ☐ I do not know

    If yes, do you think this is a positive shift?

    - ☐ Yes
    - ☐ No
    - ☐ I do not know

28. Do you believe the evaluation process of Horizon Europe can identify and provide funding for research proposals that are not considered mainstream but have significant scientific merit?

    - ☐ It very much can
    - ☐ It somewhat can
    - ☐ It can a little bit
    - ☐ No it cannot
    - ☐ I do not know

29. Do you believe there are any tendencies in evaluations of Horizon Europe proposals to favour specific organisations, countries, or disciplines?

    - ☐ No, I believe there is no tendency to favour specific types of stakeholders
    - ☐ Yes, I believe certain organisations are favoured
    - ☐ Yes, I believe certain countries are favoured
    - ☐ Yes, I believe certain disciplines are favoured

30. Do you believe the current Horizon Europe evaluation process is suitable to properly evaluate multi-disciplinary proposals?

    - ☐ It very much is
    - ☐ It somewhat is
    - ☐ It is to a limited extent
    - ☐ It is not
    - ☐ I do not know

    If no, please elaborate (max 150 words)___

31. How do you believe the introduction of remote meetings has affected the evaluation process?

- It had a positive effect on the evaluation process
- It negatively affected the evaluation process
- It had no effect on the evaluation process

[FET Flagship beneficiaries] FET Flagship

Please indicate which Flagships did you participate in*:

- The Human Brain Project (HBP)
- The Graphene Flagship
- Battery 2030+
- Quantum technologies flagship

32. Has the funding received from the Flagship programme enabled you to make significant advancements in your scientific knowledge or technological innovation?

- It has to a large extent
- It has to a fair extent
- It has to a limited extent
- It has not
- I cannot say

If you believe the funding has had any positive impact, can you please provide an example of this impact (max 150 words)?___

33. Did the funding received from the Flagship programme enable you to translate any scientific knowledge into tangible, market-ready technology?

- Yes
- No

If yes, please provide an example (max 150 words)_______

34. Did your position as a beneficiary of the FET Flagship programme help you create collaboration and engagement with other industry partners?

- Yes, I established numerous collaborations
- Yes, I established a fair number of collaborations
- Yes, I established few collaborations
- No, I did not establish any new collaborations

35. In light of the discontinuation of the FET Flagship programme, have you considered applying for calls under the Horizon Europe?

- Yes I have
- No, due to fear of increased competition for funding
- No, due to the shorter duration of the projects
- No, due to eligibility criteria
- No, due to lack of calls that are in line with my expertise
- No, due to (max 150 words)____ ______

36. Do you feel that other funding instruments under Horizon Europe align with the type of research you aim to conduct, particularly by supporting research areas aligned with FET Flagship goals?

- Completely align
- Mostly align
- Partially align
- No alignment
- I do not know

37. What do you perceive as the potential impact of discontinuing the FET Flagships program on R&I activities in Europe? (max 150 words)____

Annex 3 – Interview questions

Impact of discontinuing the FET Flagship

Questions for FET Flagship Beneficiaries

1. Did the funding received from the Flagship programme enable you to translate any scientific knowledge into tangible, market-ready technology? If yes, please provide an example. If not, what were the hurdles to do so?
2. Did the funding contribute to laying the foundation for the future development of market-ready technology, with lasting effects beyond the fellowship period?
3. In part, the FET Flagship programmes aimed to create research communities. In your experience, have you felt its impact toward creating solid and sustainable research communities? What was that effect in the long run (i.e.. Long-lasting collaborations, use of shared resources, structuring effect on the community)?
4. In your opinion, has it been challenging to establish research communities without the flagship programme, and are there any alternative approaches currently in use? If yes, do you believe Horizon Europe offers the same space to be part of a research community as a beneficiary?
5. Where you able to accomplish something in FET Flagship that is not possible to do in the current Horizon Europe? If yes, could you explain the most significant ones?
6. Do you believe Horizon Europe provides you with the necessary opportunities needed to conduct the type of research you would have done under the FET Flagships? If yes, provide an example of an attractive call under HE that you may consider. If not, what is the issue?
7. In light of the discontinuation of the FET Flagship programme, to what extent existing calls under Horizon Europe can replace the flagship funding? If you have contemplated applying for calls under Horizon Europe, what influenced your decision? If not, what are the reasons for your choice?
8. How would you assess the impact of flagships on the European research and innovation landscape in Europe? Do you think the discontinuation of the FET Flagship will have any impact on this, particularly in cases where flagships have fostered self-sustaining developments?
9. Are there specific strategies or initiatives that you recommend to maximize the impact of Horizon Europe in fostering groundbreaking scientific research and innovation in the absence of the FET Flagship programme?
10. Do you think that FET Flagship programme had a negative impact on other instruments from FET (like FET Open, bottom-up scheme), e.g. draining the money from these other instruments and reducing the success rates to the minimal?

Questions for National Contact Points

1. Do you consider certain types of stakeholders more in need of support when it comes to guidance in terms of application process?
2. Can you provide examples of concrete major progress funded by the FET Flagship programme in your country and its significance within the research landscape?
3. How has the discontinuation of the FET Flagship programme under Horizon Europe affected the landscape of research and innovation in your country?
4. How do you perceive the potential of new approaches and instruments, such as Missions or the European Innovation Council (EIC) under Horizon Europe, to fill the void left by the FET Flagship programme in your country?
5. Did you notice any important features from the FET Flagship missing in Horizon Europe? If yes, could you explain the most significant ones?

6. In your opinion, to what extent has Horizon Europe's ability to facilitate the translation of scientific discoveries into marketable innovations been affected by the absence of the FET Flagship programme?
7. Are there specific strategies or initiatives that you recommend to maximise the impact of Horizon Europe in fostering scientific progress and innovation in your country, particularly in light of the discontinuation of the FET Flagship programme?

Perception of HE evaluation system

Questions for National Contact Points

1. How can the evaluation system be better adapted to ensure that the best projects are being funded?
2. Can you identify any specific areas within the current application process that you believe could be simplified without compromising the integrity of the evaluation? Specially in terms of the application process
3. Horizon Europe introduced novelties such as Clusters, Missions, EIC, and the open science policy. Do you think any of these elements influenced the evaluation process? please elaborate.
4. How has the introduction of remote evaluation meetings affected the evaluation process?
5. Do you believe the evaluation process favours certain:
    a. Countries
    b. Disciplines
    c. Organizations
    d. Other: specify

SME barriers to participation

Questions to SMEs beneficiaries and when applicable to SME associations

1. What is in your opinion the biggest hurdle for SMEs in terms of application process (and the reporting while running the project) under Horizon Europe?
2. Do you have a dedicated department/team within your SMEs that helps guide the application process?
3. Did you need to hire external support to help carry out the application process for Horizon Europe?
4. Is it challenging to find suitable calls under Horizon Europe for the type of business you carry?
5. In your experience, do any of the rules of participation put SMEs at a disadvantage?
6. How can Horizon Europe facilitate the application process for SMEs?
7. What steps can Horizon Europe take in order to better support SMEs to produce market-ready innovations?
8. Do you find difficult the financing mechanism of HE where only part of the budget is paid in advance with the rest only at the end of the project (forcing you to cover the workforce from other sources before the final payment?

Questions to private consultancies that support grant proposal writing process

9. Can you provide an overview of your consultancy's role in supporting applicants through the Horizon Europe application process?

FOLLOW UP: What are the main items where you help: *a) project idea; b) consortium assembly; c) impact section; d) budget; e) technical parts of writing (GANTT, PERT, deliverable tables etc)*?

1. Are there instances in your work where you advise not to submit because the idea is not suitable for a particular call or the applicants do not possess the right level of expertise for a "competitive" proposal?
2. Do you cover proposal preparation for all topics considered for Horizon Europe or do you target specific topics? *a) we cover all topics; b) We cover specific topics*. (list which)
3. Do you consistently receive application support requests from particular groups of applicants, categorized by either applicant type or geographical location?
4. What are the most prevalent challenges you observe applicants encounter? Are there common trends by type of applicant?
5. In terms of simplification measures, do you have any recommendations regarding the evaluation process and the application process (a two-step applications, for example) and how they would impact the evaluation? Please elaborate.

Annex 4 – Statistics on the evolution of calls, from FP7 to Horizon Europe

Table 8 – EU contribution and participations by sub-programme (FP7)

Programmes	Total EC contribution (in million euro)	Number of projects	Number of participations	Average EU contribution per project (in 1000 euro)	Average participations per project	Average EU contribution per participation (in 1000 euro)
FP7-COOPERATION	28.336 (64%)	7.912 (31%)	87.623	3.581	11,07	323
FP7-IDEAS	7.673 (17%)	4.525 (18%)	5.405	1.696	1,19	1.420
FP7-PEOPLE	4.777 (11%)	10.715 (43%)	19.515	446	1,82	245
FP7-CAPACITIES	3.772 (8%)	2.025 (8%)	19.047	1.863	9,41	198

Source: Extracted from Commitment and coherence. Ex-Post Evaluation of the 7th EU Framework Programme. (data from eCORDA)

Table 9 – EU contribution and participations by type of action (H2020)

Type of action	EU contribution (in million euro)	Number of participations
RIA	20385 (28.9%)	56638 (29.6%)
ERC	13462 (19.1%)	10278 (5.4%)
IA	11362 (16.1%)	29214 (15.2%)
JTI	6767 (9.6%)	16237 (8.5%)
MSCA	6486 (9.2%)	31836 (16.6%)
CSA	3197 (4.5%)	19770 (10.3%)
SME	2840 (4.0%)	6220 (3.2%)
KICS	2301 (3.3%)	3548 (1.9%)
SGA-RIA	1102 (1.6%)	-
Art185	-	9228 (4.8%)
Others	2716 (3.8%)	8689 (4.5%)

Source: Own elaboration using data from Horizon Dashboard

Table 10 – EU contribution and participations by type of action (Horizon Europe)

Type of action	EU contribution (in million euro)	Number of participations
HORIZON-RIA	95982 (33.5%)	25062 (36.9%)
HORIZON-ERC	43302 (15.1%)	2670 (3.9%)
HORIZON-IA	42804 (14.9%)	10656 (15.7%)
HORIZON-CSA	19588 (6.8%)	7707 (11.3%)
HORIZON-JU-IA	15304 (5.3%)	2288 (3.4%)
HORIZON-JU-RIA	10783 (3.8%)	3022 (4.4%)
HORIZON-EIC	10182 (3.5%)	1959 (2.9%)
HORIZON-EIC-ACC-BF	9917 (3.5%)	-
HORIZON-EIT-KIC	8896 (3.1%)	-
HORIZON-TMA-MSCA-DN	-	4191 (6.2%)
HORIZON-TMA-MSCA-PF-EF	-	3396 (5.0%)
Others	30122 (10.5%)	6972 (10.3%)

Source: Own elaboration using data from Horizon Dashboard

Table 11 – Application numbers for each specific programme (FP7)

Thematic Priority	Retained Proposals	Non-Successful Eligible Proposals	Success Rate Proposals
Marie-Curie Actions	4,325	9,781	31%
European Research Council	1,314	7,968	14%
Information and Communication Technologies	1,130	6,366	15%
Health	492	2,038	19%
Research for the benefit of SMEs	406	1,774	19%
Nanosciences, Nanotechnologies, Materials and new Production Technologies - NMP	336	599	36%
Transport (including Aeronautics)	316	1,032	23%
Research Infrastructures	229	365	39%
Environment (including Climate Change)	206	1,091	16%
Food, Agriculture and Fisheries, and Biotechnology	195	967	17%
Energy	174	774	18%
Joint Technology Initiatives (Annex IV-SP1)	174	227	43%
Socio-economic sciences and Humanities	130	1,229	10%
Research Potential	124	1,393	8%
Security	122	692	15%
Space	113	267	30%
Science in Society	98	360	21%
Nuclear Fission and Radiation Protection	52	69	43%
Activities of International Cooperation	48	118	29%

Thematic Priority	Retained Proposals	Non-Successful Eligible Proposals	Success Rate Proposals
Regions of Knowledge	41	166	20%
General Activities	16	6	73%
Support for the coherent development of research policies	7	12	37%
Fusion Energy	3	6	33%

Source: Own elaboration using data from Horizon Dashboard.

Table 12 – Application numbers for each specific programme (H2020)

Thematic Priority (Programme)	Retained Proposals	Non-Successful Eligible Proposals	Success Rate Proposals
Marie-Sklodowska-Curie Actions	11,381	67,382	14%
European Research Council (ERC)	7,029	47,717	13%
Innovation in SMEs	3,122	38,661	7%
Information and Communication Technologies	1,896	19,114	9%
Smart, green and integrated transport	1,768	6,967	20%
Secure, clean and efficient energy	1,371	9,368	13%
Health, demographic change and wellbeing	1,168	10,463	10%
Food security, sustainable agriculture and forestry, marine and maritime and inland water research and the bioeconomy	908	6,149	13%
Climate action, environment, resource efficiency and raw materials	739	5,758	11%
Future and Emerging Technologies (FET)	632	6,474	9%
Europe in a changing world - inclusive, innovative and reflective societies	433	6,001	7%
Space	427	2,105	17%
Secure societies - protecting freedom and security of Europe and its citizens	412	3,716	10%
Nanotechnologies	377	4,688	7%
Research Infrastructures	312	571	35%
Advanced manufacturing and processing	273	1,547	15%
Cross Theme	226	4,570	5%
Twinning of research institutions	210	1,706	11%
Advanced materials	169	401	30%
Biotechnology	123	1,505	8%
Spreading excellence and widening participation	111	51	69%
Euratom Research and Training Programme	96	162	37%
Teaming of excellent research institutions and low performing RDI regions	79	368	18%

Thematic Priority (Programme)	Retained Proposals	Non-Successful Eligible Proposals	Success Rate Proposals
Science with and for Society	68	145	32%
Societal Challenges	67	2,016	3%
ERA chairs	56	336	14%
Make scientific and technological careers attractive to young students, and forster sustainable interaction between schools, research institutions, industry and civil society organisations	50	374	12%
Integrate society in science and innovation issues, policies and activities in order to integrate citizens' interests and values and to increase the quality, relevance, social acceptability and sustainability of research and innovation outcomes in various fields of activity from social innovation to areas such as biotechnology and nanotechnology	43	350	11%
Develop the governance for the advancement of responsible research and innovation by all stakeholders, which is sensitive to society needs and demands and promote an ethics framework for research and innovation	37	160	19%
Promote gender equality in particular by supporting structural change in the organisation of research institutions and in the content and design of research activities	30	208	13%
Encourage citizens to engage in science through formal and informal science education, and promote the diffusion of science-based activities, namely in science centres and through other appropriate channels	15	114	12%
Access to risk finance	11	40	22%
Improving knowledge on science communication in order to improve the quality and effectiveness of interactions between scientists, general media and the public	8	42	16%
Strengthening the administrative and operational capacity of transnational networks of National Contact Points	5	0	100%
Develop the accessibility and the use of the results of publicly-funded research	3	10	23%
Industrial Leadership	2	15	12%
Leadership in enabling and industrial technologies (LEIT)	2	12	14%
Take due and proportional precautions in research and innovation activities by anticipating and assessing potential environmental, health and safety impacts	1	1	50%
Supporting access to international networks for excellent researchers and innovators who lack sufficient involvement in European and international networks	1	0	100%

Source: Own elaborate using data from Horizon Dashboard

Table 13 – Application numbers for each specific programme (FP7)

Thematic Priority (Programme)	Retained Proposals	Non-Successful Eligible Proposals	Success Rate Proposals
European Research Council (ERC)	3,132	18,331	15%
Marie Skłodowska-Curie Actions (MSCA)	2,947	14,966	16%
Digital, Industry and Space	888	3,484	20%
Climate, Energy and Mobility	714	2,290	24%
Food, Bioeconomy Natural Resources, Agriculture and Environment	482	1,312	27%
Health	437	1,572	22%
The European Innovation Council (EIC)	417	4,480	9%
Widening participation and spreading excellence	257	642	29%
Culture, creativity and inclusive society	221	1,282	15%
Research infrastructures	130	102	56%
European innovation ecosystems	107	523	17%
Civil Security for Society	94	556	14%
Reforming and enhancing the European R&I System	74	120	38%
The European Institute of Innovation and Technology (EIT)	17	3	85%

Source: Own elaboration using data from CORDIS Database

Table 14 – Number of beneficiaries and projects by type of call (FP7)

Types of calls	Beneficiary entities	Projects	Calls
Mono-participant	13,240	13,240	15
Multi-participant	126,768	12,544	460
Total	140,008	25,784	475

Source: Own elaboration using data from CORDIS Database

Table 15 – Number of beneficiaries and projects by type of call (H2020)

Types of calls	Beneficiary entities	Projects	Calls
Mono-participant	20,740	20,470	91
Multi-participant	157,571	14,661	393
Total	178,311	35,131	484

Source: Own elaboration using data from CORDIS Database

Table 16 – Number of beneficiaries and projects by type of call (Horizon Europe)

Types of calls	Beneficiary entities	Projects	Calls
Mono-participant	4,331	4,331	38
Multi-participant	5,2281	4,817	245
Total	56612	9148	283

Source: Own elaboration using data from CORDIS Database

Horizon Europe is the EU's key funding programme for research and innovation, containing ambitious commitments to scientific progress, climate neutrality, and improving the EU's competitiveness and growth. This study evaluates the following selected items in Horizon Europe: the evolution of calls and funding, adoption of the Common Model Grant Agreement (CMGA), implementation of the strategic plan, discontinuation of the Future and Emerging Technologies (FET) Flagship, the participation barriers faced by SMEs, and stakeholders' views on the evaluation system. Drawing on surveys and interviews, the study aims to identify both the key strengths and shortcomings of Horizon Europe in terms of these selected items. Lastly, the study suggests ways to mitigate the shortcomings highlighted by stakeholders.

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